Multiverse analysis

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The authors made the following contributions. First Author: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; Ernst-August Doelle: Writing - Review & Editing, Supervision.

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Keywords: keywords

Word count: X

Multiverse analysis

Results

Study characteristics

Table 1 describes each study in the primary analytic sample, and Table 2 shows how each study's measure of socioeconomic status was harmonised.

Model diagnostics

The model diagnostics for the main analysis, and for all models presented in this document are in Table 3

Fixed Effect Model

In this document we show the results using another analytic technique. Namely treating study ID as a fixed effect. This is what we wrote in the protocol originally.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using fixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 4 and Figure 1.

The association between physical activity intensity and sleep. We estimated how physical activity intensity is associated with sleep across different age groups. We present the results controlling for SES, sex, BMI z-score, and the fixed effects of study ID, in Table 4 and Figure 2.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age.

Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 3.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 4.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 5.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 5 and Figure 6.

Log Transforming PA Volume

Based on preliminary modelling on a subset of the data, we chose to leave physical activity volume in its natural form to enhance interpretability. The log scale meant that the variable became unintepretable towards the edge of its range. Below, we present the results had we not made this decision. These results are all moderated by age, as this was our primary analysis. The associated figures are back-transformed to the scale range.

The association between physical activity volume and sleep. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 6 and Figure 7.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 8.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 9.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 10.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 7 and Figure 11.

All Eligible Participants Regardless of Valid Wear Days

The primary analysis required at least 4 days of valid wear time. Here we refit the primary models to every eligible participant, regardless of how many valid days they contributed. Both analyses share the same imputations and standardisation constants. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 8 and 9 and Figures 12 to 17.

Participants With at Least Seven Valid Days

Some recommend at least seven nights to estimate sleep regularity. Here we refit the primary models to the 14,981 participants with at least seven days of valid wear time. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 10 and 11 and Figures 18 to 23.

Moderation

In the following section we present models for a range of moderators which were pre-defined in our protocol.

BMI z-score

The association between physical activity volume and sleep by BMI z-score. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by BMI z-score, and the results are presented in Table 12 and Figure 24

The association between physical activity intensity and sleep by BMI z-score. We estimated how physical activity intensity is associated with sleep across BMI z-score. We present the results in Table 12 and Figure 25.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 26.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 27.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 13 and Figure 28.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by BMI z-score. Results are presented in Table 13 and Figure 29.

SES

The association between physical activity volume and sleep by SES. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by SES, and the results are presented in Table 14 and Figure 30

The association between physical activity intensity and sleep by SES. We estimated how physical activity intensity is associated with sleep across SES. We present the results in Table 14 and Figure 31.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 32.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 33.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 15 and Figure 34.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by SES. Results are presented in Table 15 and Figure 35.

sex

The association between physical activity volume and sleep by sex. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by sex, and the results are presented in Table 16 and Figure 36

The association between physical activity intensity and sleep by sex. We estimated how physical activity intensity is associated with sleep across sex. We present the results in Table 16 and Figure 37.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 38.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 39.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 17 and Figure 40.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by sex. Results are presented in Table 17 and Figure 41.

weekday

The association between physical activity volume and sleep by weekday. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by weekday, and the results are presented in Table 18 and Figure 42

The association between physical activity intensity and sleep by weekday. We estimated how physical activity intensity is associated with sleep across weekday. We present the results in Table 18 and Figure 43.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 44.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 45.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 19 and Figure 46.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by weekday. Results are presented in Table 19 and Figure 47.

season

The association between physical activity volume and sleep by season. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by season, and the results are presented in Table 20 and Figure 48

The association between physical activity intensity and sleep by season. We estimated how physical activity intensity is associated with sleep across season. We present the results in Table 20 and Figure 49.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 50.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 51.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 21 and Figure 52.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by season. Results are presented in Table 21 and Figure 53.

region

The association between physical activity volume and sleep by region. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by region, and the results are presented in Table 22 and Figure 54

The association between physical activity intensity and sleep by region. We estimated how physical activity intensity is associated with sleep across region. We present the results in Table 22 and Figure 55.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 56.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 57.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 23 and Figure 58.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by region. Results are presented in Table 23 and Figure 59.

daylight hours

The association between physical activity volume and sleep by daylight hours. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by daylight hours, and the results are presented in Table 24 and Figure 60

The association between physical activity intensity and sleep by daylight hours. We estimated how physical activity intensity is associated with sleep across daylight hours. We present the results in Table 24 and Figure 61.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 62.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 63.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 25 and Figure 64.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by daylight hours. Results are presented in Table 25 and Figure 65.

wear location

The association between physical activity volume and sleep by wear location. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by wear location, and the results are presented in Table 26 and Figure 66

The association between physical activity intensity and sleep by wear location. We estimated how physical activity intensity is associated with sleep across wear location. We present the results in Table 26 and Figure 67.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 68.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 69.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 70.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by wear location. Results are presented in Table 27 and Figure 71.

most active hour

The association between physical activity volume and sleep by most active hour. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by most active hour, and the results are presented in Table 28 and Figure 72

The association between physical activity intensity and sleep by most active hour. We estimated how physical activity intensity is associated with sleep across most active hour. We present the results in Table 28 and Figure 73.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 74.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 75.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 76.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by most active hour. Results are presented in Table 29 and Figure 77.

ethnicity

The association between physical activity volume and sleep by ethnicity. We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by ethnicity, and the results are presented in Table 30 and Figure 78

The association between physical activity intensity and sleep by ethnicity. We estimated how physical activity intensity is associated with sleep across ethnicity. We present the results in Table 30 and Figure 79.

The association between sleep duration and next-day physical activity. We estimated the association between sleep duration and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 80.

The association between sleep efficiency and next-day physical activity. We estimated the association between sleep efficiency and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 81.

The association between sleep onset and next-day physical activity. We estimated the association between sleep onset and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 82.

The association between sleep regularity and next-day physical activity. We estimated the association between sleep regularity and next-day physical activity by ethnicity. Results are presented in Table 31 and Figure 83.

Table 1

Characteristics of the included studies (primary analytic sample)

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range
100			12 countries	GT3X	Hip	6,360	10.4 (9-11)
101			Switzerland	GENEActiv	Wrist	2,928	61.9 (45-86)
102			Denmark	GT3X	Hip	691	10.0 (8-11)
103			New Zealand	Actigraph	Wrist	651	19.3 (8-58)
104			Brazil	GT3X+	Wrist	737	16.4 (14-22)
105			Czech Republic	GT3X+, GT9X Link, wGT3X-BT, wGT3X+	Wrist	1,170	14.0 (8-19)
106			Australia	GENEActiv	Wrist	301	42.0 (18-65)
107			Spain	GT3X+	Wrist	2,423	72.6 (64-94)
108			Australia	GENEActiv	Wrist	1,038	8.8 (7-11)
110			Finland	Actisleep	Wrist	801	62.6 (58-64)
111			United Kingdom	Axivity AX3	Wrist	692	44.7 (20-70)
112			Chile	GENEActiv	Wrist	144	9.5 (8-12)
113			Spain	Actisleep	Wrist	353	8.4 (8-9)
114			United States	WGT3X-BT	Wrist	154	3.5 (2-4)
115			Ecuador	wGT3X-BT	Hip	1,019	8.8 (7-13)
117			Spain	GENEActiv	Wrist	175	8.8 (4-11)
118			Australia	Actigraph	Wrist	304	16.6 (15-18)
212			Spain	WGT3X	Hip	285	4.7 (2-6)
221			United Kingdom	GENEActiv	Wrist	4,071	69.4 (59-82)
222			Brazil	GT3X	Wrist	2,939	3.8 (3-4)

Table 2

Harmonisation of socioeconomic status

Study ID	Original category
100	1
100	2
100	4
100	3
101	High school
101	Apprenticeship
101	Mandatory education
101	University education
102	Vocational basic training/vocational internship and main course
102	Short higher education
102	Long higher education/ Research degree
102	Regular secondary education and vocational secondary education
102	Medium higher education
102	Primary school
102	Unanswered
104	Incomplete elementary school
104	Complete Bachelor degree
104	Incomplete bachelor degree
104	Complete high school
104	Complete elementary school
104	Unknown
104	Has not studied
105	vocational education
105	high school education
105	master's degree or higher
105	bachelor's degree or equivalent
105	10th grade or less
105	short tertiary education
105	26-100 books
105	11-25 books
105	>200 books
105	101-200 books
105	0-10 books
106	High school or less

Table 2 continued

Study ID	Original category
106	Technical or further education institutions (e.g., TAFE)
106	University degree
107	Secondary education 2nd stage (i.e., High shool graduate, complete 12th grade or equivalent
107	Incomplete primary education (i.e., less than 6th grade or equivalent)
107	Complete primary education (i.e., complete 6th grade or equivalent)
107	Medium university studies (i.e., university diplomature or equivalent)
107	Higher university studies (i.e., Bachelor degree, Master degree, Doctoral degree or equivalen
107	Secondary education 1st stage (i.e., complete 10th grade or equivalent)
107	Illiterate (i.e., not able to read or write)
108	Enough to fill two bookcases (101-200 Books)
108	Enough to fill one shelf (11-25 Books)
108	Enough to fill 3 or more bookcases (more than 200 Books)
108	Enough to fill one bookcases (26-100 books)
108	None or very few (0-10 Books)
110	lower-level non-manual occupation
110	higher-level non-manual occupation
110	manual occupation
110	service occupation
111	10
113	primary school
113	university degree
113	bachelor's degree
113	high school
113	master's degree or higher
113	vocational training
114	poor
114	O level
114	A/S level
114	ba/bsc
114	higher degree
114	No academic quals
115	Medium
115	High
115	Low
117	secondary education

Table 2 continued

Study ID	Original category
117	higher education
117	primary education
118	2
118	7
118	5
118	3
118	10
118	8
118	9
118	6
118	4
118	1
221	O level
221	A/S level
221	ba/bsc
221	higher degree
221	No academic quals
222	q5-wealthiest
222	q4
222	q3
222	q2
222	q1-poorest

Table 3

Model diagnostics

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.41	9.98	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.43	10.01	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.73	5.49	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.72	100.00%	0.00%
Models moderated by age fixedef				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.41	9.98	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.43	10.01	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.73	5.49	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.72	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age log				
Scale sleep duration by log PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by log PA volume	-0.96	3.84	100.00%	0.00%
Scale sleep onset by log PA volume	0.42	10.00	100.00%	0.00%
Scale sleep regularity by log PA volume	-0.67	1.49	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.43	10.01	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Log PA volume by sleep duration(lagged)	-0.46	2.22	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Log PA volume by sleep efficiency(lagged)	-0.47	2.24	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Log PA volume by sleep onset(lagged)	-0.47	2.24	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Log PA volume by sleep regularity(lagged)	-0.37	2.03	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.72	100.00%	0.00%
Models moderated by age allwear				
Scale sleep duration by PA volume	-0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.98	3.92	100.00%	0.00%
Scale sleep onset by PA volume	0.41	10.00	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.98	3.92	100.00%	0.00%
Scale sleep onset by PA intensity	0.43	10.03	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.37	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.73	5.49	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.26	1.72	100.00%	0.00%
Models moderated by age 7days				

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA volume	-0.01	2.81	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.87	3.15	100.00%	0.00%
Scale sleep onset by PA volume	0.39	12.26	100.00%	0.00%
Scale sleep regularity by PA volume	-0.68	1.49	100.00%	0.00%
Scale sleep duration by PA intensity	-0.01	2.80	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.87	3.15	100.00%	0.00%
Scale sleep onset by PA intensity	0.41	12.34	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.68	1.48	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.57	5.89	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.21	1.58	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.57	5.96	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.21	1.57	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.57	5.90	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.21	1.57	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.61	6.09	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.59	100.00%	0.00%
Models moderated by bmi				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.97	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.42	9.96	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.45	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by ses				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA volume	-0.97	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.42	9.97	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.46	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.44	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by weekday				
Scale sleep duration by PA volume	-0.08	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.39	10.37	100.00%	0.00%
Scale sleep regularity by PA volume	-0.71	1.73	100.00%	0.00%
Scale sleep duration by PA intensity	-0.08	2.16	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.96	3.85	100.00%	0.00%
Scale sleep onset by PA intensity	0.41	10.28	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.72	1.73	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.74	5.48	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.28	1.77	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.73	5.50	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.28	1.76	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.74	5.49	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.30	1.79	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.53	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.29	1.75	100.00%	0.00%
Models moderated by season				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.97	3.83	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep onset by PA volume	0.42	9.96	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.46	100.00%	0.00%
Scale sleep duration by PA intensity	-0.07	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.35	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.44	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by region				
Scale sleep duration by PA volume	-0.06	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.85	100.00%	0.00%
Scale sleep onset by PA volume	0.42	10.00	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.07	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.85	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	10.01	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.35	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.72	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.69	5.39	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.33	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.73	5.47	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.72	100.00%	0.00%
Models moderated by daylight				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.97	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.42	9.96	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep regularity by PA volume	-0.67	1.46	100.00%	0.00%
Scale sleep duration by PA intensity	-0.07	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.69	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.44	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by wear location				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.41	9.99	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.83	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	10.00	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.45	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.47	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.27	1.73	100.00%	0.00%
Models moderated by PA mostactivehr				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.43	9.99	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.47	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA intensity	-0.07	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.45	10.01	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	5.37	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.39	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.46	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by sex				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.97	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.42	9.96	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.46	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.44	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.71	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.36	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.44	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%
Models moderated by ethnicity				
Scale sleep duration by PA volume	-0.06	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	-0.96	3.83	100.00%	0.00%
Scale sleep onset by PA volume	0.42	9.98	100.00%	0.00%
Scale sleep regularity by PA volume	-0.67	1.46	100.00%	0.00%
Scale sleep duration by PA intensity	-0.06	2.14	100.00%	0.00%

Table 3 continued

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA intensity	-0.97	3.84	100.00%	0.00%
Scale sleep onset by PA intensity	0.44	9.99	100.00%	0.00%
Scale sleep regularity by PA intensity	-0.67	1.43	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.37	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.72	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.38	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.35	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.70	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.72	5.45	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.25	1.71	100.00%	0.00%

Table 4

Physical activity predicting sleep controlling for SES, sex, BMI z-score, and the fixed effects of study IDs.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.69 [0.65, 0.73]	< .001	0.59 [0.54, 0.63]	< .001
Physical activity	0.09 [0.08, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.03, 0.00]	.042	-0.01 [-0.03, 0.00]	.053
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [0.00, 0.01]	.476
Physical activity × Age	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [-0.01, 0.00]	.067	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.50 [-0.54, -0.46]	< .001	-0.57 [-0.61, -0.52]	< .001
Physical activity	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Age	-0.01 [-0.03, 0.00]	.173	-0.02 [-0.03, 0.00]	.038
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Age	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.151	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-1.11 [-1.15, -1.07]	< .001	-1.01 [-1.05, -0.97]	< .001
Physical activity	-0.11 [-0.13, -0.10]	< .001	-0.06 [-0.07, -0.05]	< .001
Age	-0.01 [-0.03, 0.00]	.029	-0.01 [-0.02, 0.01]	.243
Physical activity^2	0.02 [0.01, 0.03]	< .001	-0.01 [-0.02, -0.01]	< .001
Physical activity × Age	-0.01 [-0.01, -0.01]	< .001	0.01 [0.00, 0.01]	< .001
Age × Physical activity^2	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-0.03 [-0.09, 0.04]	.456	-0.36 [-0.43, -0.29]	< .001
Physical activity	0.24 [0.22, 0.26]	< .001	0.12 [0.11, 0.13]	< .001

Table 4 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.01, 0.04]	.294	-0.05 [-0.07, -0.03]	$<$ .001
Physical activity^2	-0.10 [-0.11, -0.09]	$<$ .001	-0.08 [-0.08, -0.07]	$<$ .001
Physical activity $\times$ Age	-0.01 [-0.01, 0.00]	.010	-0.04 [-0.05, -0.04]	$<$ .001
Age $\times$ Physical activity^2	-0.02 [-0.02, -0.01]	$<$ .001	0.01 [0.01, 0.01]	$<$ .001

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 5

Sleep predicting physical activity controlling for SES, sex, BMI z-score, and the fixed effects of study IDs

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-1.08 [-1.12, -1.05]	< .001	-0.40 [-0.44, -0.37]	< .001
Sleep duration	0.02 [0.01, 0.02]	< .001	0.02 [0.02, 0.03]	< .001
Age	-0.23 [-0.24, -0.22]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.076
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.229
Sleep efficiency (z)				
(Intercept)	-1.07 [-1.10, -1.03]	< .001	-0.38 [-0.41, -0.35]	< .001
Sleep efficiency	0.00 [0.00, 0.01]	.558	0.01 [0.00, 0.01]	.039
Age	-0.23 [-0.24, -0.21]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.462
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.182
Sleep onset (z)				
(Intercept)	-1.10 [-1.14, -1.06]	< .001	-0.44 [-0.47, -0.40]	< .001
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.22 [-0.24, -0.21]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.415
Sleep onset × Age	0.00 [0.00, 0.01]	.003	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.013	0.00 [0.00, 0.00]	.051
Sleep regularity (z)				
(Intercept)	-1.02 [-1.06, -0.99]	< .001	-0.36 [-0.39, -0.33]	< .001
Sleep regularity	0.09 [0.08, 0.09]	< .001	0.05 [0.05, 0.06]	< .001

Table 5 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.22 [-0.23, -0.21]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.596	0.00 [0.00, 0.00]	.864
Sleep regularity $\times$ Age	-0.02 [-0.02, -0.01]	$< .001$	-0.01 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.619	0.00 [0.00, 0.00]	.209

Note. Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 6

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.16 [-0.10, 0.42]	.237	-0.01 [-0.15, 0.13]	.918
Log pa volume	-0.26 [-0.39, -0.13]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.04 [-0.12, 0.03]	.275	-0.02 [-0.03, 0.00]	.017
Log pa volume^2	0.06 [0.04, 0.08]	< .001	0.00 [0.00, 0.01]	.492
Log pa volume × Age	0.04 [-0.01, 0.08]	.088	-0.01 [-0.01, -0.01]	< .001
Age × Log pa volume^2	-0.01 [-0.01, 0.00]	.010	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-1.06 [-1.32, -0.81]	< .001	0.15 [-0.02, 0.31]	.088
Log pa volume	0.64 [0.52, 0.76]	< .001	0.02 [0.01, 0.03]	< .001
Age	0.19 [0.11, 0.27]	< .001	-0.01 [-0.03, 0.00]	.093
Log pa volume^2	-0.08 [-0.10, -0.07]	< .001	-0.02 [-0.02, -0.01]	< .001
Log pa volume × Age	-0.08 [-0.12, -0.04]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Log pa volume^2	0.01 [0.00, 0.01]	.030	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	-0.31 [-0.63, 0.02]	.063	0.13 [-0.13, 0.38]	.325
Log pa volume	0.44 [0.33, 0.54]	< .001	-0.06 [-0.07, -0.05]	< .001
Age	0.29 [0.22, 0.36]	< .001	-0.01 [-0.02, 0.01]	.302
Log pa volume^2	-0.09 [-0.10, -0.07]	< .001	-0.01 [-0.02, -0.01]	< .001
Log pa volume × Age	-0.15 [-0.19, -0.11]	< .001	0.01 [0.00, 0.01]	< .001
Age × Log pa volume^2	0.02 [0.01, 0.03]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	-3.57 [-3.86, -3.27]	< .001	0.08 [-0.07, 0.23]	.291
Log pa volume	1.72 [1.54, 1.89]	< .001	0.12 [0.11, 0.13]	< .001

Table 6 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.33 [-0.43, -0.23]	$< .001$	-0.05 [-0.07, -0.02]	$< .001$
Log pa volume^2	-0.19 [-0.22, -0.16]	$< .001$	-0.08 [-0.08, -0.07]	$< .001$
Log pa volume $\times$ Age	0.21 [0.15, 0.27]	$< .001$	-0.04 [-0.05, -0.04]	$< .001$
Age $\times$ Log pa volume^2	-0.03 [-0.04, -0.02]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 7

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	-0.07 [-0.19, 0.05]	.238
Sleep duration	0.00 [0.00, 0.01]	< .001	0.02 [0.02, 0.03]	< .001
Age	-0.13 [-0.14, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.076
Age × Sleep duration^2	0.00 [0.00, 0.00]	.913	0.00 [0.00, 0.00]	.227
Sleep efficiency (z)				
(Intercept)	3.65 [3.51, 3.79]	< .001	-0.07 [-0.19, 0.05]	.224
Sleep efficiency	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	.037
Age	-0.13 [-0.14, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.006	0.00 [0.00, 0.00]	.455
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.182
Sleep onset (z)				
(Intercept)	3.65 [3.51, 3.80]	< .001	-0.08 [-0.21, 0.04]	.181
Sleep onset	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.13 [-0.13, -0.12]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.425
Sleep onset × Age	0.00 [0.00, 0.00]	< .001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.052
Sleep regularity (z)				
(Intercept)	3.65 [3.51, 3.78]	< .001	-0.09 [-0.20, 0.02]	.126
Sleep regularity	0.06 [0.06, 0.06]	< .001	0.05 [0.05, 0.06]	< .001

Table 7 continued

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.13 [-0.13, -0.12]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	$< .001$	0.00 [0.00, 0.00]	.855
Sleep regularity $\times$ Age	-0.01 [-0.01, -0.01]	$< .001$	-0.01 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	.206

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised and (ln) natural-log transformed; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 8

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.01 [-0.17, 0.14]	.888	-0.01 [-0.15, 0.13]	.868
Physical activity	0.09 [0.08, 0.11]	< .001	0.05 [0.04, 0.06]	< .001
Age	-0.02 [-0.03, 0.00]	.011	-0.02 [-0.03, 0.00]	.013
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [0.00, 0.01]	.571
Physical activity × Age	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [-0.01, 0.00]	.090	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.14 [-0.01, 0.30]	.072	0.14 [-0.02, 0.31]	.095
Physical activity	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Age	-0.01 [-0.02, 0.01]	.490	-0.01 [-0.03, 0.00]	.125
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Age	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.00]	.227	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.10 [-0.16, 0.37]	.456	0.12 [-0.13, 0.37]	.334
Physical activity	-0.11 [-0.13, -0.10]	< .001	-0.06 [-0.07, -0.05]	< .001
Age	-0.01 [-0.02, 0.00]	.077	-0.01 [-0.02, 0.01]	.404
Physical activity^2	0.02 [0.01, 0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Physical activity × Age	-0.01 [-0.01, -0.01]	< .001	0.01 [0.00, 0.01]	< .001
Age × Physical activity^2	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.15 [0.03, 0.27]	.012	0.08 [-0.07, 0.23]	.285
Physical activity	0.24 [0.22, 0.26]	< .001	0.12 [0.11, 0.13]	< .001

Table 8 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.01 [-0.01, 0.04]	.209	-0.05 [-0.07, -0.02]	$< .001$
Physical activity^2	-0.10 [-0.11, -0.09]	$< .001$	-0.08 [-0.08, -0.07]	$< .001$
Physical activity $\times$ Age	-0.01 [-0.01, 0.00]	.014	-0.04 [-0.05, -0.04]	$< .001$
Age $\times$ Physical activity^2	-0.02 [-0.02, -0.02]	$< .001$	0.01 [0.01, 0.01]	$< .001$

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 9

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [-0.17, 0.43]	.393	-0.07 [-0.19, 0.05]	.232
Sleep duration	0.02 [0.01, 0.02]	< .001	0.02 [0.02, 0.03]	< .001
Age	-0.23 [-0.24, -0.22]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.01, -0.01]	< .001	0.00 [0.00, 0.00]	.068
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.263
Sleep efficiency (z)				
(Intercept)	0.12 [-0.18, 0.41]	.430	-0.08 [-0.20, 0.04]	.219
Sleep efficiency	0.00 [0.00, 0.01]	.566	0.01 [0.00, 0.01]	.041
Age	-0.23 [-0.24, -0.21]	< .001	-0.27 [-0.28, -0.26]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.467
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.159
Sleep onset (z)				
(Intercept)	0.12 [-0.18, 0.42]	.429	-0.09 [-0.21, 0.04]	.175
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age	-0.22 [-0.23, -0.21]	< .001	-0.27 [-0.29, -0.26]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.481
Sleep onset × Age	0.00 [0.00, 0.01]	.003	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	.015	0.00 [0.00, 0.00]	.044
Sleep regularity (z)				
(Intercept)	0.10 [-0.18, 0.39]	.476	-0.09 [-0.20, 0.02]	.122
Sleep regularity	0.09 [0.08, 0.09]	< .001	0.05 [0.05, 0.06]	< .001

Table 9 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.22 [-0.23, -0.21]	$< .001$	-0.27 [-0.28, -0.26]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.655	0.00 [0.00, 0.00]	.869
Sleep regularity $\times$ Age	-0.02 [-0.02, -0.01]	$< .001$	-0.01 [-0.02, -0.01]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.642	0.00 [0.00, 0.00]	.173

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 10

Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.03 [-0.28, 0.22]	.824	-0.04 [-0.27, 0.20]	.750
Physical activity	0.09 [0.07, 0.11]	< .001	0.05 [0.04, 0.07]	< .001
Age	0.03 [0.01, 0.04]	.001	0.03 [0.01, 0.05]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.690
Physical activity × Age	-0.01 [-0.01, 0.00]	.013	-0.01 [-0.01, 0.00]	.001
Age × Physical activity^2	0.00 [0.00, 0.00]	.581	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.13 [-0.16, 0.42]	.380	0.12 [-0.19, 0.43]	.447
Physical activity	0.05 [0.04, 0.06]	< .001	0.03 [0.02, 0.04]	< .001
Age	-0.04 [-0.06, -0.03]	< .001	-0.05 [-0.07, -0.03]	< .001
Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.02 [-0.03, -0.02]	< .001
Physical activity × Age	-0.03 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Age × Physical activity^2	0.00 [0.00, 0.01]	.049	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.03 [-0.34, 0.41]	.857	0.04 [-0.31, 0.40]	.806
Physical activity	-0.09 [-0.10, -0.08]	< .001	-0.05 [-0.06, -0.04]	< .001
Age	-0.07 [-0.08, -0.06]	< .001	-0.06 [-0.07, -0.05]	< .001
Physical activity^2	0.01 [0.00, 0.02]	< .001	-0.01 [-0.01, 0.00]	.005
Physical activity × Age	-0.02 [-0.03, -0.02]	< .001	0.00 [-0.01, 0.00]	.175
Age × Physical activity^2	0.01 [0.00, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.16 [0.04, 0.28]	.010	0.09 [-0.09, 0.28]	.327
Physical activity	0.23 [0.20, 0.25]	< .001	0.13 [0.11, 0.14]	< .001

Table 10 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	0.02 [-0.01, 0.05]	.210	-0.05 [-0.08, -0.01]	.006
Physical activity^2	-0.09 [-0.10, -0.08]	$<$ .001	-0.09 [-0.10, -0.08]	$<$ .001
Physical activity $\times$ Age	-0.01 [-0.02, -0.01]	$<$ .001	-0.04 [-0.05, -0.04]	$<$ .001
Age $\times$ Physical activity^2	-0.02 [-0.02, -0.01]	$<$ .001	0.02 [0.01, 0.02]	$<$ .001

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 11

Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.02 [-0.37, 0.33]	.917	-0.11 [-0.25, 0.04]	.145
Sleep duration	0.02 [0.02, 0.03]	< .001	0.04 [0.03, 0.05]	< .001
Age	-0.19 [-0.20, -0.18]	< .001	-0.23 [-0.25, -0.22]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep duration × Age	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Age × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.03 [-0.37, 0.32]	.881	-0.11 [-0.25, 0.03]	.137
Sleep efficiency	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.01]	.008
Age	-0.18 [-0.20, -0.17]	< .001	-0.23 [-0.24, -0.22]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.434	0.00 [0.00, 0.00]	.347
Age × Sleep efficiency^2	0.00 [0.00, 0.00]	.015	0.00 [0.00, 0.00]	.006
Sleep onset (z)				
(Intercept)	-0.02 [-0.38, 0.34]	.927	-0.12 [-0.27, 0.03]	.120
Sleep onset	-0.05 [-0.06, -0.04]	< .001	-0.02 [-0.03, -0.01]	< .001
Age	-0.19 [-0.20, -0.18]	< .001	-0.23 [-0.25, -0.22]	< .001
Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.069
Sleep onset × Age	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.180
Sleep regularity (z)				
(Intercept)	-0.04 [-0.37, 0.29]	.809	-0.12 [-0.25, 0.01]	.066
Sleep regularity	0.08 [0.07, 0.08]	< .001	0.06 [0.05, 0.06]	< .001

Table 11 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	-0.18 [-0.19, -0.17]	$< .001$	-0.23 [-0.24, -0.22]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.853	0.00 [0.00, 0.00]	.610
Sleep regularity $\times$ Age	-0.01 [-0.02, -0.01]	$< .001$	-0.02 [-0.02, -0.02]	$< .001$
Age $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.562	0.00 [0.00, 0.00]	.462

Note. Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 12

Physical activity predicting sleep controlling for SES, age, and sex.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [-0.15, 0.16]	.950	0.01 [-0.14, 0.15]	.930
Physical activity	0.10 [0.09, 0.12]	< .001	0.06 [0.05, 0.08]	< .001
BMI z-score	-0.01 [-0.01, 0.00]	.115	-0.02 [-0.03, -0.01]	< .001
Physical activity^2	0.00 [-0.01, 0.00]	.023	0.01 [0.00, 0.01]	< .001
Physical activity × BMI z-score	0.01 [0.00, 0.02]	.002	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.517	0.01 [0.00, 0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.17 [0.01, 0.32]	.035	0.16 [0.00, 0.32]	.049
Physical activity	0.04 [0.03, 0.06]	< .001	0.00 [0.00, 0.01]	.282
BMI z-score	-0.01 [-0.02, 0.00]	.006	-0.02 [-0.03, -0.01]	< .001
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.402
Physical activity × BMI z-score	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.02]	< .001
BMI z-score × Physical activity^2	0.00 [-0.01, 0.00]	.002	0.00 [0.00, 0.00]	.771
Sleep onset (z)				
(Intercept)	0.12 [-0.14, 0.39]	.358	0.12 [-0.13, 0.37]	.344
Physical activity	-0.11 [-0.12, -0.10]	< .001	-0.08 [-0.09, -0.08]	< .001
BMI z-score	0.00 [-0.01, 0.01]	.711	0.00 [0.00, 0.01]	.188
Physical activity^2	0.00 [0.00, 0.01]	.003	-0.01 [-0.01, 0.00]	< .001
Physical activity × BMI z-score	0.00 [-0.01, 0.00]	.415	-0.01 [-0.01, -0.01]	< .001
BMI z-score × Physical activity^2	0.00 [0.00, 0.00]	.160	-0.01 [-0.01, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	0.13 [0.01, 0.25]	.029	0.14 [0.00, 0.28]	.051
Physical activity	0.25 [0.23, 0.26]	< .001	0.08 [0.07, 0.09]	< .001

Table 12 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.04 [-0.05, -0.03]	$< .001$	-0.05 [-0.06, -0.04]	$< .001$
Physical activity^2	-0.05 [-0.05, -0.04]	$< .001$	-0.02 [-0.02, -0.01]	$< .001$
Physical activity $\times$ BMI z-score	0.06 [0.05, 0.06]	$< .001$	0.03 [0.02, 0.04]	$< .001$
BMI z-score $\times$ Physical activity^2	-0.01 [-0.01, -0.01]	$< .001$	-0.01 [-0.01, 0.00]	.001

Note. Adjusted for SES, age, and sex. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 13

Sleep predicting physical activity controlling for SES, age, and sex

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.16, 0.43]	.364	-0.07 [-0.19, 0.05]	.243
Sleep duration	0.01 [0.01, 0.02]	< .001	0.02 [0.02, 0.03]	< .001
BMI z-score	-0.05 [-0.06, -0.04]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × BMI z-score	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.01]	.281
BMI z-score × Sleep duration^2	0.00 [0.00, 0.00]	.487	0.00 [0.00, 0.00]	.162
Sleep efficiency (z)				
(Intercept)	0.12 [-0.17, 0.42]	.411	-0.07 [-0.19, 0.05]	.229
Sleep efficiency	0.00 [0.00, 0.01]	.204	0.01 [0.00, 0.01]	.027
BMI z-score	-0.05 [-0.06, -0.04]	< .001	-0.06 [-0.06, -0.05]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × BMI z-score	-0.01 [-0.01, 0.00]	< .001	0.00 [-0.01, 0.00]	.472
BMI z-score × Sleep efficiency^2	0.00 [0.00, 0.00]	.476	0.00 [0.00, 0.00]	.599
Sleep onset (z)				
(Intercept)	0.12 [-0.17, 0.42]	.416	-0.08 [-0.20, 0.04]	.196
Sleep onset	-0.02 [-0.03, -0.02]	< .001	-0.01 [-0.02, -0.01]	< .001
BMI z-score	-0.05 [-0.06, -0.05]	< .001	-0.06 [-0.07, -0.05]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.001	0.00 [0.00, 0.01]	.020
Sleep onset × BMI z-score	0.00 [-0.01, 0.00]	.625	0.00 [-0.01, 0.00]	.119
BMI z-score × Sleep onset^2	0.00 [0.00, 0.01]	.008	0.00 [0.00, 0.00]	.648
Sleep regularity (z)				
(Intercept)	0.11 [-0.18, 0.39]	.466	-0.09 [-0.20, 0.03]	.144
Sleep regularity	0.09 [0.08, 0.09]	< .001	0.05 [0.05, 0.06]	< .001

Table 13 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score	-0.04 [-0.05, -0.04]	$< .001$	-0.05 [-0.06, -0.05]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.00]	.929	0.00 [0.00, 0.00]	.564
Sleep regularity $\times$ BMI z-score	0.02 [0.02, 0.03]	$< .001$	0.01 [0.01, 0.02]	$< .001$
BMI z-score $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.739	0.00 [0.00, 0.00]	.852

Note. Adjusted for SES, age, and sex. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 14

Physical activity predicting sleep controlling for age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [-0.15, 0.16]	.939	0.01 [-0.14, 0.15]	.931
Physical activity	0.09 [0.07, 0.11]	< .001	0.06 [0.05, 0.08]	< .001
Medium	0.01 [-0.01, 0.03]	.362	0.01 [-0.01, 0.03]	.317
High	0.03 [0.00, 0.05]	.022	0.01 [-0.02, 0.03]	.501
Physical activity^2	0.00 [-0.01, 0.00]	.472	0.01 [0.00, 0.02]	.013
Physical activity × Medium	0.01 [-0.01, 0.03]	.302	0.00 [-0.01, 0.02]	.672
Physical activity × High	0.02 [0.00, 0.04]	.048	0.00 [-0.02, 0.02]	.930
Medium × Physical activity^2	0.00 [0.00, 0.01]	.510	0.00 [-0.01, 0.01]	.367
High × Physical activity^2	-0.01 [-0.01, 0.00]	.105	0.00 [-0.01, 0.01]	.553
Sleep efficiency (z)				
(Intercept)	0.17 [0.01, 0.33]	.032	0.16 [0.00, 0.32]	.054
Physical activity	0.03 [0.01, 0.05]	< .001	-0.01 [-0.02, 0.01]	.294
Medium	0.02 [0.00, 0.05]	.032	0.03 [0.01, 0.06]	.005
High	0.03 [0.01, 0.06]	.004	0.03 [0.00, 0.05]	.028
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.768
Physical activity × Medium	0.00 [-0.02, 0.02]	.872	0.01 [-0.01, 0.02]	.214
Physical activity × High	0.03 [0.01, 0.05]	.003	0.02 [0.01, 0.04]	.003
Medium × Physical activity^2	0.00 [0.00, 0.01]	.191	-0.01 [-0.02, 0.00]	.199
High × Physical activity^2	-0.01 [-0.01, 0.00]	.120	0.00 [-0.01, 0.01]	.380
Sleep onset (z)				
(Intercept)	0.12 [-0.14, 0.39]	.360	0.12 [-0.13, 0.37]	.351
Physical activity	-0.13 [-0.14, -0.11]	< .001	-0.09 [-0.10, -0.08]	< .001
Medium	0.03 [0.02, 0.05]	< .001	0.04 [0.02, 0.06]	< .001

Table 14 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	0.01 [-0.01, 0.03]	.172	0.02 [0.00, 0.04]	.018
Physical activity^2	0.00 [0.00, 0.01]	.086	-0.01 [-0.01, 0.00]	.028
Physical activity × Medium	0.01 [0.00, 0.03]	.155	0.01 [-0.01, 0.02]	.367
Physical activity × High	0.03 [0.02, 0.05]	< .001	0.02 [0.01, 0.03]	.001
Medium × Physical activity^2	0.00 [-0.01, 0.01]	.889	0.00 [-0.01, 0.01]	.951
High × Physical activity^2	0.00 [-0.01, 0.00]	.517	-0.01 [-0.01, 0.00]	.208
Sleep regularity (z)				
(Intercept)	0.14 [0.02, 0.26]	.025	0.14 [0.00, 0.28]	.053
Physical activity	0.23 [0.20, 0.26]	< .001	0.07 [0.06, 0.09]	< .001
Medium	0.02 [0.00, 0.04]	.075	0.00 [-0.02, 0.03]	.841
High	0.05 [0.02, 0.07]	< .001	0.03 [0.01, 0.06]	.015
Physical activity^2	-0.04 [-0.05, -0.03]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Medium	0.02 [0.00, 0.05]	.058	0.00 [-0.01, 0.02]	.704
Physical activity × High	0.03 [0.00, 0.05]	.029	0.03 [0.01, 0.05]	.001
Medium × Physical activity^2	-0.01 [-0.01, 0.00]	.132	0.00 [-0.01, 0.01]	.889
High × Physical activity^2	-0.01 [-0.01, 0.00]	.159	-0.01 [-0.02, 0.00]	.112

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 15

Sleep predicting physical activity controlling for age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.16, 0.43]	.361	-0.07 [-0.19, 0.05]	.250
Sleep duration	0.00 [-0.01, 0.00]	.349	0.02 [0.01, 0.03]	< .001
Medium	-0.07 [-0.09, -0.05]	< .001	0.00 [-0.02, 0.02]	.828
High	-0.09 [-0.12, -0.07]	< .001	0.01 [-0.01, 0.03]	.231
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep duration × Medium	0.02 [0.01, 0.03]	< .001	0.00 [-0.01, 0.01]	.937
Sleep duration × High	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.051
Medium × Sleep duration^2	0.00 [0.00, 0.01]	.691	0.00 [0.00, 0.01]	.499
High × Sleep duration^2	0.00 [-0.01, 0.01]	.746	0.00 [-0.01, 0.01]	.908
Sleep efficiency (z)				
(Intercept)	0.12 [-0.17, 0.42]	.425	-0.08 [-0.20, 0.04]	.209
Sleep efficiency	0.01 [0.00, 0.02]	.096	0.00 [-0.01, 0.01]	.728
Medium	-0.06 [-0.09, -0.04]	< .001	0.01 [-0.01, 0.02]	.526
High	-0.08 [-0.10, -0.06]	< .001	0.02 [0.00, 0.04]	.019
Sleep efficiency^2	0.00 [-0.01, 0.00]	.040	-0.01 [-0.01, 0.00]	.009
Sleep efficiency × Medium	-0.01 [-0.02, 0.01]	.318	0.01 [0.00, 0.02]	.218
Sleep efficiency × High	-0.01 [-0.02, 0.01]	.394	0.00 [-0.01, 0.02]	.602
Medium × Sleep efficiency^2	0.00 [-0.01, 0.00]	.567	0.00 [-0.01, 0.01]	.880
High × Sleep efficiency^2	-0.01 [-0.01, 0.00]	.058	-0.01 [-0.01, 0.00]	.015
Sleep onset (z)				
(Intercept)	0.13 [-0.17, 0.43]	.396	-0.08 [-0.20, 0.04]	.203
Sleep onset	-0.03 [-0.04, -0.02]	< .001	-0.01 [-0.02, 0.00]	.112
Medium	-0.07 [-0.09, -0.05]	< .001	0.00 [-0.01, 0.02]	.700

Table 15 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High	-0.09 [-0.12, -0.07]	< .001	0.01 [-0.01, 0.03]	.408
Sleep onset^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.01]	.497
Sleep onset × Medium	0.01 [0.00, 0.02]	.230	0.00 [-0.02, 0.01]	.518
Sleep onset × High	-0.01 [-0.02, 0.01]	.328	-0.02 [-0.03, 0.00]	.008
Medium × Sleep onset^2	0.01 [0.00, 0.01]	.027	0.00 [-0.01, 0.01]	.846
High × Sleep onset^2	0.01 [0.00, 0.01]	.167	0.00 [0.00, 0.01]	.392
Sleep regularity (z)				
(Intercept)	0.10 [-0.18, 0.39]	.476	-0.09 [-0.20, 0.03]	.135
Sleep regularity	0.07 [0.06, 0.08]	< .001	0.04 [0.03, 0.05]	< .001
Medium	-0.06 [-0.08, -0.04]	< .001	0.01 [-0.01, 0.03]	.356
High	-0.08 [-0.11, -0.06]	< .001	0.02 [0.00, 0.04]	.052
Sleep regularity^2	0.00 [-0.01, 0.01]	.856	0.00 [0.00, 0.01]	.710
Sleep regularity × Medium	0.02 [0.01, 0.03]	< .001	0.00 [-0.01, 0.01]	.890
Sleep regularity × High	0.03 [0.01, 0.04]	< .001	0.02 [0.01, 0.04]	< .001
Medium × Sleep regularity^2	0.00 [-0.01, 0.00]	.366	0.00 [-0.01, 0.00]	.459
High × Sleep regularity^2	0.00 [-0.01, 0.00]	.531	-0.01 [-0.01, 0.00]	.147

Note. Adjusted for age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 16

Physical activity predicting sleep controlling for SES, age, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [-0.15, 0.16]	.924	0.01 [-0.14, 0.15]	.940
Physical activity	0.11 [0.10, 0.13]	< .001	0.06 [0.05, 0.08]	< .001
Male	-0.13 [-0.14, -0.11]	< .001	-0.13 [-0.14, -0.11]	< .001
Physical activity^2	0.00 [-0.01, 0.00]	.107	0.01 [0.00, 0.02]	.001
Physical activity × Male	-0.02 [-0.04, -0.01]	.003	0.00 [-0.01, 0.01]	.914
Male × Physical activity^2	0.00 [0.00, 0.01]	.434	0.00 [-0.01, 0.01]	.507
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.33]	.026	0.16 [0.00, 0.32]	.051
Physical activity	0.05 [0.03, 0.06]	< .001	-0.01 [-0.02, 0.00]	.057
Male	-0.17 [-0.19, -0.16]	< .001	-0.16 [-0.18, -0.14]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.731
Physical activity × Male	-0.01 [-0.02, 0.01]	.293	0.03 [0.02, 0.04]	< .001
Male × Physical activity^2	0.01 [0.00, 0.02]	< .001	0.00 [-0.01, 0.00]	.451
Sleep onset (z)				
(Intercept)	0.13 [-0.14, 0.39]	.352	0.12 [-0.13, 0.37]	.363
Physical activity	-0.11 [-0.13, -0.10]	< .001	-0.09 [-0.10, -0.08]	< .001
Male	0.01 [-0.01, 0.02]	.372	0.02 [0.00, 0.03]	.009
Physical activity^2	0.00 [0.00, 0.00]	.707	0.00 [-0.01, 0.00]	.102
Physical activity × Male	0.01 [0.00, 0.02]	.127	0.02 [0.01, 0.03]	< .001
Male × Physical activity^2	0.00 [0.00, 0.01]	.310	-0.01 [-0.01, 0.00]	.025
Sleep regularity (z)				
(Intercept)	0.15 [0.03, 0.27]	.015	0.14 [0.00, 0.28]	.045
Physical activity	0.26 [0.24, 0.28]	< .001	0.07 [0.06, 0.08]	< .001

Table 16 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	-0.13 [-0.15, -0.12]	$< .001$	-0.12 [-0.14, -0.10]	$< .001$
Physical activity^2	-0.06 [-0.06, -0.05]	$< .001$	-0.02 [-0.03, -0.02]	$< .001$
Physical activity $\times$ Male	-0.01 [-0.03, 0.01]	.192	0.02 [0.01, 0.04]	$< .001$
Male $\times$ Physical activity^2	0.02 [0.01, 0.02]	$< .001$	0.00 [0.00, 0.01]	.318

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 17

Sleep predicting physical activity controlling for SES, age, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [-0.16, 0.43]	.372	-0.07 [-0.19, 0.05]	.237
Sleep duration	0.01 [0.00, 0.01]	.062	0.02 [0.02, 0.03]	< .001
Male	0.16 [0.14, 0.17]	< .001	0.22 [0.21, 0.24]	< .001
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Sleep duration × Male	0.02 [0.01, 0.02]	< .001	0.00 [-0.01, 0.01]	.683
Male × Sleep duration^2	0.00 [-0.01, 0.00]	.779	0.00 [-0.01, 0.00]	.650
Sleep efficiency (z)				
(Intercept)	0.12 [-0.17, 0.41]	.425	-0.07 [-0.19, 0.05]	.229
Sleep efficiency	0.02 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.002
Male	0.16 [0.14, 0.17]	< .001	0.22 [0.21, 0.23]	< .001
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Male	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.051
Male × Sleep efficiency^2	0.00 [-0.01, 0.00]	.110	0.00 [0.00, 0.01]	.594
Sleep onset (z)				
(Intercept)	0.12 [-0.17, 0.42]	.414	-0.08 [-0.20, 0.04]	.196
Sleep onset	0.00 [-0.01, 0.01]	.733	-0.01 [-0.02, 0.00]	.003
Male	0.15 [0.14, 0.16]	< .001	0.22 [0.20, 0.23]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	< .001	0.00 [0.00, 0.01]	.221
Sleep onset × Male	-0.05 [-0.06, -0.04]	< .001	-0.01 [-0.02, 0.00]	.121
Male × Sleep onset^2	0.00 [0.00, 0.01]	.121	0.00 [0.00, 0.01]	.468
Sleep regularity (z)				
(Intercept)	0.11 [-0.18, 0.40]	.453	-0.08 [-0.20, 0.03]	.155
Sleep regularity	0.07 [0.07, 0.08]	< .001	0.05 [0.04, 0.06]	< .001

Table 17 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male	0.16 [0.15, 0.17]	$< .001$	0.22 [0.21, 0.24]	$< .001$
Sleep regularity^2	0.00 [-0.01, 0.00]	.096	0.00 [-0.01, 0.00]	.238
Sleep regularity $\times$ Male	0.03 [0.02, 0.04]	$< .001$	0.00 [0.00, 0.01]	.395
Male $\times$ Sleep regularity^2	0.00 [0.00, 0.01]	.080	0.00 [0.00, 0.01]	.598

Note. Adjusted for SES, age, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 18

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.10 [-0.06, 0.25]	.214	0.11 [-0.03, 0.25]	.136
Physical activity	0.10 [0.08, 0.11]	< .001	0.03 [0.01, 0.04]	< .001
Monday	-0.11 [-0.13, -0.10]	< .001	-0.12 [-0.15, -0.10]	< .001
Saturday	-0.01 [-0.02, 0.01]	.342	-0.03 [-0.05, -0.01]	.005
Sunday	-0.14 [-0.16, -0.12]	< .001	-0.14 [-0.17, -0.12]	< .001
Thursday	-0.12 [-0.14, -0.11]	< .001	-0.13 [-0.15, -0.11]	< .001
Tuesday	-0.14 [-0.16, -0.11]	< .001	-0.15 [-0.18, -0.13]	< .001
Wednesday	-0.15 [-0.17, -0.12]	< .001	-0.16 [-0.19, -0.13]	< .001
Physical activity^2	-0.01 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.00]	.082
Physical activity × Monday	0.01 [-0.01, 0.03]	.370	0.06 [0.04, 0.08]	< .001
Physical activity × Saturday	0.03 [0.01, 0.05]	.001	0.00 [-0.01, 0.02]	.826
Physical activity × Sunday	0.00 [-0.02, 0.02]	.994	0.03 [0.02, 0.05]	< .001
Physical activity × Thursday	-0.01 [-0.03, 0.01]	.280	0.05 [0.03, 0.07]	< .001
Physical activity × Tuesday	0.01 [-0.01, 0.03]	.422	0.07 [0.05, 0.09]	< .001
Physical activity × Wednesday	-0.01 [-0.04, 0.02]	.404	0.05 [0.03, 0.07]	< .001
Monday × Physical activity^2	0.01 [0.00, 0.02]	.011	0.02 [0.00, 0.03]	.013
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.069	0.00 [-0.01, 0.02]	.480
Sunday × Physical activity^2	0.02 [0.02, 0.03]	< .001	0.02 [0.01, 0.03]	.002
Thursday × Physical activity^2	0.02 [0.01, 0.03]	< .001	0.02 [0.01, 0.03]	< .001
Tuesday × Physical activity^2	0.01 [0.01, 0.02]	.001	0.03 [0.01, 0.04]	< .001
Wednesday × Physical activity^2	0.02 [0.01, 0.03]	< .001	0.03 [0.01, 0.04]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [-0.01, 0.30]	.067	0.14 [-0.02, 0.30]	.091

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity	0.04 [0.02, 0.06]	< .001	-0.02 [-0.03, 0.00]	.021
Monday	0.05 [0.03, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Saturday	-0.03 [-0.04, -0.01]	.002	-0.06 [-0.08, -0.05]	< .001
Sunday	0.00 [-0.02, 0.02]	.883	0.00 [-0.02, 0.02]	.961
Thursday	0.05 [0.03, 0.07]	< .001	0.06 [0.04, 0.08]	< .001
Tuesday	0.04 [0.02, 0.06]	< .001	0.06 [0.03, 0.08]	< .001
Wednesday	0.05 [0.03, 0.08]	< .001	0.07 [0.04, 0.10]	< .001
Physical activity^2	-0.01 [-0.01, 0.00]	.018	0.00 [0.00, 0.01]	.398
Physical activity × Monday	-0.03 [-0.05, -0.01]	.007	0.03 [0.01, 0.04]	.004
Physical activity × Saturday	0.06 [0.04, 0.08]	< .001	0.02 [0.00, 0.03]	.028
Physical activity × Sunday	-0.01 [-0.03, 0.01]	.307	0.01 [-0.01, 0.02]	.412
Physical activity × Thursday	-0.04 [-0.06, -0.01]	.002	0.01 [0.00, 0.03]	.094
Physical activity × Tuesday	-0.03 [-0.05, 0.00]	.028	0.02 [0.00, 0.04]	.030
Physical activity × Wednesday	-0.02 [-0.05, 0.00]	.110	0.02 [0.00, 0.04]	.050
Monday × Physical activity^2	0.00 [-0.01, 0.01]	.635	-0.02 [-0.03, -0.01]	.005
Saturday × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.01 [0.00, 0.02]	.019
Sunday × Physical activity^2	0.00 [-0.01, 0.01]	.918	-0.01 [-0.02, 0.01]	.310
Thursday × Physical activity^2	0.00 [-0.01, 0.01]	.566	-0.01 [-0.02, 0.00]	.038
Tuesday × Physical activity^2	0.00 [-0.01, 0.01]	.890	-0.02 [-0.03, 0.00]	.011
Wednesday × Physical activity^2	0.00 [-0.01, 0.01]	.778	-0.02 [-0.03, 0.00]	.015
Sleep onset (z)				
(Intercept)	0.20 [-0.06, 0.47]	.136	0.19 [-0.06, 0.44]	.137
Physical activity	-0.07 [-0.08, -0.05]	< .001	-0.07 [-0.08, -0.06]	< .001
Monday	-0.15 [-0.17, -0.14]	< .001	-0.13 [-0.15, -0.11]	< .001

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	0.09 [0.08, 0.10]	< .001	0.07 [0.05, 0.08]	< .001
Sunday	-0.14 [-0.15, -0.13]	< .001	-0.13 [-0.15, -0.12]	< .001
Thursday	-0.12 [-0.13, -0.10]	< .001	-0.10 [-0.12, -0.09]	< .001
Tuesday	-0.13 [-0.14, -0.11]	< .001	-0.10 [-0.12, -0.08]	< .001
Wednesday	-0.12 [-0.14, -0.10]	< .001	-0.10 [-0.12, -0.08]	< .001
Physical activity^2	0.00 [0.00, 0.00]	.937	0.00 [-0.01, 0.00]	.332
Physical activity × Monday	-0.09 [-0.11, -0.07]	< .001	-0.01 [-0.02, 0.00]	.110
Physical activity × Saturday	0.05 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.018
Physical activity × Sunday	-0.06 [-0.07, -0.04]	< .001	-0.01 [-0.02, 0.01]	.411
Physical activity × Thursday	-0.07 [-0.09, -0.06]	< .001	-0.02 [-0.03, -0.01]	.003
Physical activity × Tuesday	-0.08 [-0.10, -0.06]	< .001	-0.01 [-0.03, 0.00]	.073
Physical activity × Wednesday	-0.08 [-0.10, -0.06]	< .001	-0.01 [-0.03, 0.01]	.234
Monday × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.00 [-0.02, 0.01]	.387
Saturday × Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.718
Sunday × Physical activity^2	0.00 [0.00, 0.01]	.178	0.01 [0.00, 0.02]	.220
Thursday × Physical activity^2	0.01 [0.00, 0.01]	.046	-0.01 [-0.02, 0.00]	.192
Tuesday × Physical activity^2	0.01 [0.00, 0.02]	.004	-0.01 [-0.02, 0.00]	.064
Wednesday × Physical activity^2	0.01 [0.00, 0.02]	.007	-0.01 [-0.02, 0.00]	.274
Sleep regularity (z)				
(Intercept)	-0.02 [-0.15, 0.10]	.720	-0.02 [-0.16, 0.13]	.808
Physical activity	0.25 [0.22, 0.27]	< .001	0.00 [-0.01, 0.01]	.990
Monday	0.33 [0.30, 0.35]	< .001	0.35 [0.32, 0.38]	< .001
Saturday	-0.01 [-0.03, 0.01]	.266	-0.04 [-0.06, -0.02]	< .001
Sunday	0.01 [-0.01, 0.04]	.261	-0.02 [-0.05, 0.00]	.096

Table 18 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	0.25 [0.23, 0.27]	< .001	0.28 [0.25, 0.30]	< .001
Tuesday	0.31 [0.29, 0.34]	< .001	0.34 [0.31, 0.37]	< .001
Wednesday	0.29 [0.27, 0.32]	< .001	0.33 [0.30, 0.36]	< .001
Physical activity^2	-0.04 [-0.05, -0.03]	< .001	-0.01 [-0.02, 0.00]	.255
Physical activity × Monday	-0.06 [-0.08, -0.03]	< .001	0.10 [0.09, 0.12]	< .001
Physical activity × Saturday	0.01 [-0.01, 0.03]	.406	0.01 [-0.01, 0.03]	.260
Physical activity × Sunday	0.01 [-0.01, 0.03]	.407	0.05 [0.03, 0.07]	< .001
Physical activity × Thursday	-0.04 [-0.07, -0.01]	.003	0.07 [0.06, 0.09]	< .001
Physical activity × Tuesday	-0.06 [-0.09, -0.03]	< .001	0.11 [0.09, 0.12]	< .001
Physical activity × Wednesday	-0.07 [-0.10, -0.05]	< .001	0.08 [0.06, 0.10]	< .001
Monday × Physical activity^2	-0.01 [-0.02, 0.00]	.249	-0.04 [-0.06, -0.03]	< .001
Saturday × Physical activity^2	-0.01 [-0.02, 0.00]	.125	0.00 [-0.02, 0.01]	.474
Sunday × Physical activity^2	-0.01 [-0.02, 0.00]	.218	0.00 [-0.02, 0.01]	.567
Thursday × Physical activity^2	0.00 [-0.01, 0.01]	.636	-0.03 [-0.04, -0.01]	< .001
Tuesday × Physical activity^2	0.01 [0.00, 0.02]	.121	-0.03 [-0.04, -0.01]	< .001
Wednesday × Physical activity^2	0.00 [-0.01, 0.01]	.463	-0.04 [-0.05, -0.02]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 19

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.20 [-0.10, 0.50]	.182	-0.04 [-0.16, 0.08]	.539
Sleep duration	0.03 [0.02, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Monday	-0.07 [-0.08, -0.06]	< .001	-0.01 [-0.03, 0.00]	.036
Saturday	-0.12 [-0.13, -0.11]	< .001	-0.12 [-0.13, -0.11]	< .001
Sunday	-0.18 [-0.20, -0.17]	< .001	-0.13 [-0.14, -0.11]	< .001
Thursday	0.00 [-0.01, 0.01]	.860	0.03 [0.01, 0.04]	< .001
Tuesday	-0.02 [-0.03, -0.01]	.003	0.02 [0.01, 0.03]	.005
Wednesday	-0.04 [-0.05, -0.02]	< .001	0.02 [0.01, 0.04]	< .001
Sleep duration^2	-0.02 [-0.03, -0.02]	< .001	0.00 [-0.01, 0.00]	.444
Sleep duration × Monday	-0.01 [-0.02, 0.00]	.043	-0.02 [-0.03, -0.01]	< .001
Sleep duration × Saturday	-0.05 [-0.06, -0.04]	< .001	-0.07 [-0.09, -0.06]	< .001
Sleep duration × Sunday	-0.03 [-0.04, -0.02]	< .001	-0.07 [-0.08, -0.06]	< .001
Sleep duration × Thursday	0.01 [0.00, 0.02]	.049	0.01 [0.00, 0.02]	.197
Sleep duration × Tuesday	0.01 [-0.01, 0.02]	.342	-0.01 [-0.02, 0.00]	.125
Sleep duration × Wednesday	0.00 [-0.01, 0.02]	.412	0.01 [0.00, 0.02]	.108
Monday × Sleep duration^2	0.01 [0.00, 0.02]	.004	0.00 [-0.01, 0.01]	.859
Saturday × Sleep duration^2	0.00 [0.00, 0.01]	.363	-0.01 [-0.01, 0.00]	.092
Sunday × Sleep duration^2	0.01 [0.00, 0.02]	.038	0.00 [-0.01, 0.01]	.559
Thursday × Sleep duration^2	0.00 [-0.01, 0.01]	.945	-0.01 [-0.02, 0.00]	.027
Tuesday × Sleep duration^2	0.00 [-0.01, 0.01]	.977	-0.01 [-0.02, 0.00]	.044
Wednesday × Sleep duration^2	0.01 [0.00, 0.02]	.104	0.00 [-0.01, 0.01]	.626
Sleep efficiency (z)				
(Intercept)	0.18 [-0.11, 0.48]	.223	-0.03 [-0.16, 0.09]	.575

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency	-0.01 [-0.02, 0.00]	.044	-0.02 [-0.03, -0.01]	< .001
Monday	-0.06 [-0.08, -0.05]	< .001	-0.02 [-0.03, 0.00]	.015
Saturday	-0.12 [-0.13, -0.11]	< .001	-0.14 [-0.15, -0.12]	< .001
Sunday	-0.17 [-0.18, -0.15]	< .001	-0.13 [-0.14, -0.11]	< .001
Thursday	0.01 [-0.01, 0.02]	.431	0.02 [0.00, 0.03]	.014
Tuesday	-0.01 [-0.02, 0.00]	.109	0.02 [0.00, 0.03]	.022
Wednesday	-0.03 [-0.04, -0.02]	< .001	0.02 [0.01, 0.03]	.005
Sleep efficiency^2	0.00 [-0.01, 0.00]	.112	-0.01 [-0.01, 0.00]	.003
Sleep efficiency × Monday	0.02 [0.01, 0.03]	.001	0.02 [0.01, 0.03]	.001
Sleep efficiency × Saturday	0.02 [0.01, 0.04]	< .001	0.06 [0.05, 0.07]	< .001
Sleep efficiency × Sunday	0.01 [-0.01, 0.02]	.406	0.06 [0.05, 0.08]	< .001
Sleep efficiency × Thursday	0.00 [-0.01, 0.02]	.653	0.00 [-0.01, 0.01]	.956
Sleep efficiency × Tuesday	0.01 [-0.01, 0.02]	.460	0.01 [0.00, 0.03]	.036
Sleep efficiency × Wednesday	0.01 [0.00, 0.02]	.205	0.01 [0.00, 0.02]	.180
Monday × Sleep efficiency^2	0.00 [0.00, 0.01]	.274	0.00 [-0.01, 0.01]	.727
Saturday × Sleep efficiency^2	0.00 [0.00, 0.01]	.189	0.01 [0.00, 0.01]	.028
Sunday × Sleep efficiency^2	-0.01 [-0.01, 0.00]	.031	0.00 [0.00, 0.01]	.561
Thursday × Sleep efficiency^2	0.00 [-0.01, 0.00]	.217	0.00 [-0.01, 0.01]	.868
Tuesday × Sleep efficiency^2	-0.01 [-0.02, 0.00]	.020	0.00 [-0.01, 0.00]	.424
Wednesday × Sleep efficiency^2	0.00 [-0.01, 0.01]	.998	0.00 [-0.01, 0.01]	.553
Sleep onset (z)				
(Intercept)	0.18 [-0.12, 0.48]	.232	-0.06 [-0.18, 0.06]	.360
Sleep onset	-0.04 [-0.05, -0.03]	< .001	-0.05 [-0.06, -0.04]	< .001
Monday	-0.06 [-0.08, -0.05]	< .001	-0.01 [-0.02, 0.00]	.066

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday	-0.11 [-0.12, -0.10]	< .001	-0.10 [-0.12, -0.09]	< .001
Sunday	-0.17 [-0.18, -0.16]	< .001	-0.11 [-0.13, -0.10]	< .001
Thursday	0.00 [-0.02, 0.01]	.774	0.01 [0.00, 0.03]	.034
Tuesday	-0.02 [-0.03, 0.00]	.023	0.02 [0.01, 0.04]	< .001
Wednesday	-0.03 [-0.05, -0.02]	< .001	0.02 [0.01, 0.03]	.002
Sleep onset^2	0.00 [-0.01, 0.00]	.293	0.01 [0.01, 0.02]	< .001
Sleep onset × Monday	0.03 [0.02, 0.04]	< .001	0.04 [0.03, 0.05]	< .001
Sleep onset × Saturday	0.07 [0.06, 0.08]	< .001	0.12 [0.11, 0.13]	< .001
Sleep onset × Sunday	0.05 [0.04, 0.06]	< .001	0.13 [0.12, 0.14]	< .001
Sleep onset × Thursday	0.00 [-0.02, 0.01]	.542	-0.01 [-0.02, 0.00]	.159
Sleep onset × Tuesday	0.01 [0.00, 0.02]	.099	0.02 [0.01, 0.03]	< .001
Sleep onset × Wednesday	0.01 [0.00, 0.02]	.154	0.00 [-0.01, 0.02]	.497
Monday × Sleep onset^2	0.01 [0.00, 0.01]	.142	0.00 [-0.01, 0.01]	.884
Saturday × Sleep onset^2	-0.01 [-0.01, 0.00]	.064	-0.03 [-0.03, -0.02]	< .001
Sunday × Sleep onset^2	-0.01 [-0.01, 0.00]	.111	-0.03 [-0.03, -0.02]	< .001
Thursday × Sleep onset^2	0.00 [-0.01, 0.01]	.707	0.00 [-0.01, 0.01]	.919
Tuesday × Sleep onset^2	-0.01 [-0.01, 0.00]	.172	-0.01 [-0.02, 0.00]	.004
Wednesday × Sleep onset^2	0.00 [-0.01, 0.01]	.741	0.00 [-0.01, 0.00]	.347
Sleep regularity (z)				
(Intercept)	0.16 [-0.13, 0.45]	.279	-0.05 [-0.16, 0.07]	.441
Sleep regularity	0.09 [0.08, 0.10]	< .001	0.03 [0.02, 0.04]	< .001
Monday	-0.04 [-0.05, -0.02]	< .001	0.00 [-0.02, 0.01]	.546
Saturday	-0.08 [-0.09, -0.07]	< .001	-0.11 [-0.13, -0.10]	< .001
Sunday	-0.13 [-0.15, -0.12]	< .001	-0.11 [-0.13, -0.10]	< .001

Table 19 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	0.00 [-0.02, 0.01]	.662	0.02 [0.01, 0.03]	.006
Tuesday	-0.03 [-0.04, -0.01]	< .001	0.00 [-0.01, 0.02]	.585
Wednesday	-0.03 [-0.04, -0.01]	< .001	0.02 [0.00, 0.03]	.026
Sleep regularity^2	0.00 [-0.01, 0.01]	.706	0.00 [-0.01, 0.00]	.309
Sleep regularity × Monday	-0.03 [-0.05, -0.02]	< .001	-0.02 [-0.03, 0.00]	.013
Sleep regularity × Saturday	-0.01 [-0.02, 0.00]	.170	0.02 [0.00, 0.03]	.013
Sleep regularity × Sunday	-0.03 [-0.04, -0.01]	< .001	0.02 [0.00, 0.03]	.020
Sleep regularity × Thursday	-0.01 [-0.03, 0.00]	.035	0.00 [-0.01, 0.01]	.865
Sleep regularity × Tuesday	-0.01 [-0.02, 0.01]	.248	0.01 [0.00, 0.02]	.093
Sleep regularity × Wednesday	-0.02 [-0.03, -0.01]	.003	0.01 [-0.01, 0.02]	.228
Monday × Sleep regularity^2	0.00 [-0.01, 0.01]	.882	0.00 [-0.01, 0.01]	.632
Saturday × Sleep regularity^2	-0.01 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.560
Sunday × Sleep regularity^2	-0.02 [-0.03, -0.01]	< .001	0.00 [-0.01, 0.01]	.955
Thursday × Sleep regularity^2	0.00 [-0.01, 0.01]	.727	0.00 [-0.01, 0.00]	.291
Tuesday × Sleep regularity^2	0.01 [0.00, 0.01]	.213	0.00 [0.00, 0.01]	.310
Wednesday × Sleep regularity^2	0.00 [-0.01, 0.00]	.347	0.00 [-0.01, 0.01]	.774

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 20

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [-0.14, 0.18]	.805	0.01 [-0.13, 0.16]	.855
Physical activity	0.12 [0.09, 0.14]	< .001	0.06 [0.05, 0.08]	< .001
Spring	-0.03 [-0.05, 0.00]	.020	-0.01 [-0.04, 0.01]	.250
Summer	-0.10 [-0.12, -0.08]	< .001	-0.09 [-0.11, -0.06]	< .001
Winter	0.07 [0.05, 0.09]	< .001	0.07 [0.05, 0.09]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.840	0.02 [0.01, 0.03]	< .001
Physical activity × Spring	-0.02 [-0.05, 0.00]	.028	0.00 [-0.02, 0.01]	.689
Physical activity × Summer	-0.02 [-0.05, 0.00]	.066	0.00 [-0.02, 0.02]	.894
Physical activity × Winter	-0.01 [-0.03, 0.01]	.518	0.01 [-0.01, 0.03]	.190
Spring × Physical activity^2	0.00 [-0.01, 0.01]	.502	-0.01 [-0.02, 0.00]	.032
Summer × Physical activity^2	0.00 [-0.01, 0.01]	.653	-0.01 [-0.02, 0.00]	.149
Winter × Physical activity^2	-0.01 [-0.01, 0.00]	.050	-0.01 [-0.02, 0.00]	.147
Sleep efficiency (z)				
(Intercept)	0.16 [0.01, 0.32]	.038	0.15 [-0.01, 0.31]	.065
Physical activity	0.05 [0.03, 0.07]	< .001	0.01 [0.00, 0.02]	.053
Spring	0.01 [-0.02, 0.03]	.574	0.01 [-0.01, 0.03]	.413
Summer	-0.02 [-0.05, 0.00]	.060	-0.01 [-0.04, 0.01]	.353
Winter	0.04 [0.02, 0.06]	< .001	0.05 [0.02, 0.07]	< .001
Physical activity^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.01]	.943
Physical activity × Spring	-0.02 [-0.05, 0.00]	.022	-0.02 [-0.04, -0.01]	.007
Physical activity × Summer	-0.02 [-0.04, 0.01]	.196	-0.01 [-0.03, 0.01]	.240
Physical activity × Winter	0.00 [-0.02, 0.02]	.847	0.00 [-0.02, 0.02]	.993
Spring × Physical activity^2	0.01 [0.00, 0.01]	.194	0.00 [-0.01, 0.01]	.759

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Physical activity^2	0.01 [0.00, 0.01]	.203	0.00 [-0.02, 0.01]	.551
Winter × Physical activity^2	0.00 [-0.01, 0.01]	.658	-0.01 [-0.02, 0.00]	.173
Sleep onset (z)				
(Intercept)	0.10 [-0.16, 0.37]	.440	0.10 [-0.15, 0.35]	.428
Physical activity	-0.12 [-0.14, -0.10]	< .001	-0.08 [-0.09, -0.07]	< .001
Spring	0.02 [0.00, 0.04]	.022	0.02 [0.00, 0.04]	.034
Summer	0.07 [0.05, 0.09]	< .001	0.06 [0.04, 0.08]	< .001
Winter	-0.01 [-0.02, 0.01]	.396	0.00 [-0.01, 0.02]	.638
Physical activity^2	0.00 [-0.01, 0.00]	.100	-0.01 [-0.02, -0.01]	< .001
Physical activity × Spring	0.01 [-0.01, 0.02]	.478	0.00 [-0.02, 0.01]	.582
Physical activity × Summer	0.04 [0.02, 0.06]	< .001	0.02 [0.01, 0.04]	.002
Physical activity × Winter	0.00 [-0.02, 0.02]	.893	-0.01 [-0.02, 0.00]	.102
Spring × Physical activity^2	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	.061
Summer × Physical activity^2	0.01 [0.00, 0.01]	.120	0.02 [0.01, 0.03]	< .001
Winter × Physical activity^2	0.01 [0.00, 0.02]	.002	0.00 [-0.01, 0.01]	.968
Sleep regularity (z)				
(Intercept)	0.16 [0.04, 0.28]	.010	0.18 [0.03, 0.32]	.016
Physical activity	0.22 [0.20, 0.24]	< .001	0.09 [0.08, 0.11]	< .001
Spring	-0.04 [-0.07, -0.02]	< .001	-0.08 [-0.10, -0.05]	< .001
Summer	0.02 [-0.01, 0.04]	.158	0.00 [-0.03, 0.03]	.882
Winter	-0.02 [-0.04, 0.01]	.130	-0.03 [-0.06, -0.01]	.008
Physical activity^2	-0.04 [-0.04, -0.03]	< .001	-0.03 [-0.04, -0.02]	< .001
Physical activity × Spring	0.05 [0.03, 0.08]	< .001	-0.02 [-0.04, -0.01]	.007
Physical activity × Summer	0.04 [0.01, 0.06]	.010	-0.01 [-0.03, 0.01]	.435

Table 20 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Winter	0.04 [0.01, 0.06]	.002	0.00 [-0.02, 0.02]	.795
Spring × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.02 [0.00, 0.03]	.013
Summer × Physical activity^2	-0.01 [-0.02, 0.00]	.267	0.02 [0.01, 0.04]	.005
Winter × Physical activity^2	-0.01 [-0.02, 0.00]	.015	0.00 [-0.01, 0.01]	.853

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 21

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.13 [-0.16, 0.43]	.377	-0.06 [-0.18, 0.07]	.364
Sleep duration	0.04 [0.03, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Spring	0.01 [-0.01, 0.03]	.453	-0.01 [-0.03, 0.01]	.251
Summer	0.03 [0.01, 0.05]	.009	-0.03 [-0.05, -0.01]	< .001
Winter	-0.04 [-0.06, -0.02]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep duration^2	-0.02 [-0.02, -0.02]	< .001	-0.01 [-0.01, 0.00]	.006
Sleep duration × Spring	-0.05 [-0.06, -0.04]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep duration × Summer	-0.05 [-0.06, -0.03]	< .001	-0.03 [-0.04, -0.02]	< .001
Sleep duration × Winter	-0.01 [-0.02, 0.00]	.282	-0.01 [-0.02, 0.01]	.329
Spring × Sleep duration^2	0.00 [-0.01, 0.01]	.833	0.00 [-0.01, 0.00]	.145
Summer × Sleep duration^2	0.00 [-0.01, 0.01]	.745	-0.01 [-0.02, 0.00]	.024
Winter × Sleep duration^2	0.00 [-0.01, 0.01]	.733	0.00 [-0.01, 0.01]	.829
Sleep efficiency (z)				
(Intercept)	0.13 [-0.17, 0.42]	.401	-0.05 [-0.17, 0.07]	.421
Sleep efficiency	0.01 [0.00, 0.01]	.184	0.01 [0.00, 0.02]	.036
Spring	0.00 [-0.02, 0.02]	.919	-0.02 [-0.04, 0.00]	.013
Summer	0.02 [0.00, 0.04]	.026	-0.05 [-0.07, -0.03]	< .001
Winter	-0.05 [-0.06, -0.03]	< .001	-0.04 [-0.05, -0.02]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency × Spring	0.00 [-0.01, 0.01]	.746	-0.01 [-0.02, 0.01]	.311
Sleep efficiency × Summer	0.00 [-0.02, 0.01]	.541	0.00 [-0.01, 0.01]	.849
Sleep efficiency × Winter	0.00 [-0.02, 0.01]	.589	-0.01 [-0.02, 0.00]	.210
Spring × Sleep efficiency^2	0.01 [0.00, 0.01]	.068	0.01 [0.00, 0.01]	.057

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Sleep efficiency^2	0.00 [0.00, 0.01]	.208	0.01 [0.00, 0.01]	.065
Winter × Sleep efficiency^2	0.01 [0.00, 0.01]	.076	0.00 [0.00, 0.01]	.139
Sleep onset (z)				
(Intercept)	0.13 [-0.17, 0.43]	.408	-0.06 [-0.19, 0.06]	.322
Sleep onset	-0.02 [-0.03, -0.01]	< .001	-0.02 [-0.02, -0.01]	< .001
Spring	0.01 [-0.01, 0.03]	.223	-0.01 [-0.03, 0.00]	.149
Summer	0.02 [0.00, 0.04]	.062	-0.04 [-0.06, -0.02]	< .001
Winter	-0.04 [-0.06, -0.02]	< .001	-0.03 [-0.05, -0.01]	< .001
Sleep onset^2	0.00 [-0.01, 0.00]	.344	0.00 [0.00, 0.01]	.099
Sleep onset × Spring	-0.02 [-0.03, -0.01]	.001	0.00 [-0.01, 0.01]	.954
Sleep onset × Summer	0.03 [0.01, 0.04]	< .001	0.02 [0.01, 0.04]	.002
Sleep onset × Winter	-0.03 [-0.04, -0.02]	< .001	-0.01 [-0.02, 0.00]	.150
Spring × Sleep onset^2	0.00 [-0.01, 0.00]	.150	0.00 [-0.01, 0.00]	.502
Summer × Sleep onset^2	0.00 [-0.01, 0.01]	.703	0.00 [-0.01, 0.00]	.384
Winter × Sleep onset^2	0.00 [-0.01, 0.00]	.551	0.00 [-0.01, 0.01]	.831
Sleep regularity (z)				
(Intercept)	0.11 [-0.18, 0.40]	.468	-0.07 [-0.19, 0.05]	.256
Sleep regularity	0.08 [0.08, 0.09]	< .001	0.06 [0.05, 0.07]	< .001
Spring	0.01 [-0.01, 0.03]	.283	-0.01 [-0.03, 0.01]	.295
Summer	0.03 [0.01, 0.05]	.010	-0.04 [-0.05, -0.02]	< .001
Winter	-0.04 [-0.06, -0.02]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep regularity^2	0.00 [-0.01, 0.00]	.116	0.00 [-0.01, 0.00]	.681
Sleep regularity × Spring	0.01 [0.00, 0.02]	.216	-0.01 [-0.03, 0.00]	.014
Sleep regularity × Summer	-0.02 [-0.04, -0.01]	< .001	-0.03 [-0.04, -0.01]	< .001

Table 21 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity × Winter	0.01 [0.00, 0.03]	.015	0.00 [-0.01, 0.01]	.956
Spring × Sleep regularity^2	0.00 [0.00, 0.01]	.237	0.00 [-0.01, 0.00]	.541
Summer × Sleep regularity^2	0.00 [-0.01, 0.01]	.758	-0.01 [-0.01, 0.00]	.188
Winter × Sleep regularity^2	0.01 [0.00, 0.01]	.067	0.00 [0.00, 0.01]	.170

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 22

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.15 [-0.01, 0.32]	.074	0.19 [0.03, 0.35]	.018
Physical activity	0.02 [-0.03, 0.07]	.526	0.01 [-0.04, 0.06]	.775
Europe	-0.15 [-0.21, -0.09]	< .001	-0.18 [-0.24, -0.11]	< .001
Africa	-0.14 [-0.22, -0.06]	< .001	-0.18 [-0.25, -0.10]	< .001
Asia	-0.40 [-0.49, -0.32]	< .001	-0.47 [-0.54, -0.40]	< .001
North America	-0.02 [-0.10, 0.05]	.535	-0.06 [-0.13, 0.01]	.114
South America	-0.32 [-0.39, -0.26]	< .001	-0.42 [-0.49, -0.36]	< .001
Physical activity^2	0.01 [0.00, 0.02]	.083	0.04 [0.02, 0.06]	< .001
Physical activity × Europe	0.07 [0.02, 0.11]	.003	0.05 [0.00, 0.10]	.062
Physical activity × Africa	0.07 [0.01, 0.14]	.035	0.02 [-0.05, 0.09]	.598
Physical activity × Asia	0.09 [-0.02, 0.19]	.118	0.06 [-0.01, 0.13]	.078
Physical activity × North America	0.08 [0.02, 0.14]	.010	0.03 [-0.03, 0.09]	.306
Physical activity × South America	0.17 [0.12, 0.21]	< .001	0.06 [0.02, 0.11]	.009
Europe × Physical activity^2	-0.01 [-0.02, 0.00]	.019	-0.04 [-0.06, -0.02]	< .001
Africa × Physical activity^2	-0.04 [-0.10, 0.02]	.187	-0.02 [-0.06, 0.02]	.347
Asia × Physical activity^2	-0.05 [-0.13, 0.03]	.219	-0.04 [-0.09, 0.01]	.095
North America × Physical activity^2	0.00 [-0.03, 0.02]	.765	-0.02 [-0.06, 0.01]	.226
South America × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.00 [-0.02, 0.02]	.987
Sleep efficiency (z)				
(Intercept)	0.29 [0.12, 0.45]	< .001	0.33 [0.16, 0.50]	< .001
Physical activity	-0.02 [-0.05, 0.01]	.198	-0.01 [-0.04, 0.02]	.520
Europe	-0.15 [-0.22, -0.08]	< .001	-0.19 [-0.25, -0.12]	< .001
Africa	0.01 [-0.07, 0.08]	.887	-0.05 [-0.13, 0.02]	.172

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.21 [-0.30, -0.12]	< .001	-0.34 [-0.42, -0.26]	< .001
North America	0.02 [-0.06, 0.10]	.667	-0.11 [-0.18, -0.03]	.005
South America	-0.22 [-0.28, -0.15]	< .001	-0.31 [-0.38, -0.24]	< .001
Physical activity^2	0.00 [0.00, 0.01]	.506	-0.01 [-0.03, 0.00]	.092
Physical activity × Europe	0.04 [0.00, 0.07]	.028	0.01 [-0.02, 0.04]	.555
Physical activity × Africa	0.06 [0.00, 0.12]	.063	0.01 [-0.05, 0.06]	.840
Physical activity × Asia	0.02 [-0.08, 0.12]	.724	0.07 [0.02, 0.13]	.012
Physical activity × North America	0.15 [0.10, 0.21]	< .001	0.07 [0.02, 0.12]	.005
Physical activity × South America	0.14 [0.10, 0.17]	< .001	0.08 [0.04, 0.11]	< .001
Europe × Physical activity^2	-0.01 [-0.02, 0.00]	.011	0.02 [0.00, 0.03]	.050
Africa × Physical activity^2	-0.06 [-0.11, 0.00]	.051	0.01 [-0.02, 0.05]	.474
Asia × Physical activity^2	-0.19 [-0.28, -0.11]	< .001	-0.04 [-0.09, 0.01]	.105
North America × Physical activity^2	-0.04 [-0.06, -0.02]	< .001	-0.02 [-0.05, 0.02]	.292
South America × Physical activity^2	-0.03 [-0.04, -0.02]	< .001	-0.03 [-0.06, -0.01]	.003
Sleep onset (z)				
(Intercept)	0.09 [-0.19, 0.36]	.529	0.10 [-0.16, 0.35]	.454
Physical activity	-0.08 [-0.11, -0.05]	< .001	-0.06 [-0.09, -0.03]	< .001
Europe	0.04 [-0.01, 0.10]	.116	0.03 [-0.02, 0.09]	.221
Africa	0.04 [-0.03, 0.10]	.266	-0.03 [-0.10, 0.03]	.304
Asia	0.21 [0.14, 0.28]	< .001	0.14 [0.08, 0.21]	< .001
North America	0.03 [-0.04, 0.09]	.403	-0.07 [-0.13, -0.01]	.019
South America	0.02 [-0.04, 0.07]	.509	0.05 [0.00, 0.11]	.066
Physical activity^2	0.00 [0.00, 0.01]	.786	-0.05 [-0.06, -0.03]	< .001
Physical activity × Europe	-0.03 [-0.06, 0.00]	.035	-0.01 [-0.04, 0.02]	.515

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity × Africa	0.05 [0.00, 0.10]	.037	0.02 [-0.03, 0.07]	.372
Physical activity × Asia	0.00 [-0.07, 0.08]	.964	0.02 [-0.03, 0.06]	.520
Physical activity × North America	0.10 [0.06, 0.14]	< .001	0.06 [0.02, 0.10]	.007
Physical activity × South America	-0.07 [-0.10, -0.03]	< .001	-0.02 [-0.05, 0.02]	.310
Europe × Physical activity^2	0.02 [0.01, 0.02]	< .001	0.05 [0.04, 0.07]	< .001
Africa × Physical activity^2	-0.02 [-0.06, 0.02]	.380	0.04 [0.01, 0.07]	.013
Asia × Physical activity^2	-0.06 [-0.13, 0.00]	.044	0.02 [-0.02, 0.06]	.391
North America × Physical activity^2	-0.02 [-0.03, 0.00]	.037	0.01 [-0.02, 0.04]	.455
South America × Physical activity^2	0.00 [-0.01, 0.01]	.919	-0.01 [-0.03, 0.00]	.144
Sleep regularity (z)				
(Intercept)	0.01 [-0.13, 0.14]	.910	0.05 [-0.10, 0.21]	.481
Physical activity	0.27 [0.24, 0.30]	< .001	0.17 [0.13, 0.21]	< .001
Europe	0.22 [0.16, 0.28]	< .001	0.16 [0.09, 0.22]	< .001
Africa	0.38 [0.31, 0.46]	< .001	0.34 [0.26, 0.41]	< .001
Asia	0.41 [0.32, 0.50]	< .001	0.38 [0.30, 0.45]	< .001
North America	0.16 [0.08, 0.23]	< .001	0.05 [-0.03, 0.12]	.211
South America	0.05 [-0.01, 0.11]	.120	-0.04 [-0.11, 0.02]	.220
Physical activity^2	-0.04 [-0.05, -0.04]	< .001	-0.09 [-0.12, -0.07]	< .001
Physical activity × Europe	-0.04 [-0.07, 0.00]	.061	-0.11 [-0.16, -0.07]	< .001
Physical activity × Africa	-0.27 [-0.34, -0.20]	< .001	0.10 [0.03, 0.18]	.004
Physical activity × Asia	-0.32 [-0.43, -0.21]	< .001	0.03 [-0.04, 0.10]	.404
Physical activity × North America	0.03 [-0.03, 0.09]	.333	0.04 [-0.02, 0.11]	.175
Physical activity × South America	0.03 [-0.01, 0.06]	.172	0.07 [0.02, 0.12]	.006
Europe × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	0.09 [0.06, 0.11]	< .001

Table 22 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Physical activity^2	-0.21 [-0.30, -0.12]	< .001	-0.06 [-0.11, -0.02]	.009
Asia × Physical activity^2	-0.23 [-0.32, -0.14]	< .001	0.01 [-0.05, 0.07]	.815
North America × Physical activity^2	-0.03 [-0.06, -0.01]	.005	-0.04 [-0.08, 0.01]	.100
South America × Physical activity^2	-0.01 [-0.02, 0.00]	.189	-0.01 [-0.04, 0.02]	.492

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 23

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.28 [-0.01, 0.58]	.057	0.08 [-0.04, 0.21]	.186
Sleep duration	0.10 [0.08, 0.11]	< .001	0.06 [0.04, 0.07]	< .001
Europe	-0.15 [-0.21, -0.10]	< .001	-0.16 [-0.21, -0.11]	< .001
Africa	-0.13 [-0.19, -0.06]	< .001	-0.03 [-0.09, 0.03]	.341
Asia	-0.37 [-0.44, -0.30]	< .001	-0.43 [-0.49, -0.37]	< .001
North America	-0.29 [-0.35, -0.23]	< .001	-0.25 [-0.31, -0.19]	< .001
South America	-0.20 [-0.26, -0.14]	< .001	-0.26 [-0.31, -0.21]	< .001
Sleep duration^2	-0.04 [-0.05, -0.03]	< .001	-0.01 [-0.02, 0.00]	< .001
Sleep duration × Europe	-0.10 [-0.12, -0.08]	< .001	-0.02 [-0.04, -0.01]	.001
Sleep duration × Africa	-0.06 [-0.09, -0.02]	.001	-0.05 [-0.09, -0.02]	.003
Sleep duration × Asia	-0.09 [-0.13, -0.05]	< .001	0.01 [-0.02, 0.05]	.451
Sleep duration × North America	-0.04 [-0.07, -0.01]	.009	-0.02 [-0.05, 0.01]	.261
Sleep duration × South America	-0.08 [-0.10, -0.06]	< .001	-0.07 [-0.08, -0.05]	< .001
Europe × Sleep duration^2	0.03 [0.02, 0.03]	< .001	0.01 [0.00, 0.01]	.125
Africa × Sleep duration^2	0.02 [0.00, 0.04]	.075	-0.03 [-0.05, -0.01]	.008
Asia × Sleep duration^2	0.03 [0.01, 0.06]	.019	-0.05 [-0.08, -0.02]	< .001
North America × Sleep duration^2	0.02 [0.01, 0.04]	.006	0.00 [-0.01, 0.02]	.558
South America × Sleep duration^2	0.00 [-0.01, 0.01]	.673	-0.01 [-0.02, 0.00]	.148
Sleep efficiency (z)				
(Intercept)	0.31 [0.02, 0.60]	.038	0.11 [-0.02, 0.23]	.095
Sleep efficiency	-0.03 [-0.06, -0.01]	.007	0.00 [-0.02, 0.02]	.652
Europe	-0.20 [-0.26, -0.14]	< .001	-0.17 [-0.22, -0.12]	< .001
Africa	-0.17 [-0.22, -0.11]	< .001	-0.12 [-0.18, -0.07]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia	-0.42 [-0.48, -0.35]	< .001	-0.47 [-0.53, -0.41]	< .001
North America	-0.31 [-0.37, -0.25]	< .001	-0.28 [-0.33, -0.22]	< .001
South America	-0.28 [-0.33, -0.22]	< .001	-0.33 [-0.38, -0.28]	< .001
Sleep efficiency^2	-0.03 [-0.04, -0.02]	< .001	-0.02 [-0.03, -0.01]	< .001
Sleep efficiency × Europe	0.05 [0.03, 0.07]	< .001	0.02 [-0.01, 0.04]	.143
Sleep efficiency × Africa	0.03 [0.00, 0.07]	.055	-0.01 [-0.04, 0.02]	.468
Sleep efficiency × Asia	0.05 [0.01, 0.09]	.013	0.01 [-0.02, 0.05]	.456
Sleep efficiency × North America	0.03 [0.00, 0.07]	.054	0.01 [-0.02, 0.04]	.583
Sleep efficiency × South America	0.00 [-0.02, 0.03]	.731	0.00 [-0.02, 0.02]	.917
Europe × Sleep efficiency^2	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.083
Africa × Sleep efficiency^2	0.02 [0.01, 0.04]	< .001	0.02 [0.00, 0.03]	.011
Asia × Sleep efficiency^2	0.03 [0.02, 0.05]	< .001	0.01 [0.00, 0.03]	.084
North America × Sleep efficiency^2	0.02 [0.01, 0.04]	< .001	0.02 [0.01, 0.03]	.005
South America × Sleep efficiency^2	0.02 [0.01, 0.03]	< .001	0.02 [0.01, 0.03]	< .001
Sleep onset (z)				
(Intercept)	0.15 [-0.14, 0.45]	.315	0.01 [-0.11, 0.14]	.815
Sleep onset	-0.16 [-0.18, -0.14]	< .001	-0.09 [-0.11, -0.07]	< .001
Europe	-0.03 [-0.10, 0.03]	.288	-0.08 [-0.13, -0.02]	.005
Africa	0.05 [-0.04, 0.13]	.277	-0.06 [-0.14, 0.03]	.175
Asia	-0.22 [-0.29, -0.14]	< .001	-0.37 [-0.44, -0.29]	< .001
North America	-0.09 [-0.16, -0.01]	.022	-0.18 [-0.25, -0.11]	< .001
South America	-0.01 [-0.07, 0.06]	.807	-0.19 [-0.25, -0.14]	< .001
Sleep onset^2	0.01 [0.00, 0.02]	.007	0.01 [0.00, 0.01]	.039
Sleep onset × Europe	0.13 [0.11, 0.15]	< .001	0.08 [0.06, 0.10]	< .001

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset × Africa	0.14 [0.01, 0.27]	.032	-0.04 [-0.17, 0.09]	.531
Sleep onset × Asia	0.12 [0.01, 0.22]	.025	0.00 [-0.11, 0.10]	.947
Sleep onset × North America	0.20 [0.14, 0.25]	< .001	0.08 [0.02, 0.13]	.005
Sleep onset × South America	0.20 [0.18, 0.23]	< .001	0.10 [0.08, 0.12]	< .001
Europe × Sleep onset^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.343
Africa × Sleep onset^2	-0.03 [-0.10, 0.04]	.382	-0.06 [-0.13, 0.01]	.081
Asia × Sleep onset^2	-0.04 [-0.11, 0.03]	.246	-0.08 [-0.15, -0.01]	.017
North America × Sleep onset^2	0.01 [-0.02, 0.04]	.508	0.00 [-0.03, 0.03]	.840
South America × Sleep onset^2	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, 0.00]	.033
Sleep regularity (z)				
(Intercept)	0.31 [0.03, 0.60]	.032	0.09 [-0.03, 0.21]	.123
Sleep regularity	0.15 [0.13, 0.16]	< .001	0.07 [0.05, 0.08]	< .001
Europe	-0.25 [-0.31, -0.20]	< .001	-0.19 [-0.24, -0.14]	< .001
Africa	-0.20 [-0.26, -0.14]	< .001	-0.13 [-0.18, -0.07]	< .001
Asia	-0.45 [-0.52, -0.39]	< .001	-0.49 [-0.55, -0.43]	< .001
North America	-0.31 [-0.37, -0.26]	< .001	-0.24 [-0.29, -0.18]	< .001
South America	-0.27 [-0.32, -0.21]	< .001	-0.31 [-0.36, -0.26]	< .001
Sleep regularity^2	0.01 [0.00, 0.01]	.078	0.01 [0.00, 0.01]	.126
Sleep regularity × Europe	-0.08 [-0.10, -0.07]	< .001	-0.03 [-0.05, -0.02]	< .001
Sleep regularity × Africa	-0.09 [-0.11, -0.06]	< .001	0.03 [0.01, 0.06]	.009
Sleep regularity × Asia	-0.09 [-0.12, -0.06]	< .001	0.06 [0.03, 0.09]	< .001
Sleep regularity × North America	-0.05 [-0.08, -0.02]	< .001	0.03 [0.00, 0.05]	.066
Sleep regularity × South America	-0.01 [-0.03, 0.01]	.231	0.01 [-0.01, 0.03]	.193
Europe × Sleep regularity^2	-0.01 [-0.01, 0.00]	.149	-0.01 [-0.02, 0.00]	.064

Table 23 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Sleep regularity^2	-0.03 [-0.04, -0.01]	< .001	0.00 [-0.02, 0.01]	.655
Asia × Sleep regularity^2	-0.02 [-0.04, 0.00]	.034	-0.01 [-0.03, 0.01]	.333
North America × Sleep regularity^2	-0.01 [-0.03, 0.00]	.136	-0.02 [-0.03, 0.00]	.022
South America × Sleep regularity^2	0.00 [-0.01, 0.01]	.982	0.00 [-0.01, 0.01]	.964

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 24

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.37 [0.21, 0.53]	< .001	0.33 [0.18, 0.48]	< .001
Physical activity	0.19 [0.14, 0.24]	< .001	0.12 [0.08, 0.16]	< .001
Daylight hours	-0.03 [-0.03, -0.03]	< .001	-0.03 [-0.03, -0.02]	< .001
Physical activity^2	-0.02 [-0.04, -0.01]	.010	0.01 [-0.01, 0.03]	.275
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	< .001	0.00 [-0.01, 0.00]	.001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.029	0.00 [0.00, 0.00]	.707
Sleep efficiency (z)				
(Intercept)	0.26 [0.10, 0.42]	.002	0.24 [0.07, 0.41]	.005
Physical activity	0.12 [0.08, 0.17]	< .001	0.04 [0.01, 0.08]	.011
Daylight hours	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, 0.00]	.001
Physical activity^2	-0.03 [-0.05, -0.01]	.001	-0.01 [-0.03, 0.01]	.292
Physical activity × Daylight hours	-0.01 [-0.01, 0.00]	< .001	0.00 [-0.01, 0.00]	.017
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.066	0.00 [0.00, 0.00]	.394
Sleep onset (z)				
(Intercept)	-0.04 [-0.31, 0.23]	.764	0.00 [-0.25, 0.25]	.996
Physical activity	-0.19 [-0.23, -0.15]	< .001	-0.14 [-0.16, -0.11]	< .001
Daylight hours	0.01 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity^2	-0.01 [-0.03, 0.00]	.151	-0.04 [-0.06, -0.02]	< .001
Physical activity × Daylight hours	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Daylight hours × Physical activity^2	0.00 [0.00, 0.00]	.051	0.00 [0.00, 0.00]	.001
Sleep regularity (z)				
(Intercept)	0.06 [-0.08, 0.19]	.400	0.06 [-0.09, 0.21]	.450
Physical activity	0.26 [0.20, 0.31]	< .001	0.14 [0.11, 0.18]	< .001

Table 24 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.01 [0.00, 0.01]	$< .001$	0.01 [0.00, 0.01]	.004
Physical activity^2	-0.04 [-0.06, -0.02]	$< .001$	-0.05 [-0.07, -0.02]	$< .001$
Physical activity $\times$ Daylight hours	0.00 [-0.01, 0.00]	.710	0.00 [-0.01, 0.00]	.001
Daylight hours $\times$ Physical activity^2	0.00 [0.00, 0.00]	.406	0.00 [0.00, 0.00]	.039

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 25

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.08 [-0.38, 0.22]	.614	-0.11 [-0.23, 0.02]	.089
Sleep duration	0.11 [0.09, 0.13]	< .001	0.09 [0.07, 0.11]	< .001
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.044
Sleep duration^2	-0.01 [-0.02, 0.00]	.083	0.02 [0.00, 0.03]	.015
Sleep duration × Daylight hours	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, 0.00]	< .001
Daylight hours × Sleep duration^2	0.00 [0.00, 0.00]	.182	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	-0.09 [-0.39, 0.21]	.568	-0.07 [-0.20, 0.05]	.270
Sleep efficiency	0.03 [0.01, 0.06]	.010	0.01 [-0.01, 0.04]	.291
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.826
Sleep efficiency^2	-0.02 [-0.03, 0.00]	.007	-0.03 [-0.04, -0.02]	< .001
Sleep efficiency × Daylight hours	0.00 [0.00, 0.00]	.019	0.00 [0.00, 0.00]	.521
Daylight hours × Sleep efficiency^2	0.00 [0.00, 0.00]	.095	0.00 [0.00, 0.00]	.002
Sleep onset (z)				
(Intercept)	-0.09 [-0.39, 0.22]	.580	-0.10 [-0.22, 0.03]	.135
Sleep onset	-0.05 [-0.08, -0.02]	< .001	-0.04 [-0.06, -0.01]	.007
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.00]	.391
Sleep onset^2	-0.01 [-0.03, 0.00]	.160	0.01 [-0.01, 0.02]	.372
Sleep onset × Daylight hours	0.00 [0.00, 0.00]	.106	0.00 [0.00, 0.00]	.104
Daylight hours × Sleep onset^2	0.00 [0.00, 0.00]	.374	0.00 [0.00, 0.00]	.614
Sleep regularity (z)				
(Intercept)	-0.10 [-0.39, 0.19]	.493	-0.10 [-0.23, 0.02]	.095
Sleep regularity	0.13 [0.11, 0.15]	< .001	0.11 [0.09, 0.14]	< .001

Table 25 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours	0.02 [0.01, 0.02]	$< .001$	0.00 [0.00, 0.00]	.307
Sleep regularity^2	0.00 [-0.01, 0.02]	.688	0.01 [-0.01, 0.02]	.322
Sleep regularity $\times$ Daylight hours	0.00 [-0.01, 0.00]	$< .001$	0.00 [-0.01, 0.00]	$< .001$
Daylight hours $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.560	0.00 [0.00, 0.00]	.229

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 26

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.54 [0.31, 0.77]	< .001	0.44 [0.20, 0.67]	< .001
Physical activity	0.11 [0.09, 0.14]	< .001	0.08 [0.06, 0.09]	< .001
Wrist	-0.66 [-0.92, -0.41]	< .001	-0.54 [-0.80, -0.27]	< .001
Physical activity^2	-0.03 [-0.05, -0.01]	.003	0.01 [-0.01, 0.02]	.391
Physical activity × Wrist	-0.02 [-0.05, 0.00]	.064	-0.02 [-0.04, 0.00]	.077
Wrist × Physical activity^2	0.03 [0.01, 0.05]	.004	0.00 [-0.01, 0.02]	.696
Sleep efficiency (z)				
(Intercept)	-0.36 [-0.55, -0.16]	< .001	-0.46 [-0.66, -0.26]	< .001
Physical activity	0.11 [0.08, 0.14]	< .001	0.08 [0.07, 0.10]	< .001
Wrist	0.67 [0.45, 0.89]	< .001	0.77 [0.55, 0.99]	< .001
Physical activity^2	-0.11 [-0.13, -0.08]	< .001	-0.05 [-0.06, -0.03]	< .001
Physical activity × Wrist	-0.13 [-0.16, -0.10]	< .001	-0.10 [-0.12, -0.09]	< .001
Wrist × Physical activity^2	0.11 [0.09, 0.13]	< .001	0.04 [0.03, 0.06]	< .001
Sleep onset (z)				
(Intercept)	-0.82 [-1.17, -0.47]	< .001	-0.78 [-1.13, -0.43]	< .001
Physical activity	-0.03 [-0.05, -0.01]	.003	-0.03 [-0.05, -0.02]	< .001
Wrist	1.19 [0.80, 1.58]	< .001	1.12 [0.73, 1.52]	< .001
Physical activity^2	-0.03 [-0.04, -0.01]	< .001	-0.02 [-0.03, -0.01]	< .001
Physical activity × Wrist	-0.13 [-0.15, -0.11]	< .001	-0.07 [-0.08, -0.06]	< .001
Wrist × Physical activity^2	0.04 [0.03, 0.06]	< .001	0.01 [0.00, 0.02]	.021
Sleep regularity (z)				
(Intercept)	0.03 [-0.23, 0.30]	.809	-0.25 [-0.52, 0.03]	.075
Physical activity	0.17 [0.15, 0.20]	< .001	0.26 [0.24, 0.28]	< .001

Table 26 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	0.14 [-0.15, 0.43]	.340	0.47 [0.17, 0.77]	.002
Physical activity^2	-0.21 [-0.24, -0.17]	$<$ .001	-0.13 [-0.14, -0.12]	$<$ .001
Physical activity $\times$ Wrist	0.07 [0.04, 0.10]	$<$ .001	-0.22 [-0.24, -0.20]	$<$ .001
Wrist $\times$ Physical activity^2	0.16 [0.13, 0.20]	$<$ .001	0.11 [0.10, 0.13]	$<$ .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 27

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	-0.98 [-1.36, -0.59]	< .001	-0.40 [-0.62, -0.17]	< .001
Sleep duration	0.06 [0.05, 0.07]	< .001	0.07 [0.06, 0.08]	< .001
Wrist	1.38 [0.95, 1.82]	< .001	0.40 [0.15, 0.65]	.002
Sleep duration^2	-0.02 [-0.03, -0.01]	< .001	-0.03 [-0.03, -0.02]	< .001
Sleep duration × Wrist	-0.06 [-0.07, -0.05]	< .001	-0.05 [-0.06, -0.04]	< .001
Wrist × Sleep duration^2	0.00 [-0.01, 0.01]	.642	0.02 [0.01, 0.03]	< .001
Sleep efficiency (z)				
(Intercept)	-0.95 [-1.34, -0.56]	< .001	-0.37 [-0.60, -0.14]	.002
Sleep efficiency	0.02 [0.01, 0.03]	< .001	0.01 [0.01, 0.02]	.001
Wrist	1.33 [0.90, 1.77]	< .001	0.36 [0.11, 0.62]	.005
Sleep efficiency^2	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Wrist	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.03, 0.00]	.006
Wrist × Sleep efficiency^2	0.00 [0.00, 0.01]	.049	0.01 [0.00, 0.01]	.024
Sleep onset (z)				
(Intercept)	-0.97 [-1.36, -0.58]	< .001	-0.41 [-0.63, -0.18]	< .001
Sleep onset	-0.03 [-0.06, -0.01]	.006	-0.07 [-0.09, -0.04]	< .001
Wrist	1.37 [0.94, 1.80]	< .001	0.40 [0.15, 0.65]	.002
Sleep onset^2	-0.02 [-0.03, 0.00]	.032	-0.03 [-0.04, -0.01]	< .001
Sleep onset × Wrist	0.01 [-0.02, 0.03]	.637	0.05 [0.03, 0.08]	< .001
Wrist × Sleep onset^2	0.01 [0.00, 0.03]	.075	0.03 [0.01, 0.04]	< .001
Sleep regularity (z)				
(Intercept)	-0.93 [-1.31, -0.55]	< .001	-0.36 [-0.58, -0.13]	.002
Sleep regularity	0.11 [0.10, 0.12]	< .001	0.13 [0.13, 0.14]	< .001

Table 27 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist	1.29 [0.87, 1.72]	$< .001$	0.34 [0.09, 0.60]	.007
Sleep regularity^2	-0.01 [-0.01, 0.00]	.019	0.00 [0.00, 0.01]	.458
Sleep regularity $\times$ Wrist	-0.04 [-0.05, -0.03]	$< .001$	-0.11 [-0.12, -0.10]	$< .001$
Wrist $\times$ Sleep regularity^2	0.01 [0.00, 0.01]	.009	0.00 [0.00, 0.01]	.701

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 28

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.08 [-0.08, 0.24]	.320	0.10 [-0.04, 0.25]	.174
Physical activity	0.06 [0.04, 0.09]	< .001	0.06 [0.04, 0.08]	< .001
Most active hour	-0.01 [-0.01, 0.00]	< .001	-0.01 [-0.01, -0.01]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.785	-0.01 [-0.03, 0.00]	.061
Physical activity × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.711
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.283	0.00 [0.00, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.17 [0.01, 0.33]	.034	0.19 [0.03, 0.35]	.020
Physical activity	-0.04 [-0.06, -0.01]	.003	-0.02 [-0.04, 0.00]	.017
Most active hour	0.00 [0.00, 0.00]	.611	0.00 [0.00, 0.00]	< .001
Physical activity^2	0.00 [-0.01, 0.01]	.848	-0.02 [-0.03, 0.00]	.007
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.002
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.016
Sleep onset (z)				
(Intercept)	-0.05 [-0.32, 0.21]	.694	-0.05 [-0.30, 0.20]	.684
Physical activity	-0.18 [-0.20, -0.16]	< .001	-0.10 [-0.11, -0.09]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity^2	0.01 [0.00, 0.02]	.022	0.01 [0.00, 0.02]	.263
Physical activity × Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.009
Most active hour × Physical activity^2	0.00 [0.00, 0.00]	.047	0.00 [0.00, 0.00]	.006
Sleep regularity (z)				
(Intercept)	0.20 [0.07, 0.32]	.002	0.19 [0.05, 0.33]	.008
Physical activity	0.19 [0.16, 0.22]	< .001	0.05 [0.03, 0.07]	< .001

Table 28 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.00 [-0.01, 0.00]	$< .001$	0.00 [-0.01, 0.00]	$< .001$
Physical activity^2	-0.05 [-0.06, -0.03]	$< .001$	-0.03 [-0.04, -0.01]	$< .001$
Physical activity $\times$ Most active hour	0.00 [0.00, 0.01]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Most active hour $\times$ Physical activity^2	0.00 [0.00, 0.00]	.744	0.00 [0.00, 0.00]	.369

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor.

Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 29

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.02 [-0.27, 0.32]	.873	-0.09 [-0.22, 0.03]	.127
Sleep duration	0.00 [-0.02, 0.01]	.602	0.06 [0.05, 0.07]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep duration^2	0.00 [-0.01, 0.01]	.802	0.00 [-0.01, 0.01]	.605
Sleep duration × Most active hour	0.00 [0.00, 0.00]	.007	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep duration^2	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.058
Sleep efficiency (z)				
(Intercept)	0.02 [-0.27, 0.32]	.868	-0.10 [-0.22, 0.02]	.107
Sleep efficiency	-0.01 [-0.02, 0.00]	.089	-0.04 [-0.05, -0.03]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep efficiency^2	0.00 [-0.01, 0.01]	.810	0.00 [-0.01, 0.00]	.177
Sleep efficiency × Most active hour	0.00 [0.00, 0.00]	.015	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep efficiency^2	0.00 [0.00, 0.00]	.030	0.00 [0.00, 0.00]	.458
Sleep onset (z)				
(Intercept)	0.02 [-0.27, 0.32]	.870	-0.12 [-0.24, 0.01]	.063
Sleep onset	-0.04 [-0.05, -0.03]	< .001	-0.06 [-0.07, -0.05]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep onset^2	0.00 [0.00, 0.01]	.496	0.01 [0.01, 0.02]	< .001
Sleep onset × Most active hour	0.00 [0.00, 0.00]	.052	0.00 [0.00, 0.00]	< .001
Most active hour × Sleep onset^2	0.00 [0.00, 0.00]	.025	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	0.01 [-0.28, 0.29]	.959	-0.11 [-0.23, 0.00]	.056
Sleep regularity	0.02 [0.01, 0.03]	.004	0.01 [-0.01, 0.02]	.342

Table 29 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour	0.01 [0.01, 0.01]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Sleep regularity^2	0.00 [0.00, 0.01]	.408	0.00 [-0.01, 0.01]	.970
Sleep regularity $\times$ Most active hour	0.01 [0.00, 0.01]	$< .001$	0.00 [0.00, 0.00]	$< .001$
Most active hour $\times$ Sleep regularity^2	0.00 [0.00, 0.00]	.269	0.00 [0.00, 0.00]	.771

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and $\times$ an interaction.

Table 30

Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.07 [-0.09, 0.23]	.399	0.07 [-0.08, 0.22]	.369
Physical activity	0.10 [0.09, 0.12]	< .001	0.07 [0.06, 0.08]	< .001
Non-white	-0.13 [-0.16, -0.10]	< .001	-0.16 [-0.19, -0.13]	< .001
Unclear	-0.11 [-0.14, -0.08]	< .001	-0.10 [-0.12, -0.07]	< .001
Physical activity^2	0.00 [0.00, 0.00]	.908	0.00 [0.00, 0.01]	.128
Physical activity × Non-white	0.06 [0.03, 0.08]	< .001	0.03 [0.01, 0.05]	.003
Physical activity × Unclear	-0.02 [-0.05, 0.00]	.036	-0.03 [-0.04, -0.01]	.006
Non-white × Physical activity^2	0.00 [-0.01, 0.01]	.388	0.01 [-0.01, 0.02]	.401
Unclear × Physical activity^2	0.00 [-0.01, 0.00]	.387	0.01 [0.00, 0.02]	.148
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.33]	.025	0.17 [0.01, 0.33]	.038
Physical activity	0.04 [0.02, 0.05]	< .001	0.00 [-0.01, 0.01]	.402
Non-white	-0.06 [-0.09, -0.04]	< .001	-0.10 [-0.13, -0.07]	< .001
Unclear	-0.01 [-0.04, 0.02]	.610	0.00 [-0.03, 0.03]	.779
Physical activity^2	-0.01 [-0.01, -0.01]	< .001	0.00 [-0.01, 0.00]	.300
Physical activity × Non-white	0.07 [0.04, 0.09]	< .001	0.06 [0.04, 0.08]	< .001
Physical activity × Unclear	-0.01 [-0.03, 0.01]	.540	-0.02 [-0.03, 0.00]	.037
Non-white × Physical activity^2	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.03, 0.01]	.209
Unclear × Physical activity^2	0.00 [0.00, 0.01]	.630	0.00 [-0.01, 0.01]	.580
Sleep onset (z)				
(Intercept)	0.10 [-0.17, 0.37]	.478	0.10 [-0.15, 0.35]	.448
Physical activity	-0.12 [-0.14, -0.11]	< .001	-0.08 [-0.09, -0.07]	< .001
Non-white	0.07 [0.04, 0.09]	< .001	0.06 [0.03, 0.08]	< .001

Table 30 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	0.05 [0.02, 0.07]	< .001	0.04 [0.02, 0.06]	.002
Physical activity^2	0.00 [0.00, 0.01]	.034	-0.01 [-0.01, 0.00]	.003
Physical activity × Non-white	0.02 [0.00, 0.04]	.039	0.00 [-0.02, 0.02]	.834
Physical activity × Unclear	0.04 [0.02, 0.05]	< .001	0.00 [-0.02, 0.01]	.484
Non-white × Physical activity^2	-0.02 [-0.02, -0.01]	< .001	-0.01 [-0.02, 0.01]	.297
Unclear × Physical activity^2	0.00 [-0.01, 0.00]	.843	0.00 [-0.01, 0.01]	.884
Sleep regularity (z)				
(Intercept)	0.19 [0.06, 0.31]	.004	0.17 [0.03, 0.32]	.021
Physical activity	0.24 [0.22, 0.26]	< .001	0.08 [0.07, 0.09]	< .001
Non-white	0.01 [-0.02, 0.04]	.385	0.04 [0.01, 0.08]	.006
Unclear	-0.09 [-0.12, -0.06]	< .001	-0.07 [-0.10, -0.03]	< .001
Physical activity^2	-0.04 [-0.05, -0.04]	< .001	0.00 [-0.01, 0.00]	.352
Physical activity × Non-white	-0.03 [-0.06, -0.01]	.021	0.10 [0.08, 0.13]	< .001
Physical activity × Unclear	0.03 [0.01, 0.06]	.005	-0.01 [-0.03, 0.00]	.126
Non-white × Physical activity^2	0.01 [0.00, 0.02]	.274	-0.08 [-0.10, -0.06]	< .001
Unclear × Physical activity^2	-0.01 [-0.02, 0.00]	.012	-0.03 [-0.04, -0.02]	< .001

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

Table 31

Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.12 [-0.17, 0.42]	.410	-0.05 [-0.17, 0.06]	.372
Sleep duration	0.02 [0.01, 0.02]	< .001	0.04 [0.03, 0.04]	< .001
Non-white	-0.03 [-0.06, -0.01]	.016	-0.03 [-0.05, -0.01]	.015
Unclear	0.03 [0.00, 0.06]	.027	-0.03 [-0.05, -0.01]	.013
Sleep duration^2	-0.02 [-0.02, -0.01]	< .001	0.00 [-0.01, 0.00]	.088
Sleep duration × Non-white	-0.01 [-0.02, 0.00]	.179	-0.04 [-0.05, -0.02]	< .001
Sleep duration × Unclear	-0.01 [-0.02, 0.00]	.003	-0.02 [-0.03, -0.01]	< .001
Non-white × Sleep duration^2	-0.01 [-0.01, 0.00]	.123	-0.01 [-0.02, -0.01]	< .001
Unclear × Sleep duration^2	-0.01 [-0.01, 0.00]	.002	-0.01 [-0.02, -0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.12 [-0.17, 0.42]	.425	-0.04 [-0.16, 0.08]	.500
Sleep efficiency	0.01 [0.00, 0.01]	.017	0.01 [0.01, 0.02]	< .001
Non-white	-0.05 [-0.07, -0.03]	< .001	-0.06 [-0.08, -0.04]	< .001
Unclear	0.01 [-0.01, 0.04]	.340	-0.06 [-0.08, -0.03]	< .001
Sleep efficiency^2	-0.01 [-0.01, -0.01]	< .001	-0.01 [-0.01, -0.01]	< .001
Sleep efficiency × Non-white	-0.02 [-0.03, 0.00]	.043	-0.02 [-0.03, 0.00]	.029
Sleep efficiency × Unclear	-0.01 [-0.02, 0.00]	.271	-0.02 [-0.03, 0.00]	.004
Non-white × Sleep efficiency^2	0.00 [0.00, 0.01]	.352	0.00 [0.00, 0.01]	.361
Unclear × Sleep efficiency^2	0.00 [0.00, 0.01]	.845	0.00 [0.00, 0.01]	.143
Sleep onset (z)				
(Intercept)	0.13 [-0.17, 0.43]	.398	-0.05 [-0.17, 0.07]	.398
Sleep onset	-0.02 [-0.03, -0.01]	< .001	-0.01 [-0.02, -0.01]	< .001
Non-white	-0.03 [-0.06, 0.00]	.022	-0.04 [-0.06, -0.01]	.002

Table 31 continued

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear	-0.01 [-0.04, 0.02]	.575	-0.05 [-0.08, -0.03]	< .001
Sleep onset^2	-0.01 [-0.01, 0.00]	.004	0.01 [0.00, 0.01]	.003
Sleep onset × Non-white	0.05 [0.04, 0.07]	< .001	0.03 [0.01, 0.04]	< .001
Sleep onset × Unclear	-0.03 [-0.04, -0.01]	< .001	-0.01 [-0.03, 0.00]	.016
Non-white × Sleep onset^2	0.01 [0.00, 0.02]	.026	0.00 [-0.01, 0.01]	.482
Unclear × Sleep onset^2	0.00 [0.00, 0.01]	.344	0.00 [-0.01, 0.00]	.118
Sleep regularity (z)				
(Intercept)	0.09 [-0.20, 0.38]	.527	-0.06 [-0.18, 0.05]	.296
Sleep regularity	0.07 [0.07, 0.08]	< .001	0.04 [0.04, 0.05]	< .001
Non-white	-0.03 [-0.05, -0.01]	.016	-0.04 [-0.06, -0.02]	< .001
Unclear	0.03 [0.00, 0.06]	.027	-0.04 [-0.07, -0.02]	< .001
Sleep regularity^2	0.00 [-0.01, 0.00]	.832	0.00 [0.00, 0.00]	.707
Sleep regularity × Non-white	0.01 [0.00, 0.03]	.083	0.04 [0.03, 0.06]	< .001
Sleep regularity × Unclear	0.04 [0.03, 0.05]	< .001	0.01 [0.00, 0.02]	.038
Non-white × Sleep regularity^2	-0.01 [-0.02, 0.00]	.013	-0.01 [-0.02, 0.00]	.088
Unclear × Sleep regularity^2	0.00 [0.00, 0.01]	.459	0.00 [0.00, 0.01]	.714

Note. Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ^{2} denotes a quadratic term and × an interaction.

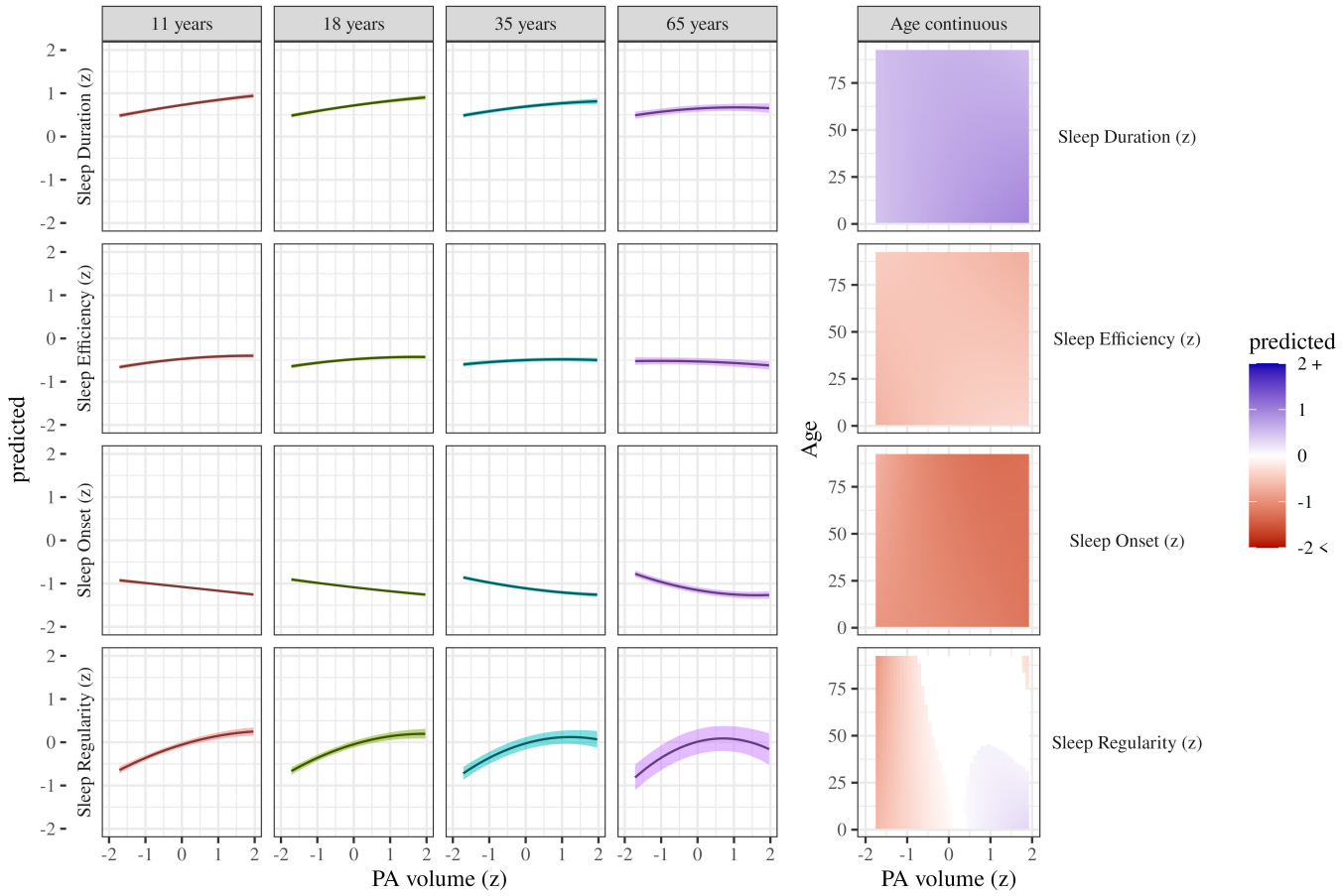

Figure 1. Sleep metrics on Physical activity volume

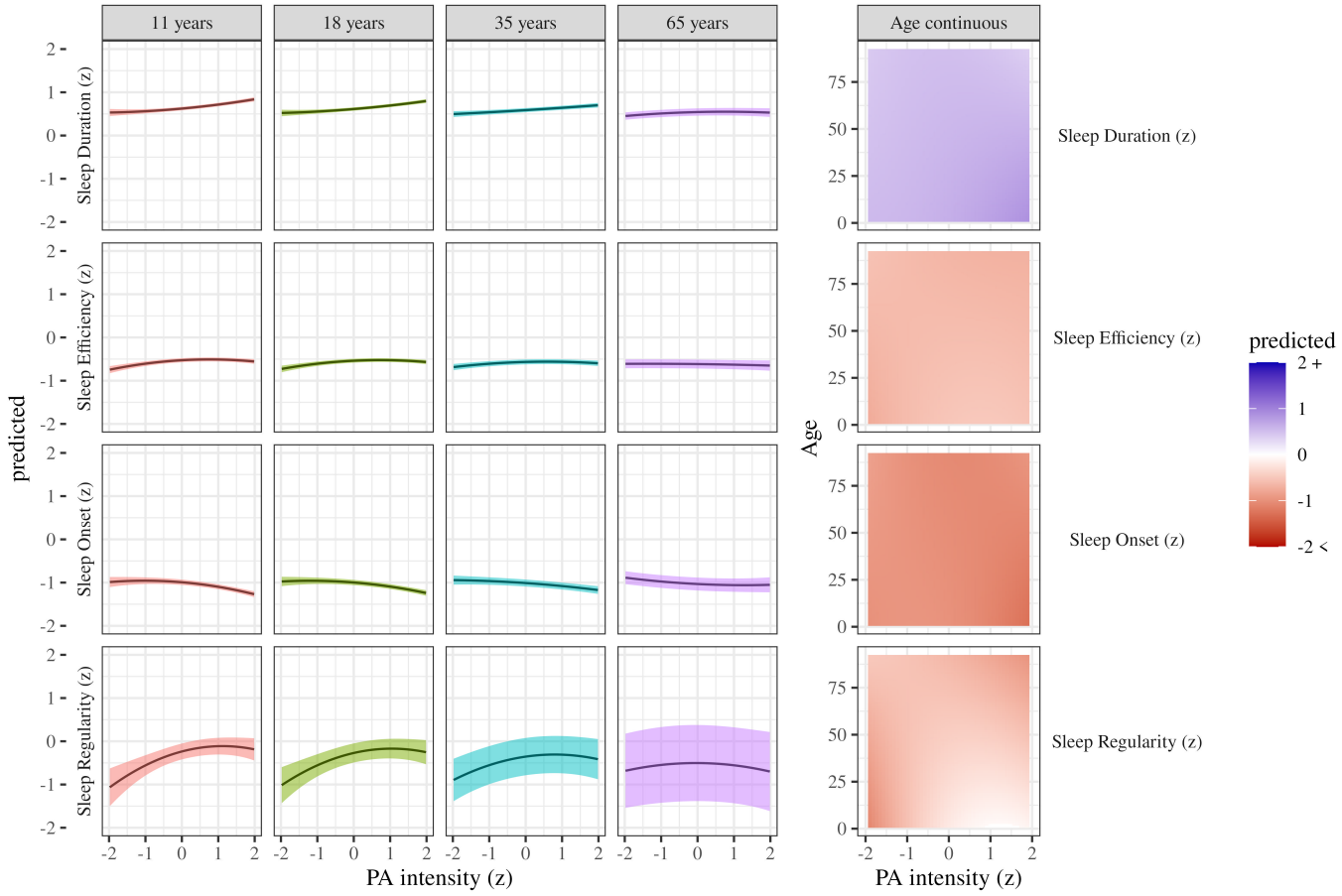

Figure 2. Sleep metrics on Physical activity intensity

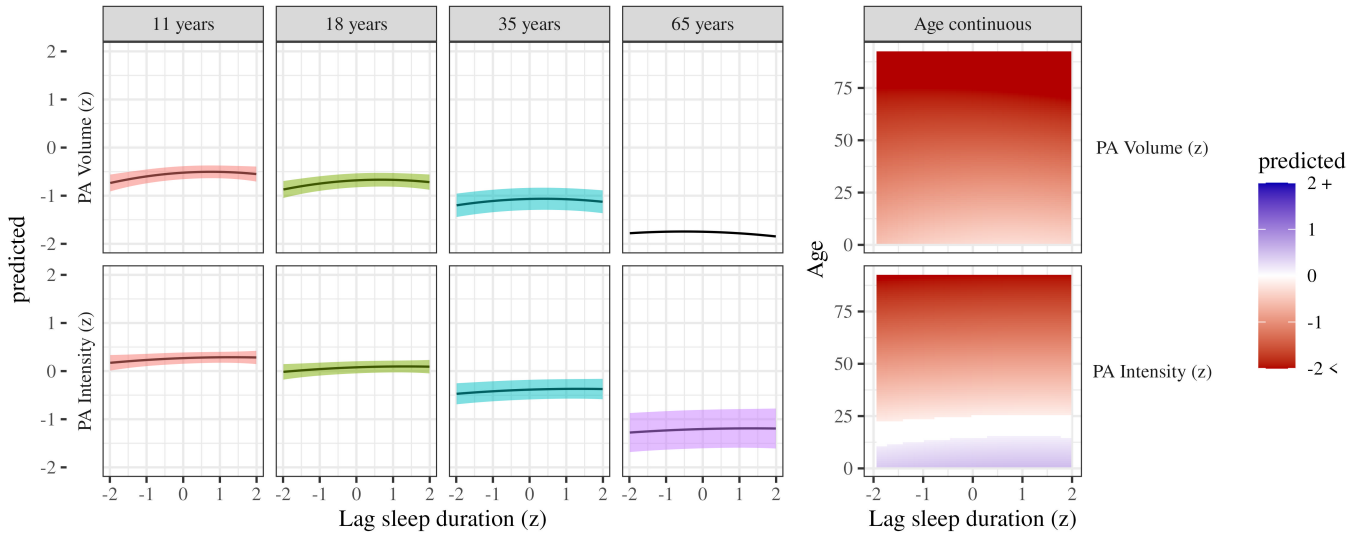

Figure 3. Physical activity by sleep duration

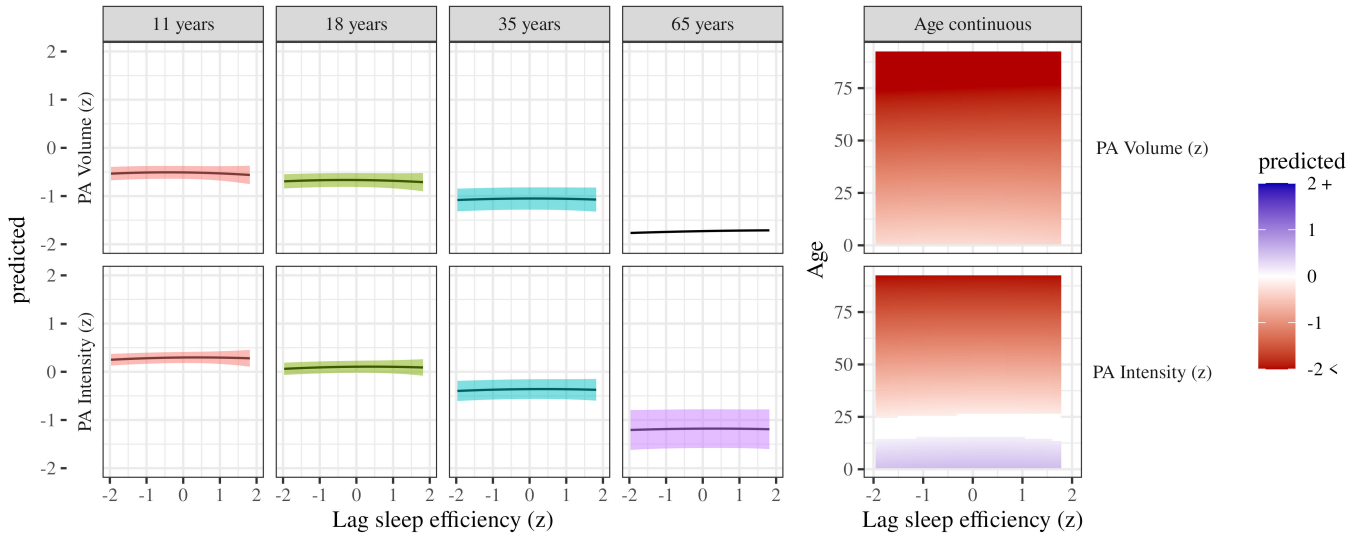

Figure 4. Physical activity by sleep efficiency

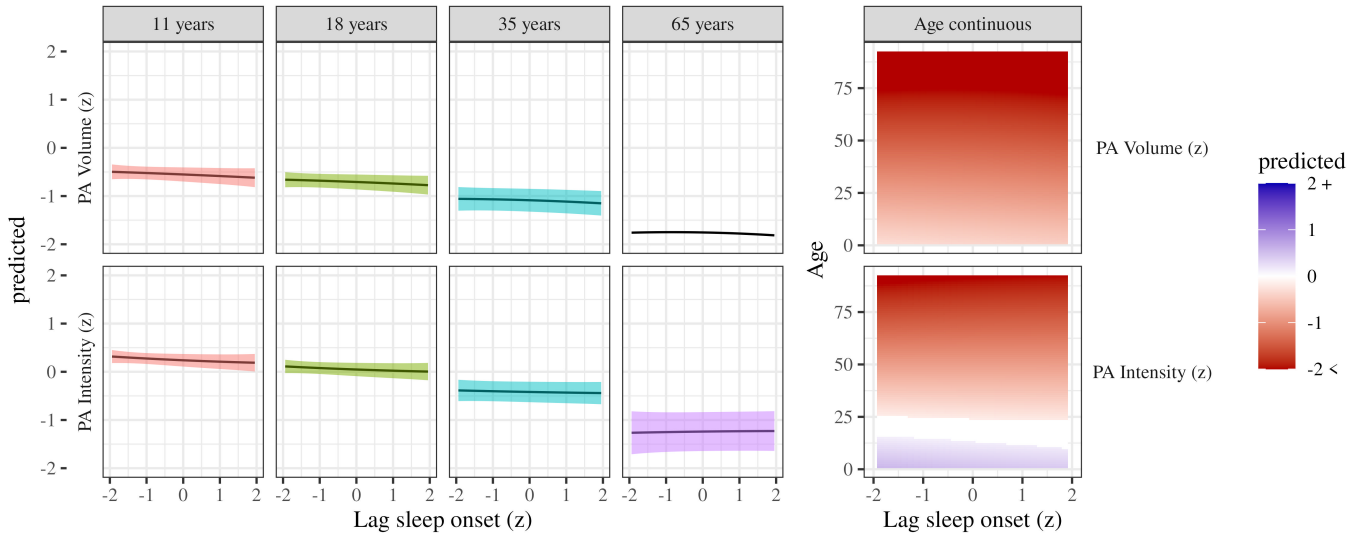

Figure 5. Physical activity by sleep onset

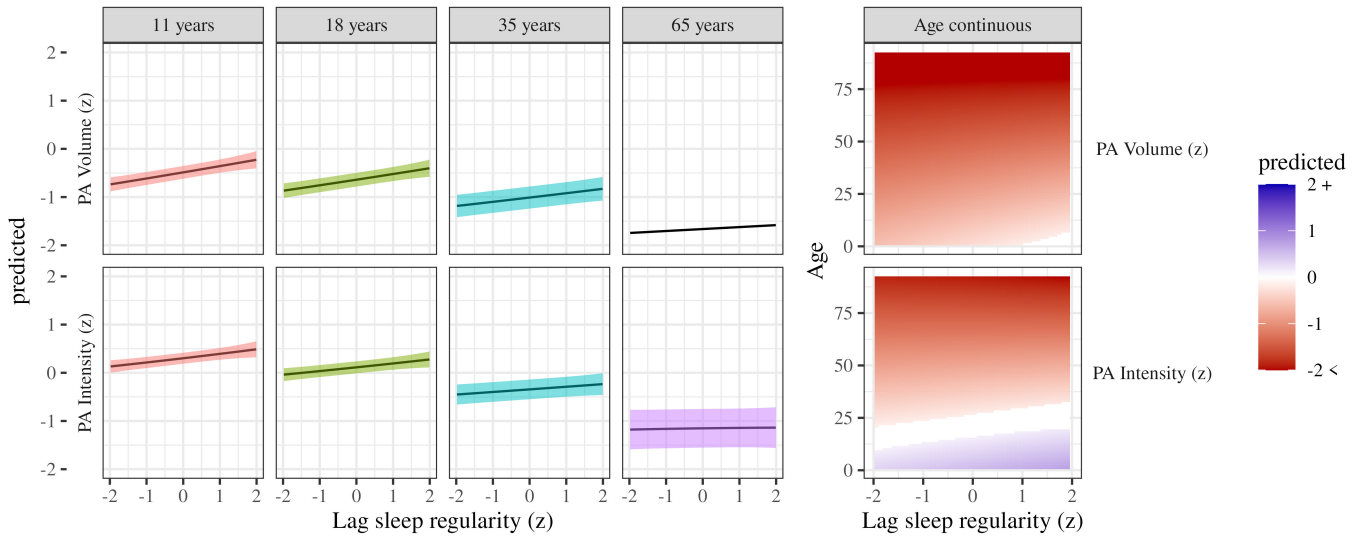

Figure 6. Physical activity by sleep regularity

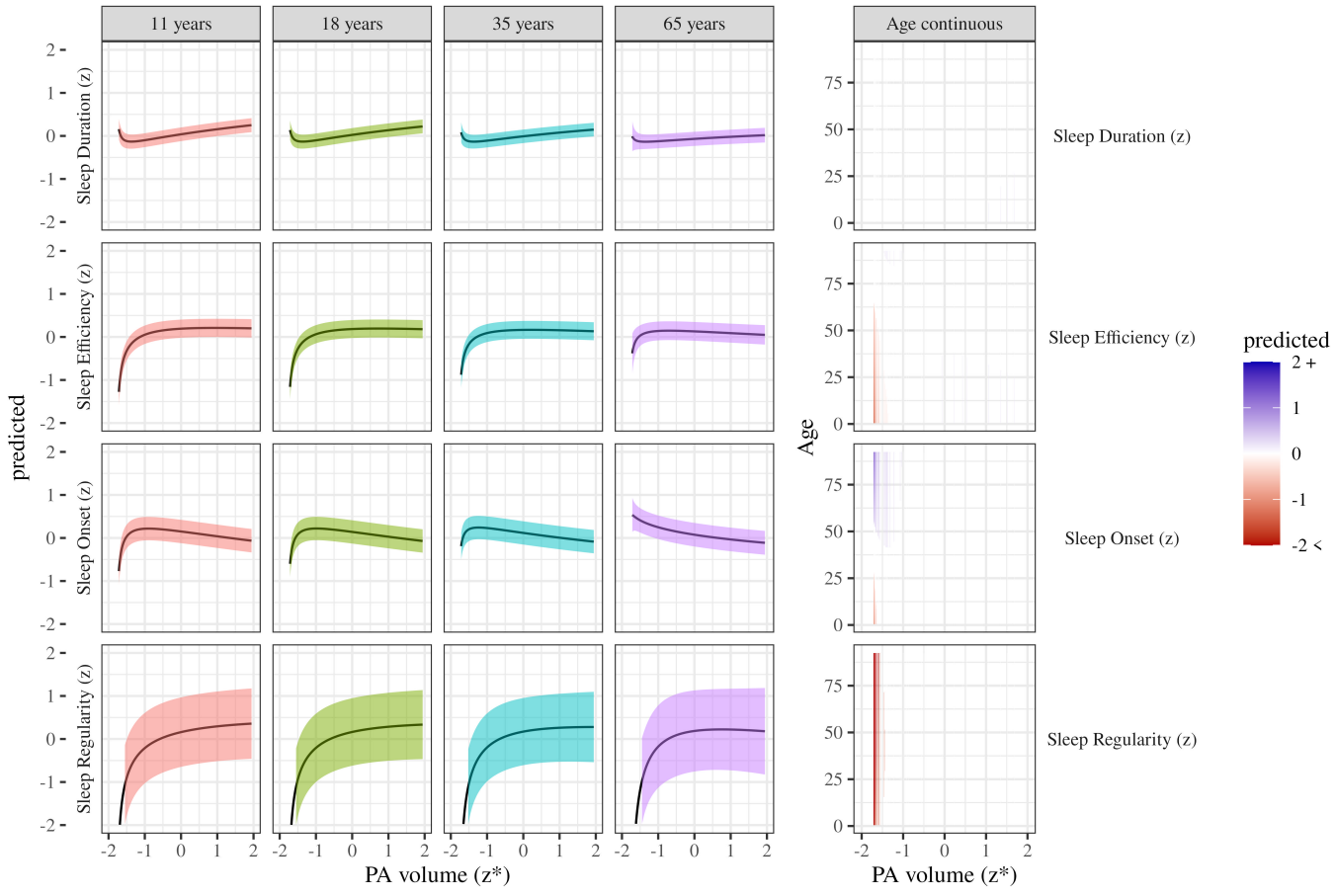

Figure 7. Sleep metrics on Physical activity volume

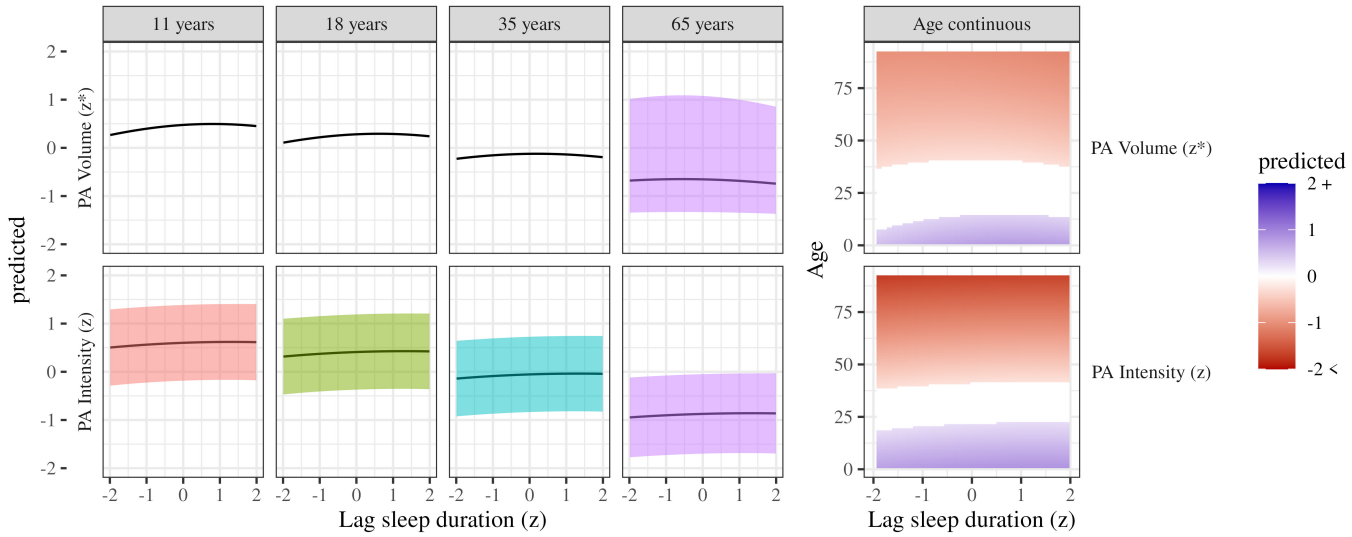

Figure 8. Physical activity by sleep duration

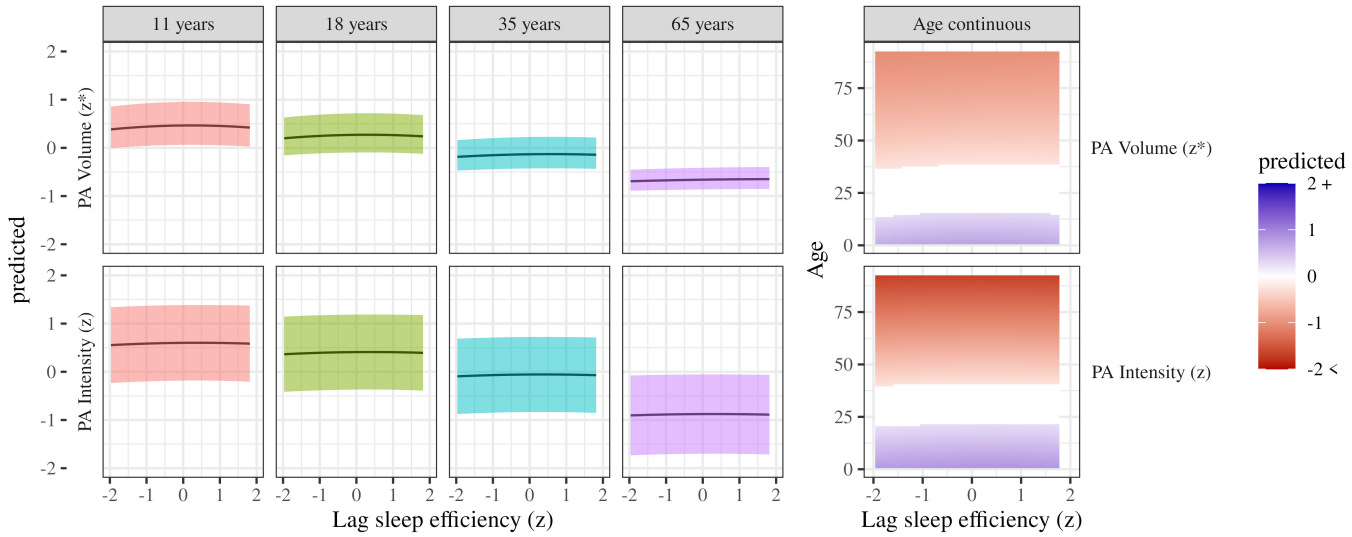

Figure 9. Physical activity by sleep efficiency

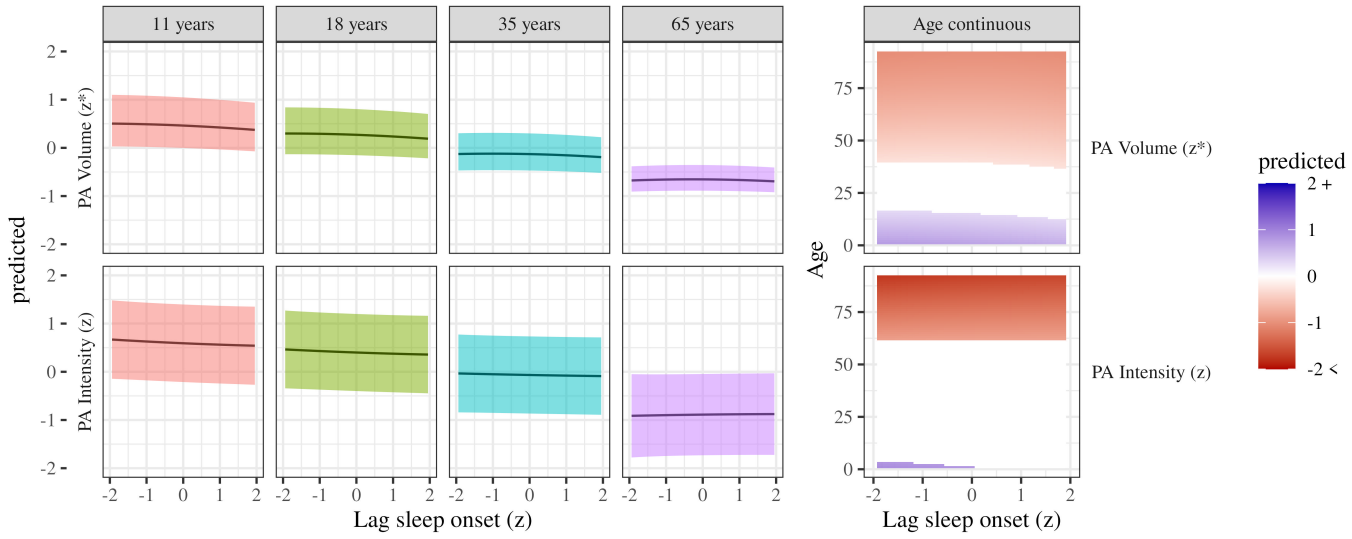

Figure 10. Physical activity by sleep onset

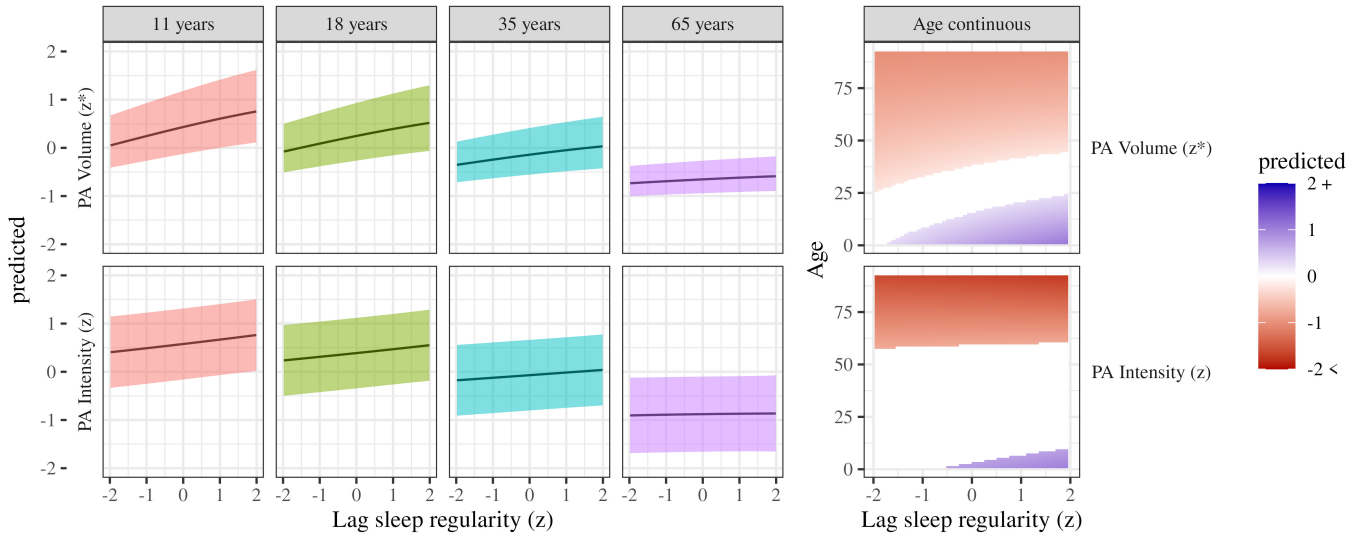

Figure 11. Physical activity by sleep regularity

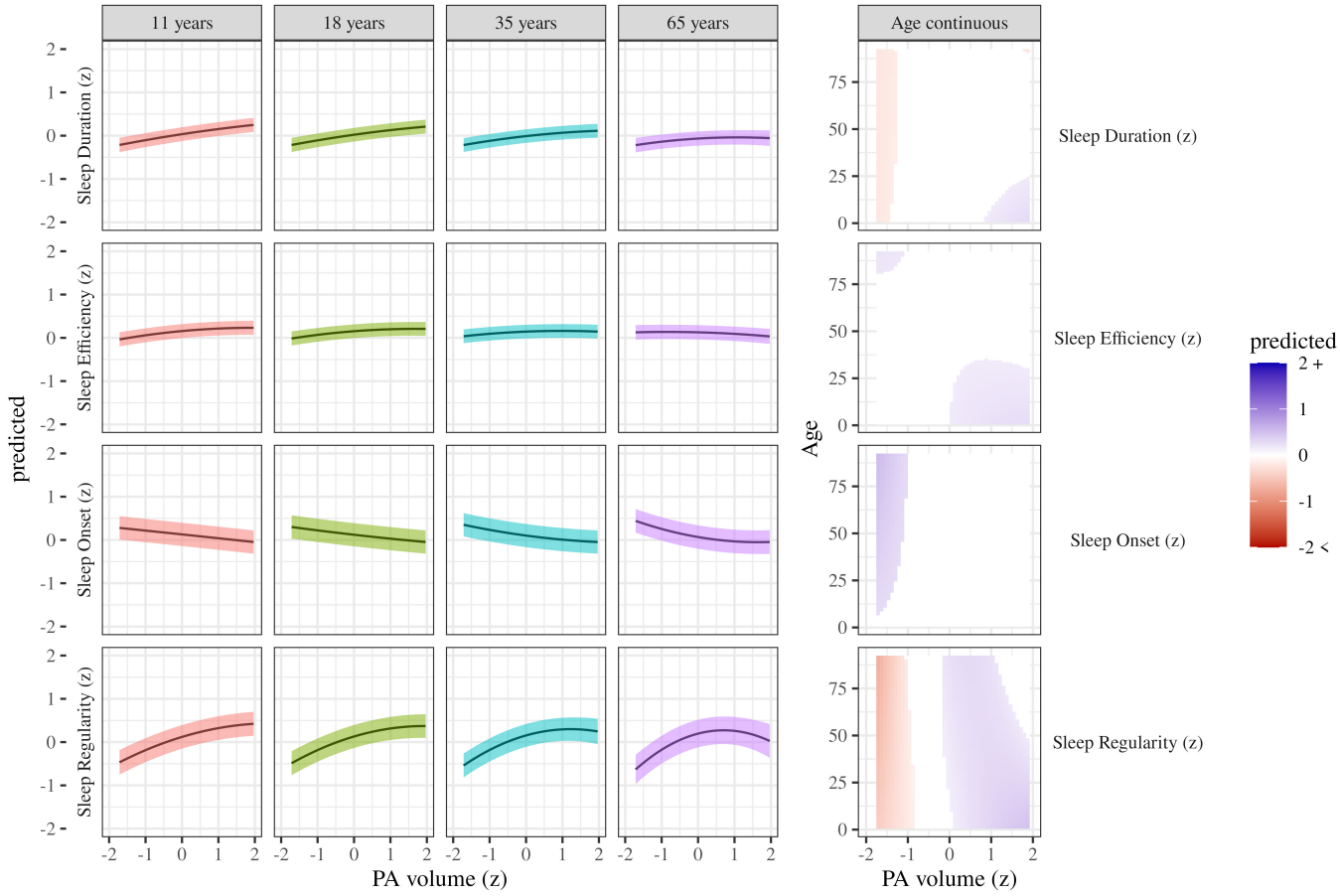

Figure 12. Sleep metrics on physical activity volume (all wear days)

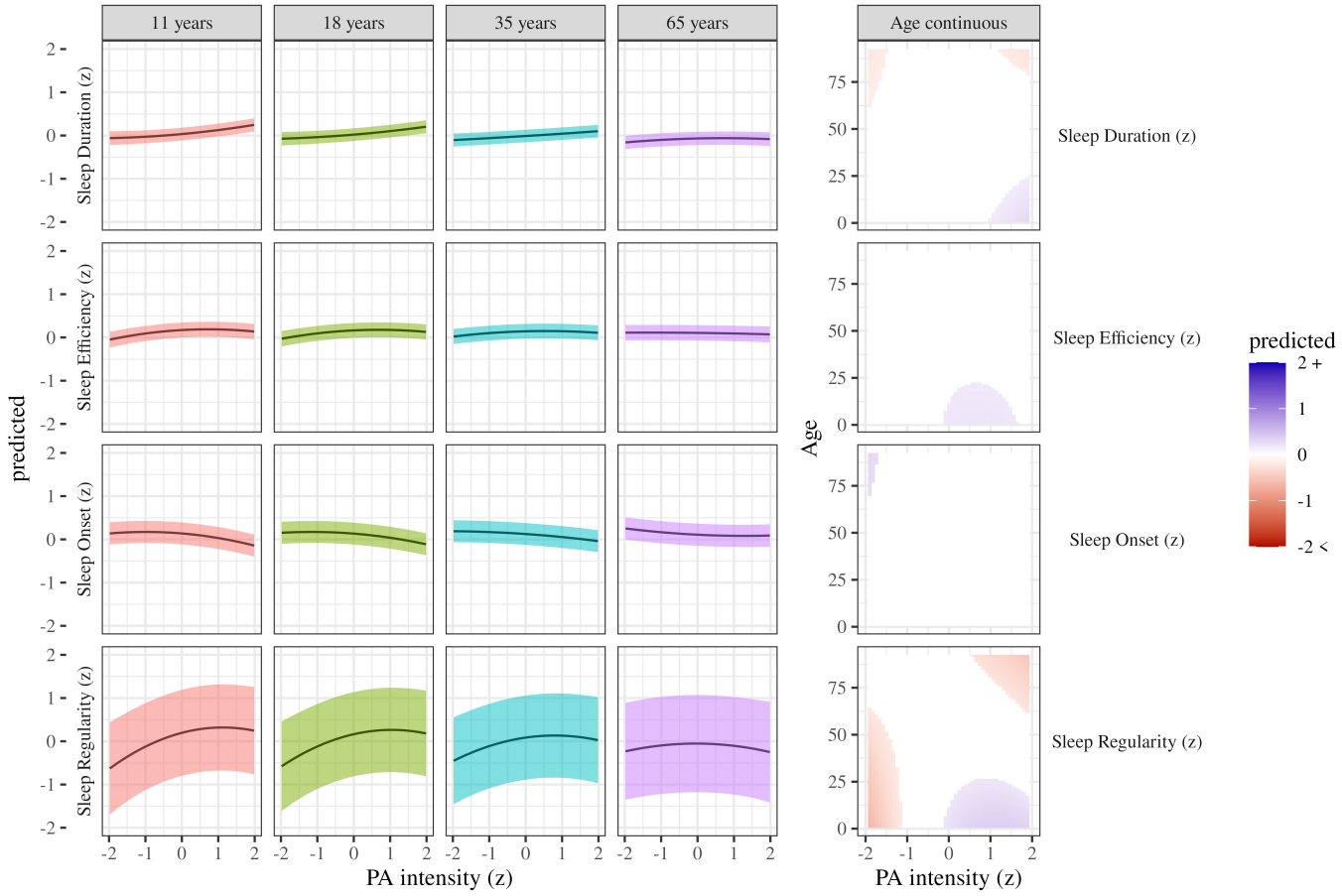

Figure 13. Sleep metrics on physical activity intensity (all wear days)

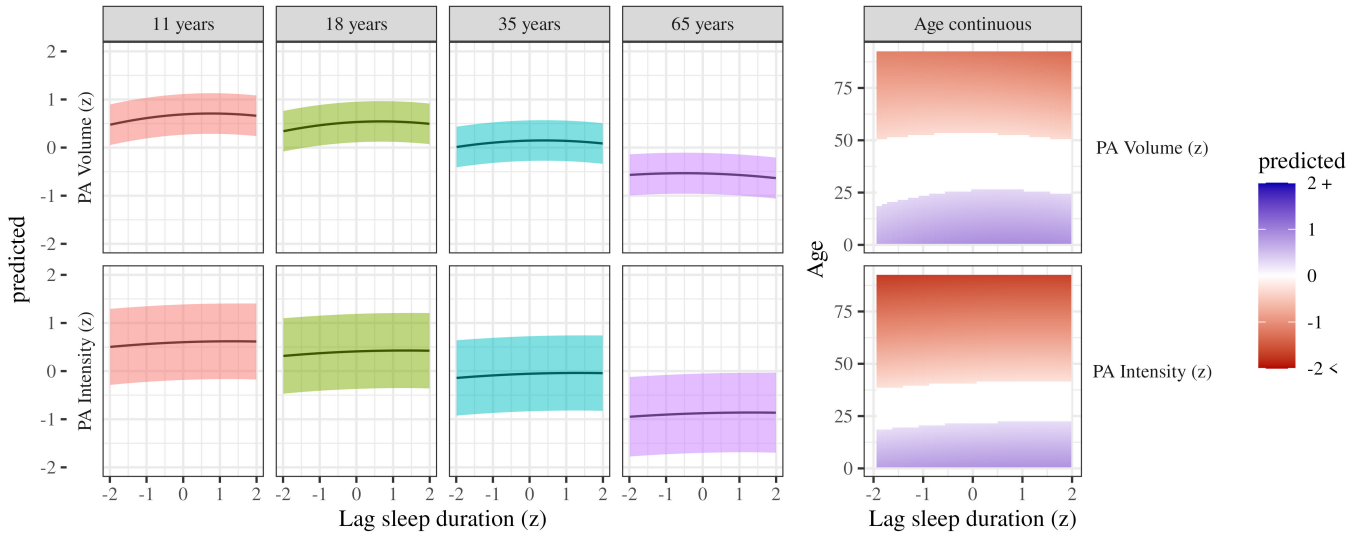

Figure 14. Physical activity by previous night sleep duration (all wear days)

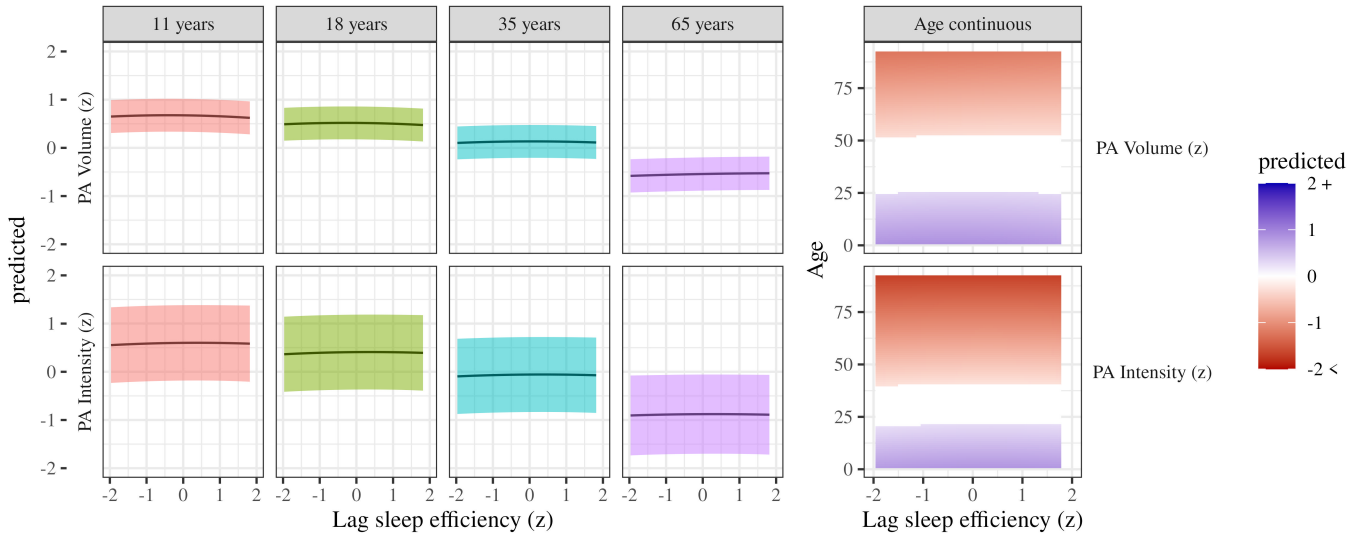

Figure 15. Physical activity by previous night sleep efficiency (all wear days)

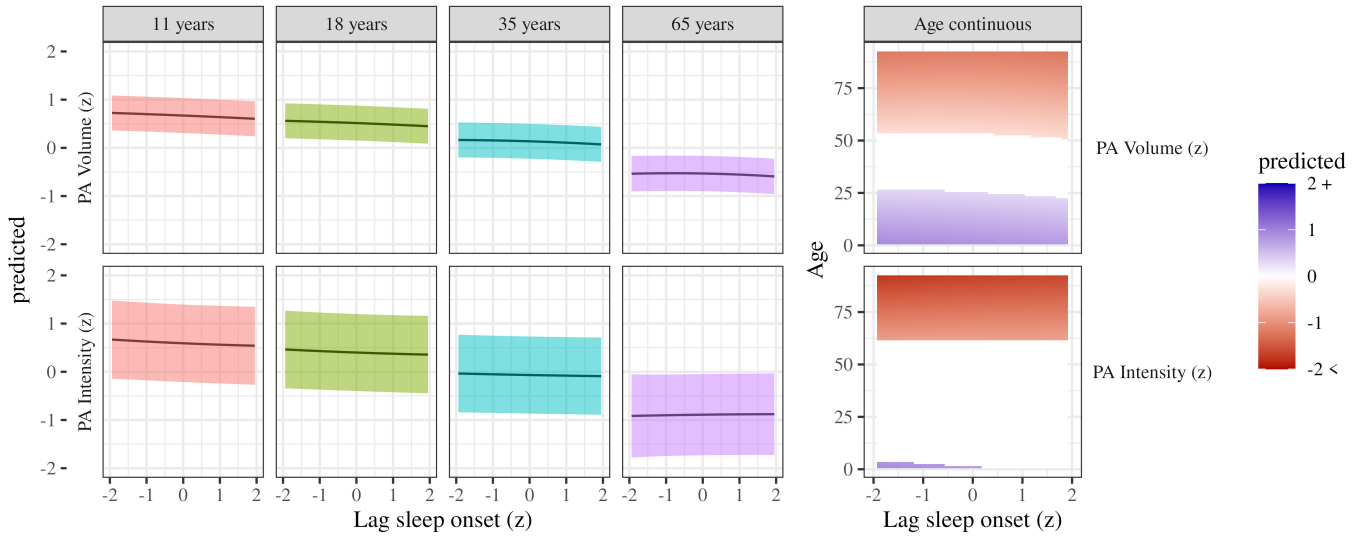

Figure 16. Physical activity by previous night sleep onset (all wear days)

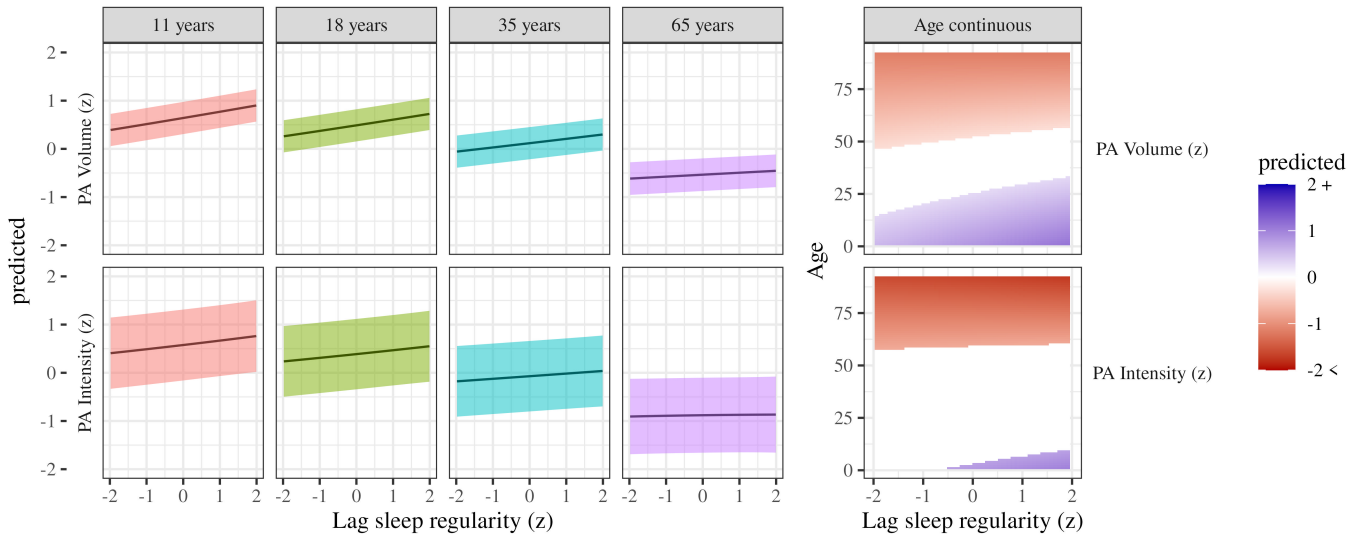

Figure 17. Physical activity by previous night sleep regularity (all wear days)

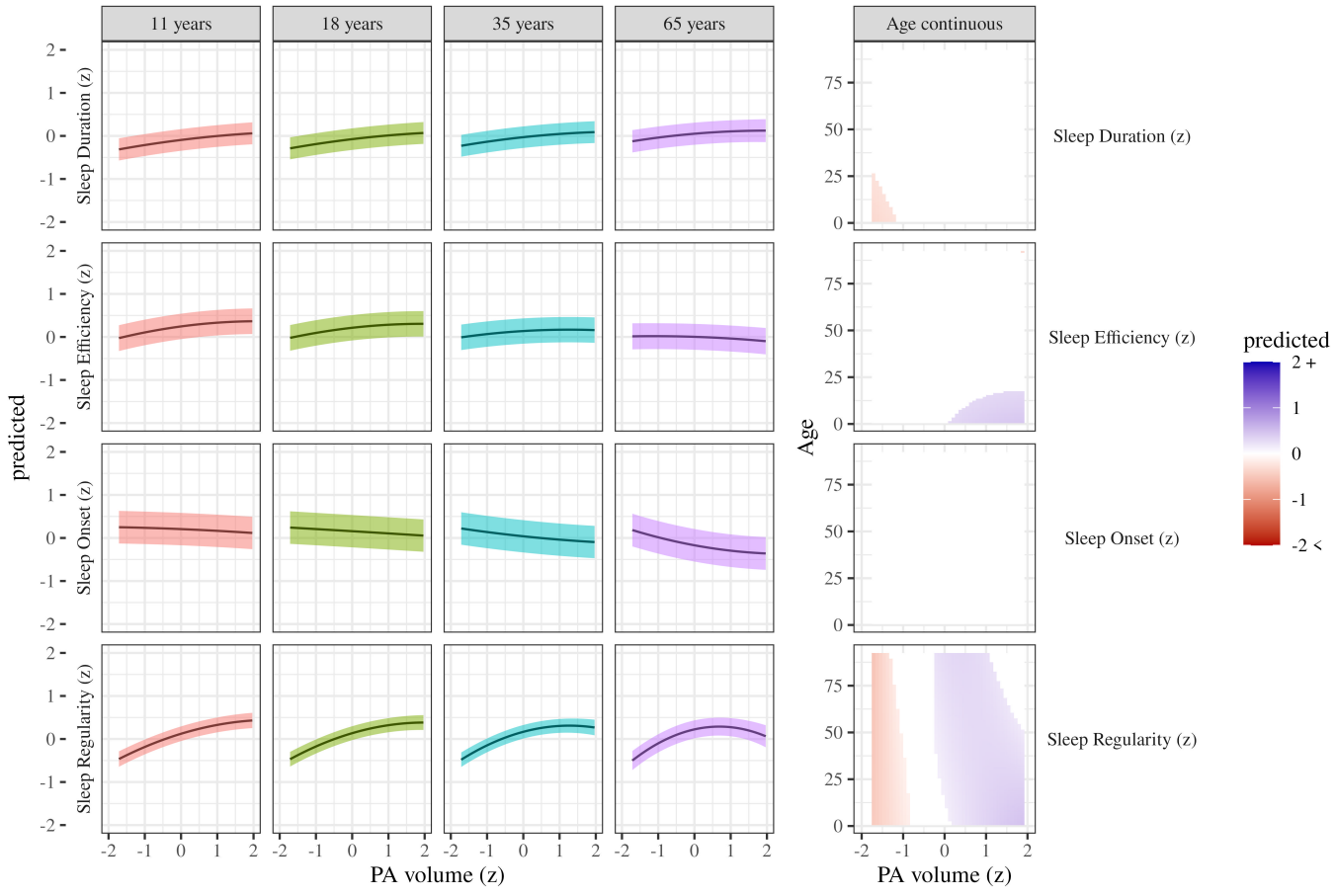

Figure 18. Sleep metrics on physical activity volume (at least seven valid days)

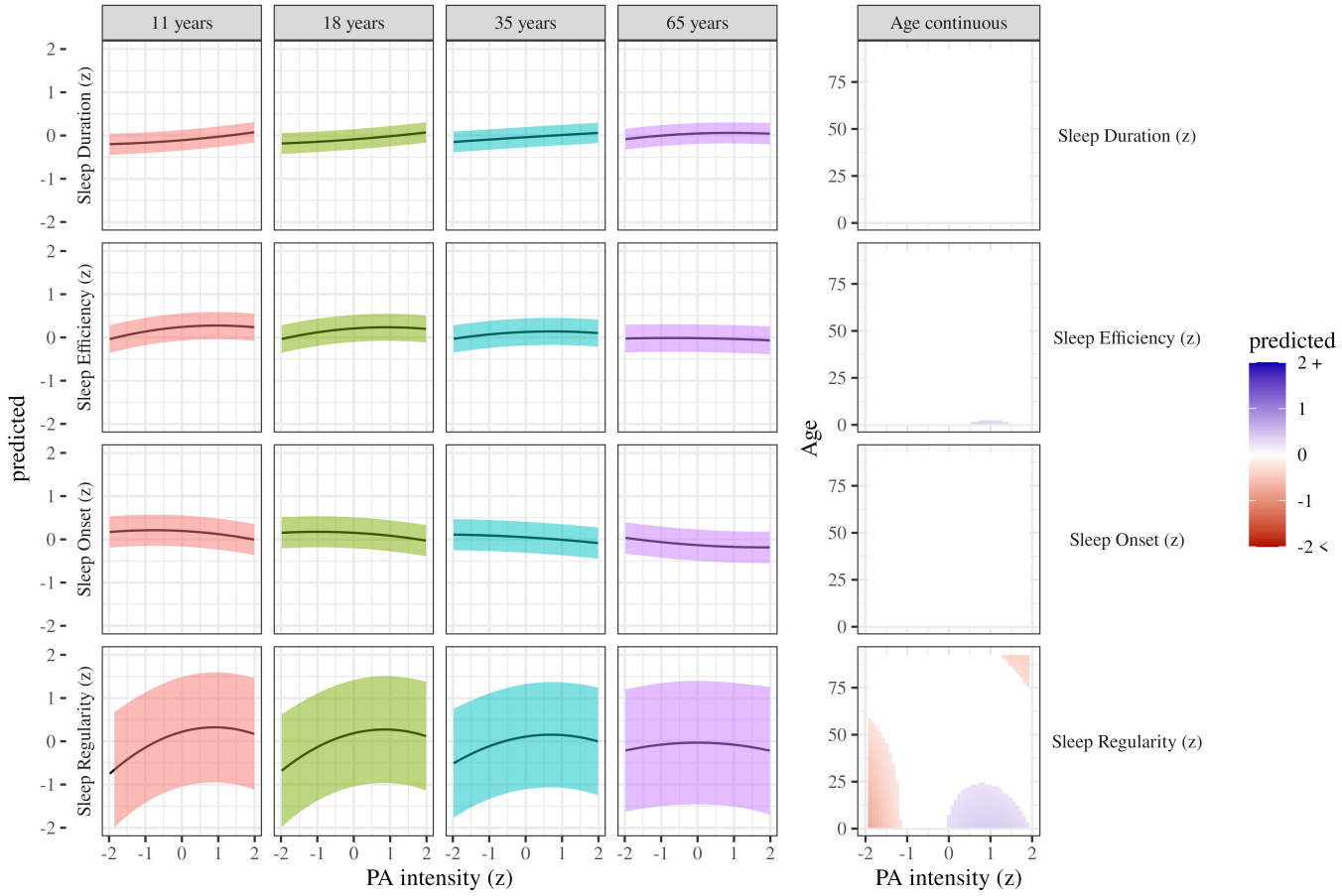

Figure 19. Sleep metrics on physical activity intensity (at least seven valid days)

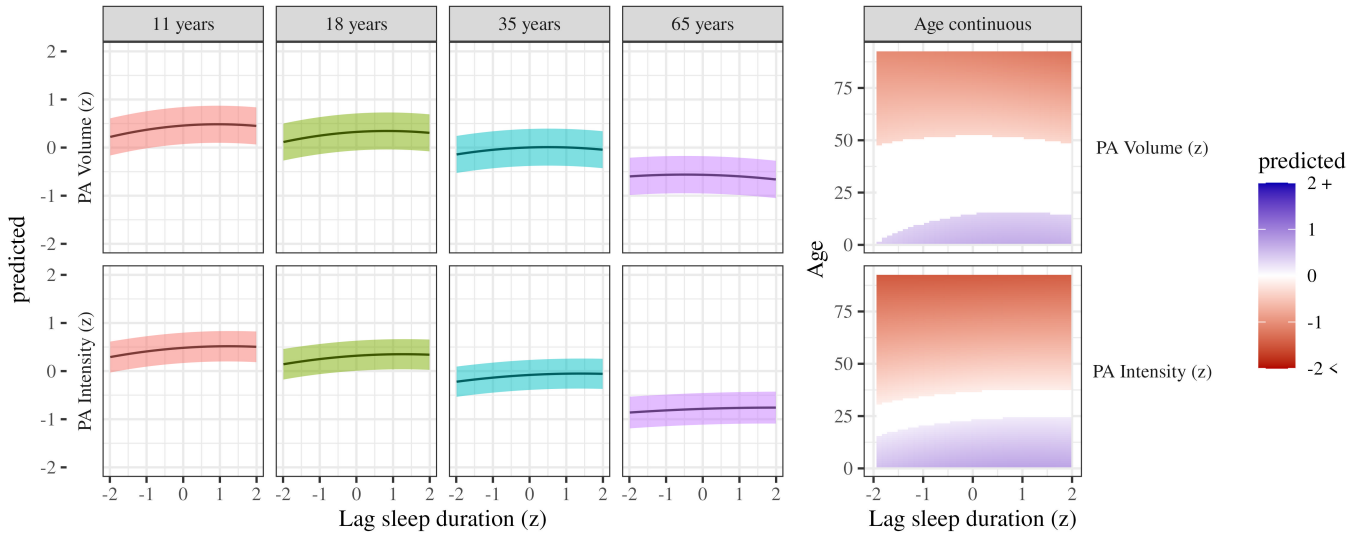

Figure 20. Physical activity by previous night sleep duration (at least seven valid days)

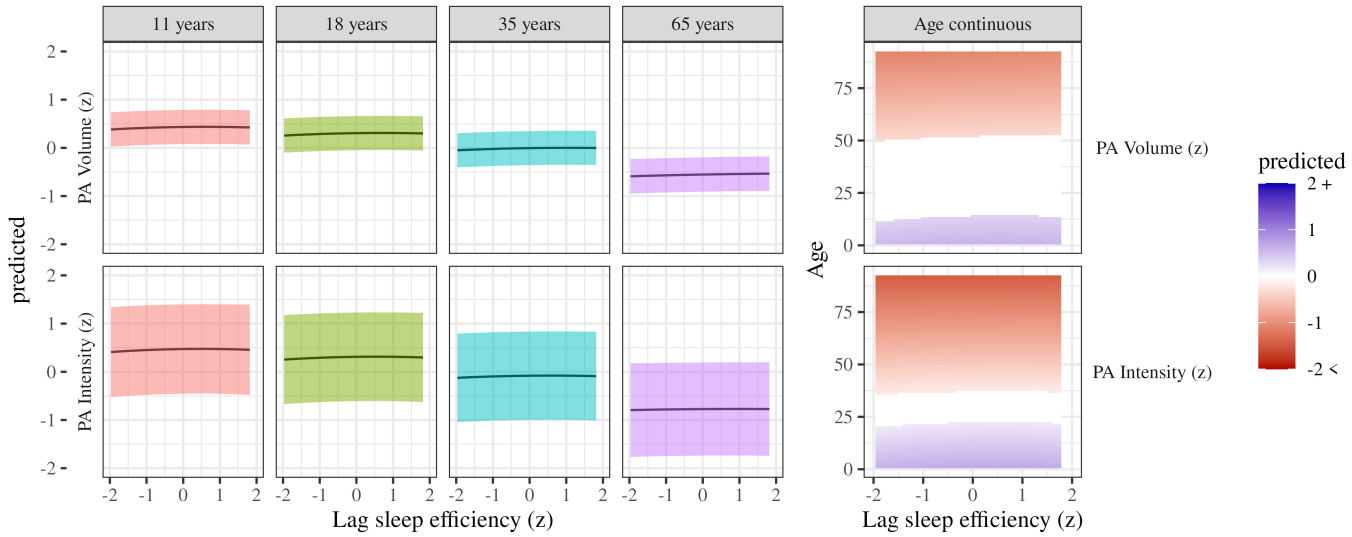

Figure 21. Physical activity by previous night sleep efficiency (at least seven valid days)

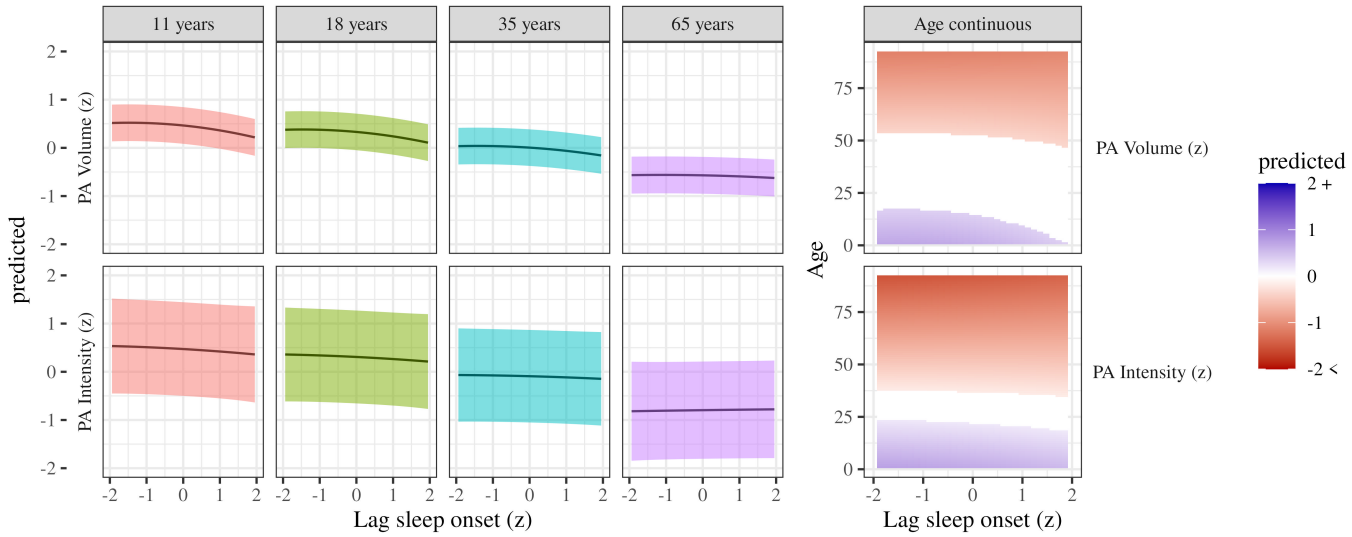

Figure 22. Physical activity by previous night sleep onset (at least seven valid days)

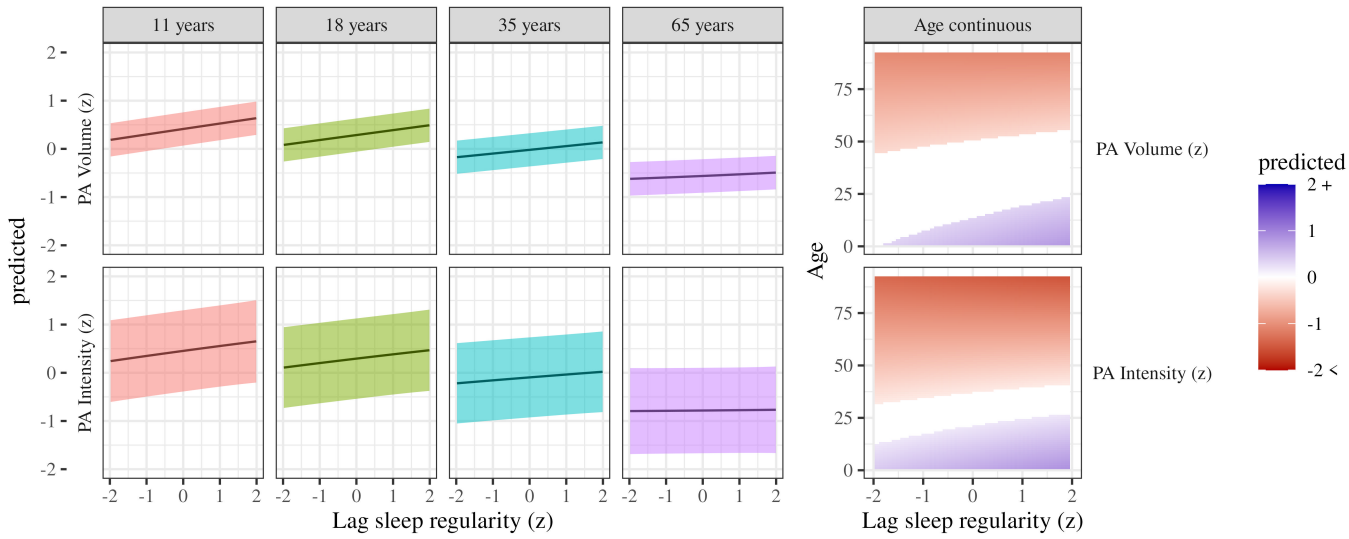

Figure 23. Physical activity by previous night sleep regularity (at least seven valid days)

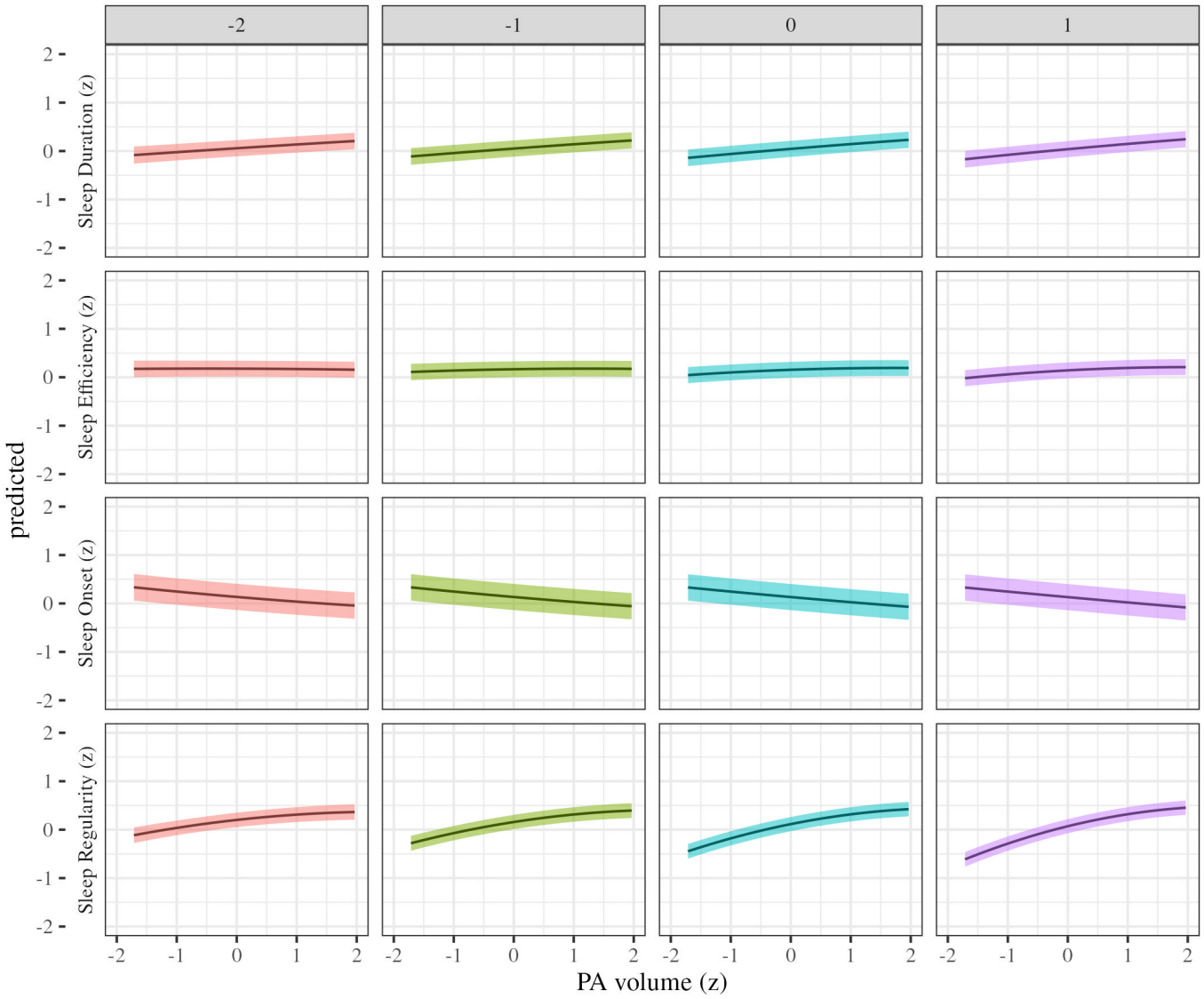

Figure 24. Sleep metrics on Physical activity volume by BMI z-score

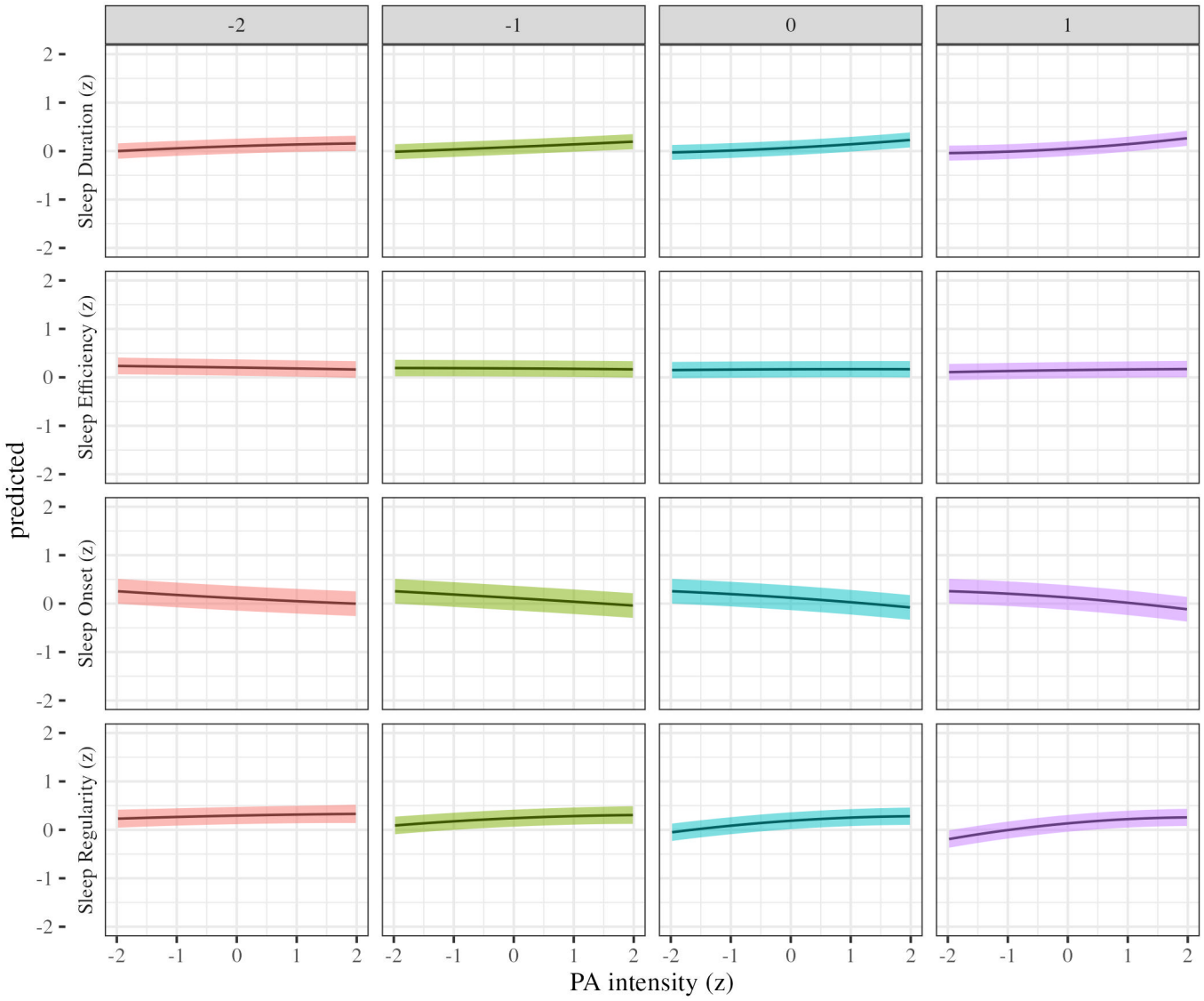

Figure 25. Sleep metrics on Physical activity intensity moderated by BMI z-score

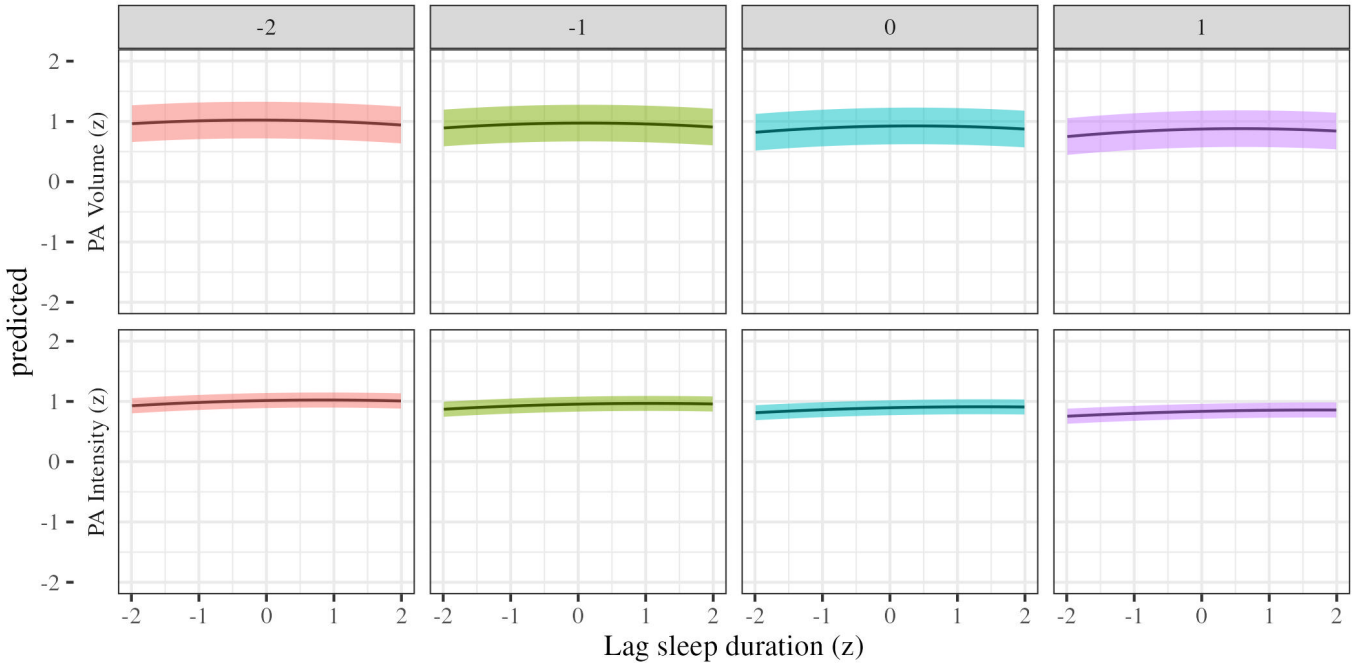

Figure 26. Physical activity by sleep duration moderated by BMI z-score

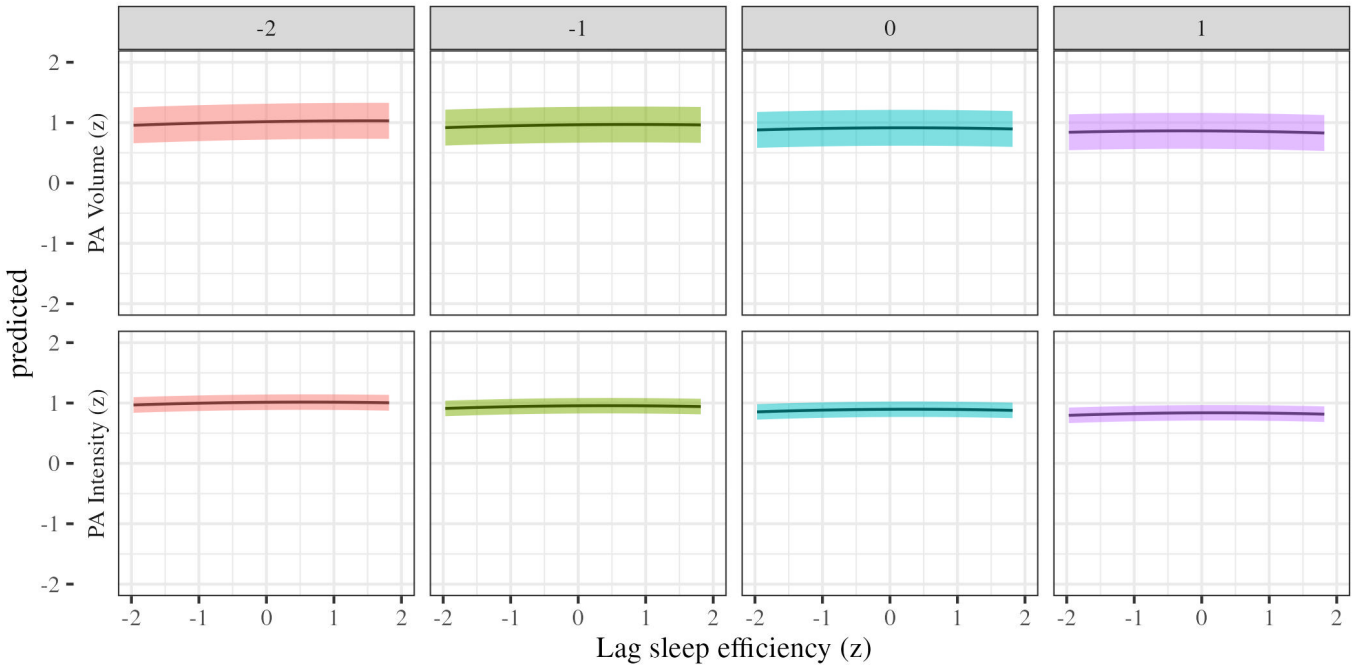

Figure 27. Physical activity by sleep efficiency moderated by BMI z-score

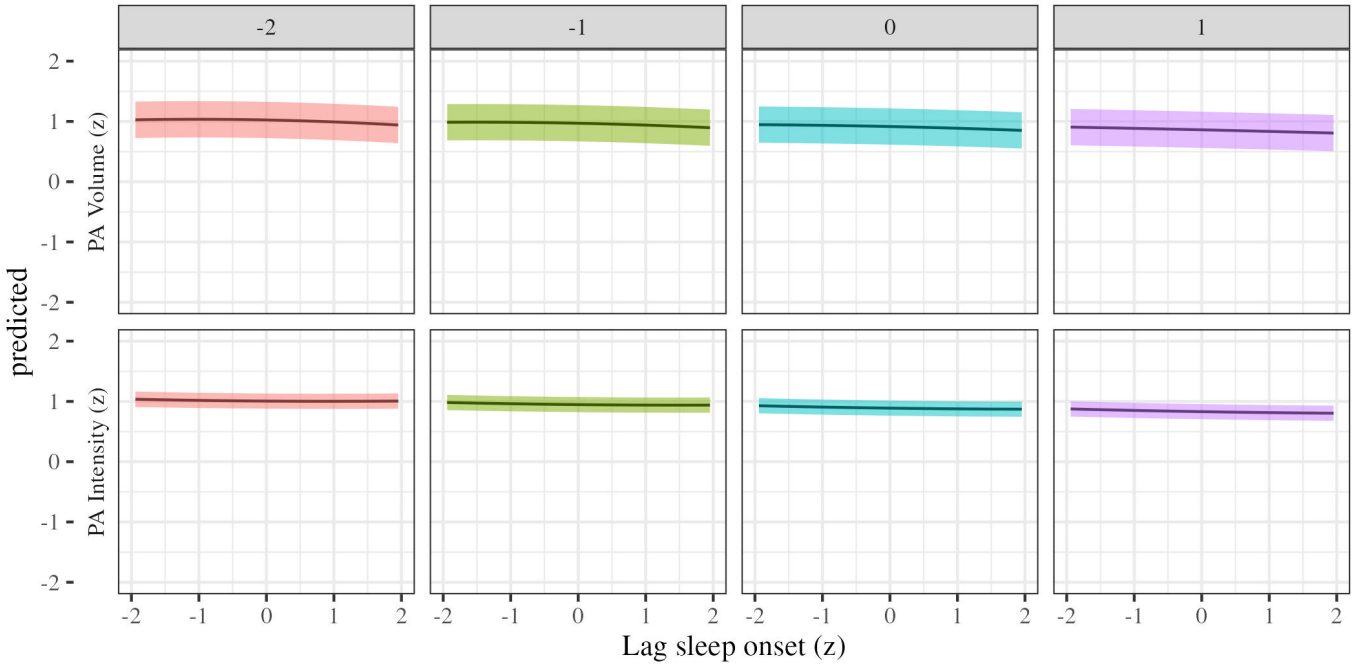

Figure 28. Physical activity by sleep onset moderated by BMI z-score

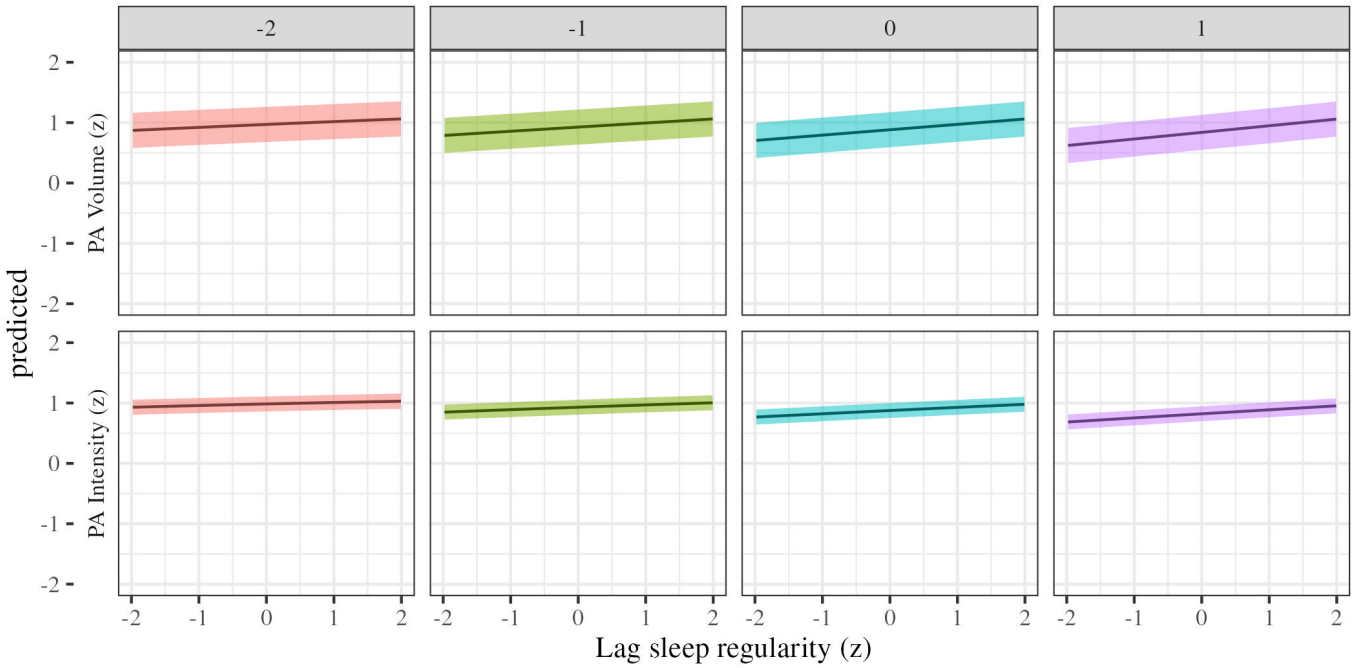

Figure 29. Physical activity by sleep regularity moderated by BMI z-score

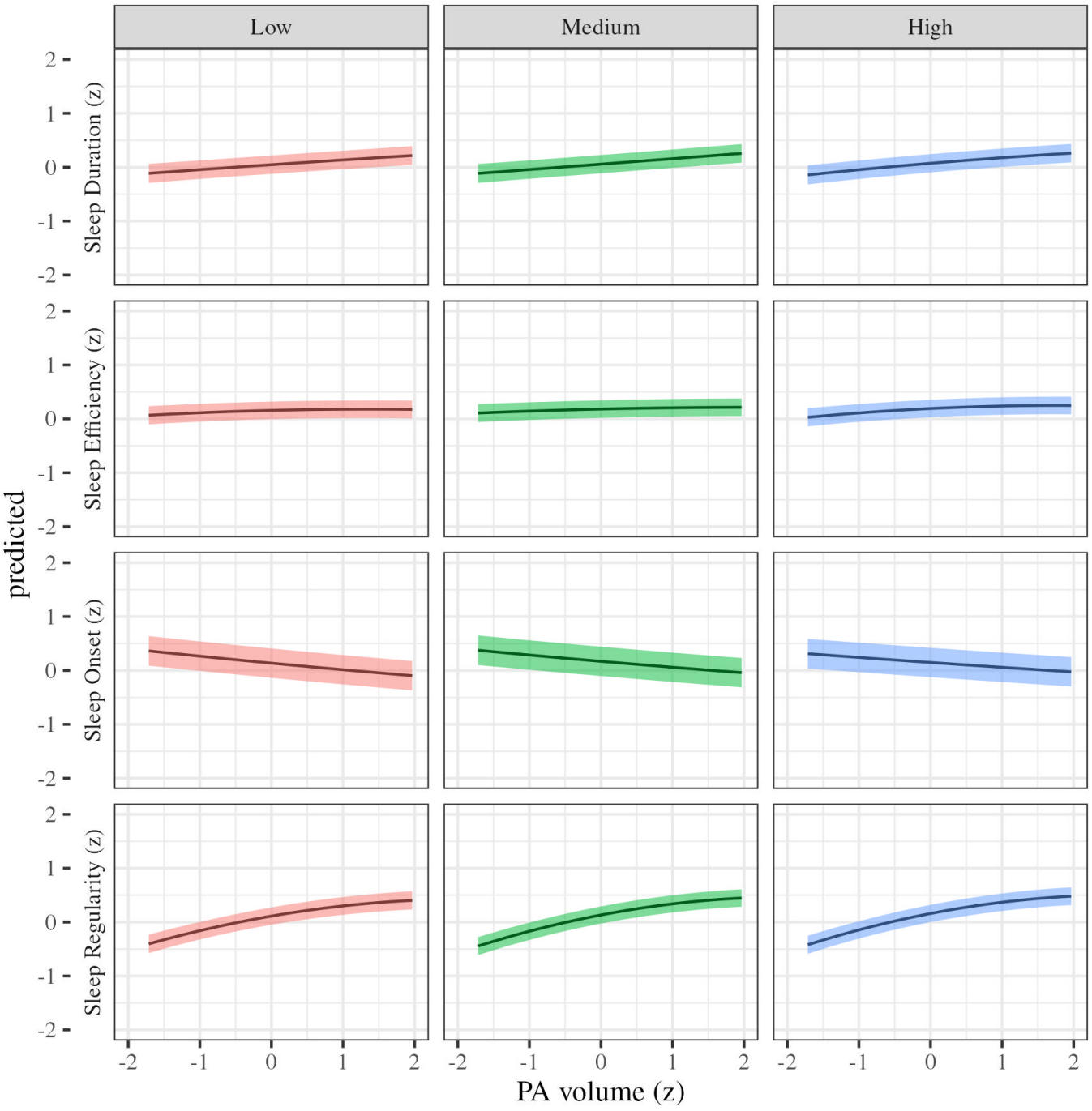

Figure 30. Sleep metrics on Physical activity volume by SES

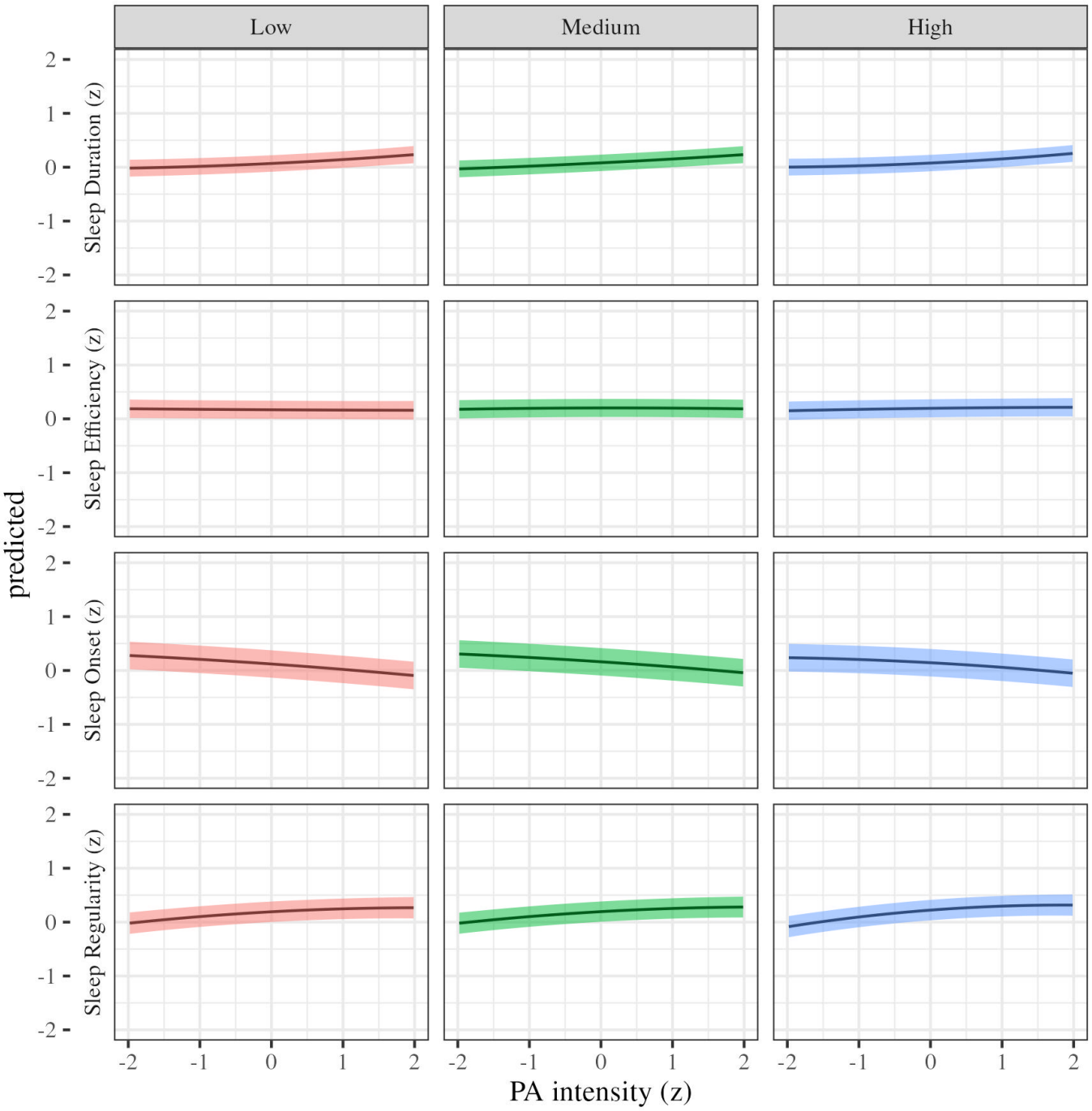

Figure 31. Sleep metrics on Physical activity intensity moderated by SES

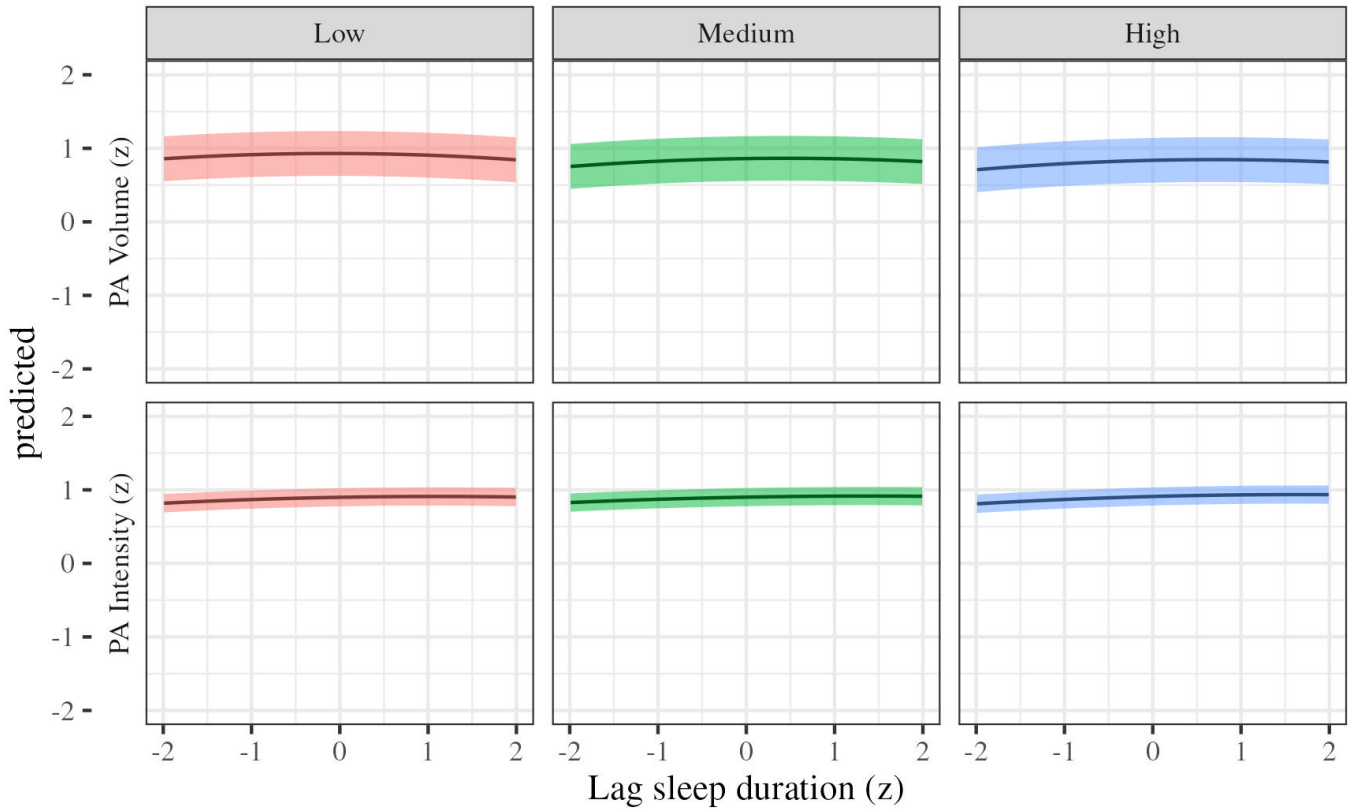

Figure 32. Physical activity by sleep duration moderated by SES

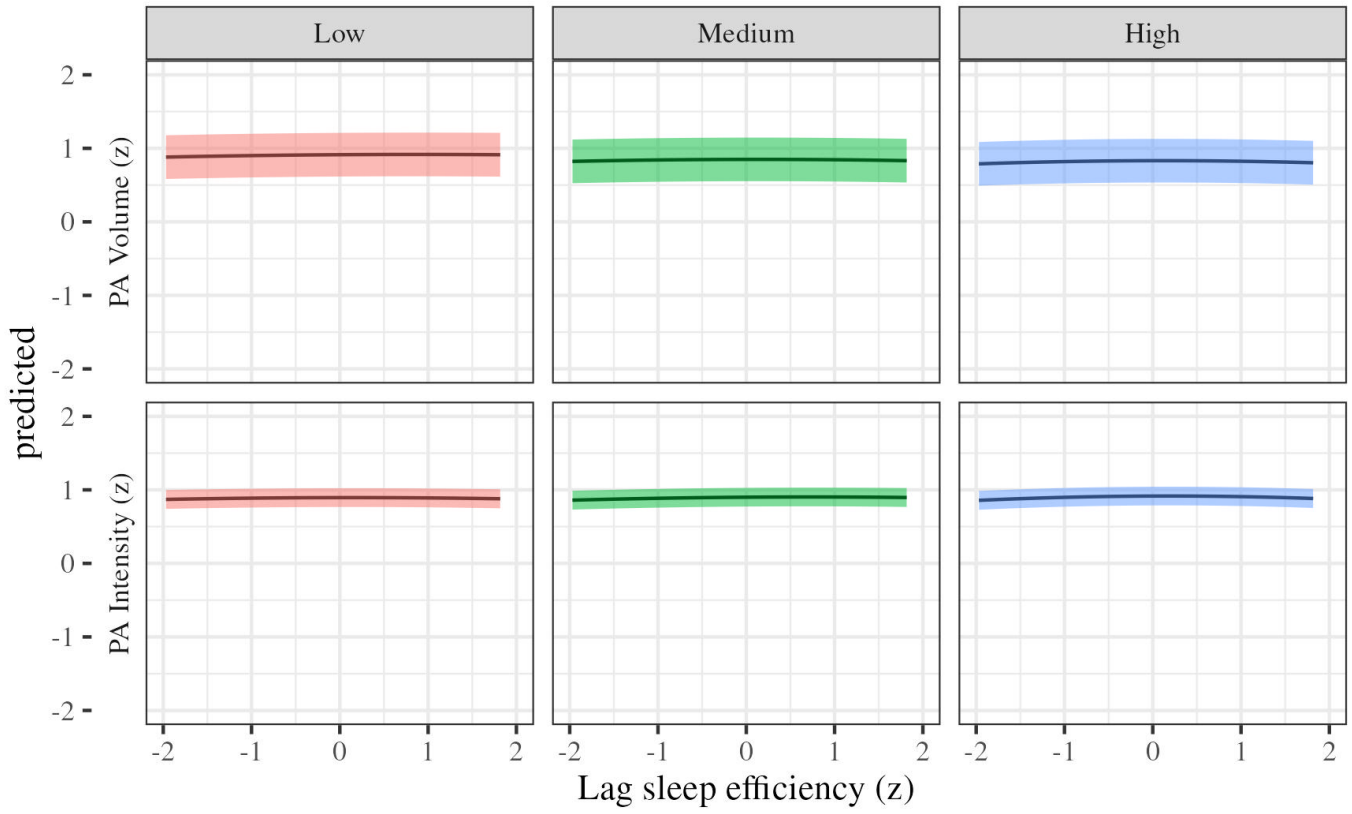

Figure 33. Physical activity by sleep efficiency moderated by SES

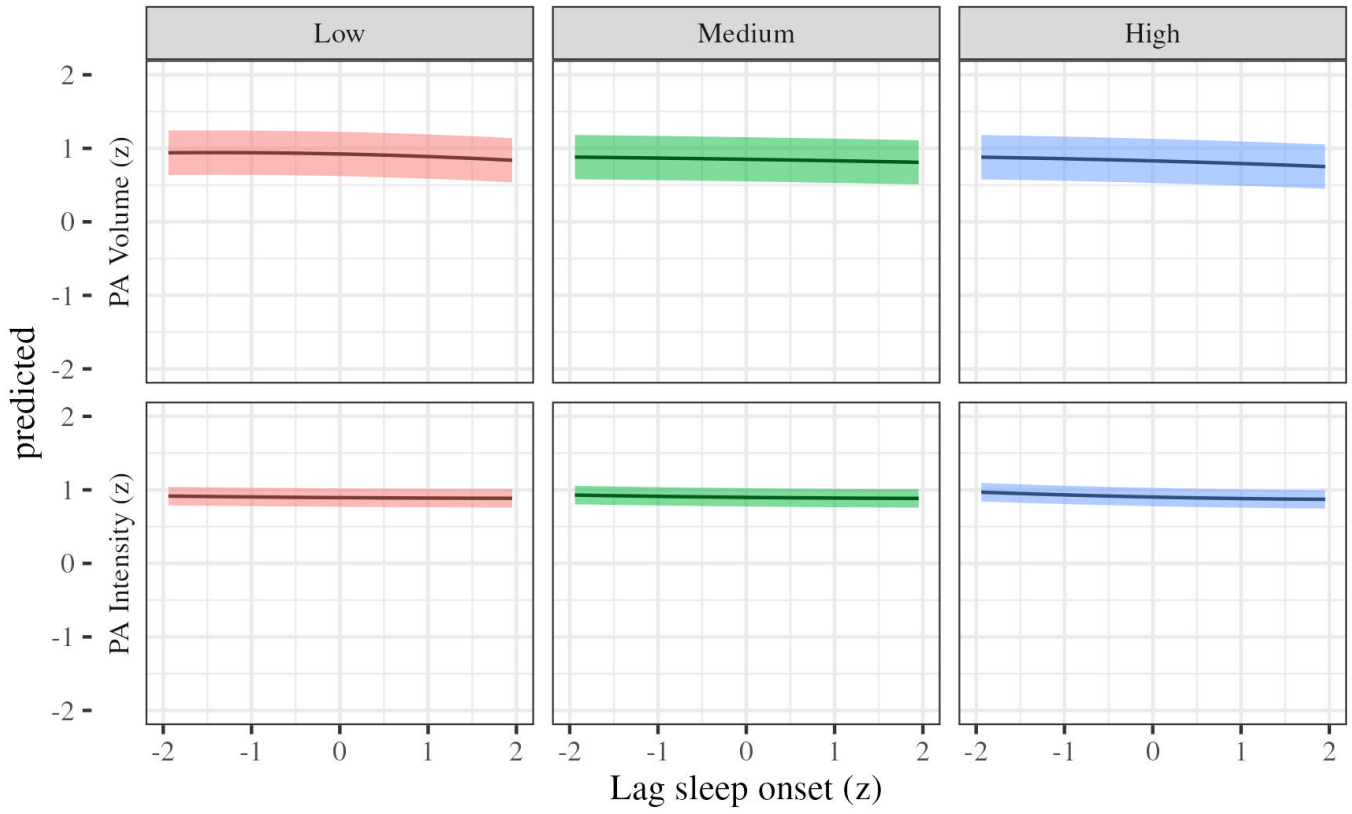

Figure 34. Physical activity by sleep onset moderated by SES

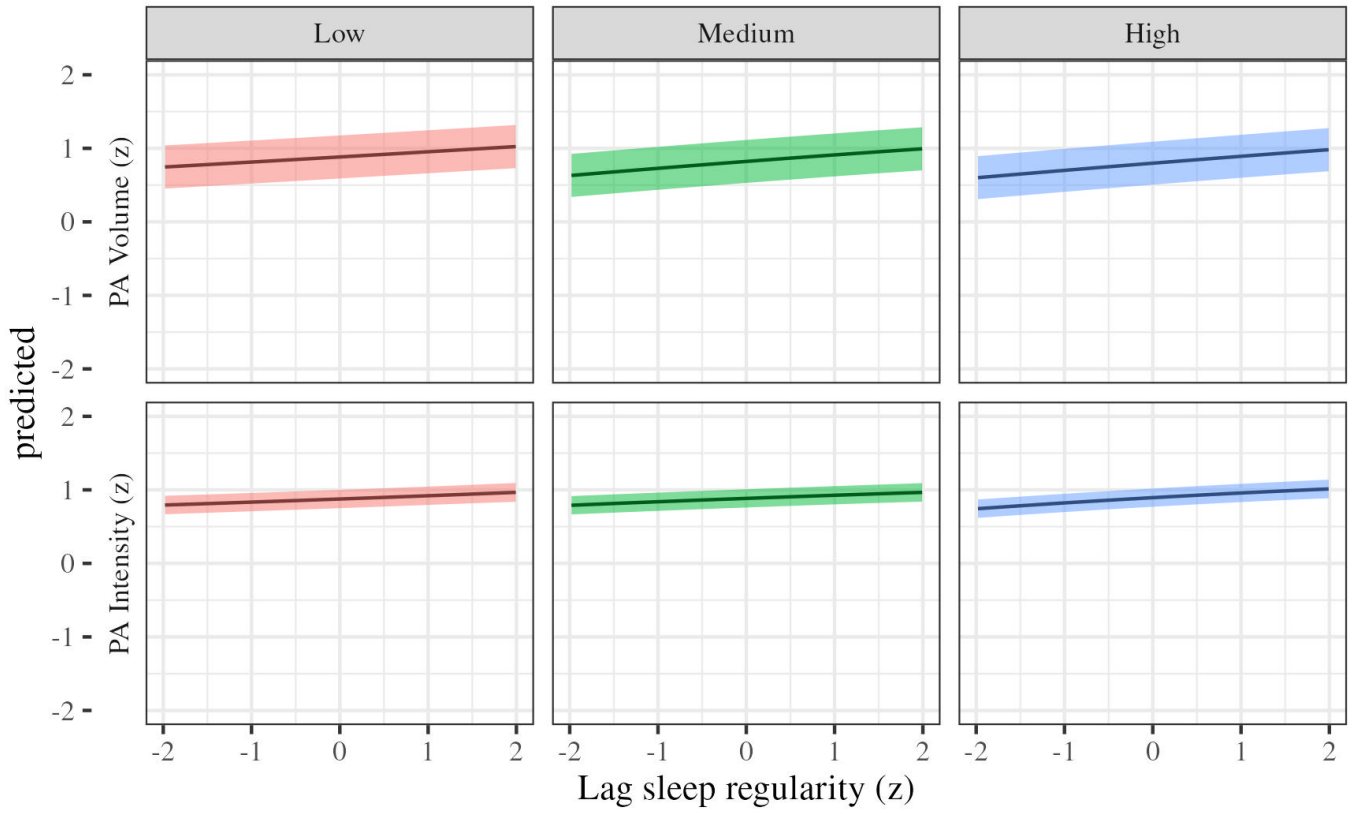

Figure 35. Physical activity by sleep regularity moderated by SES

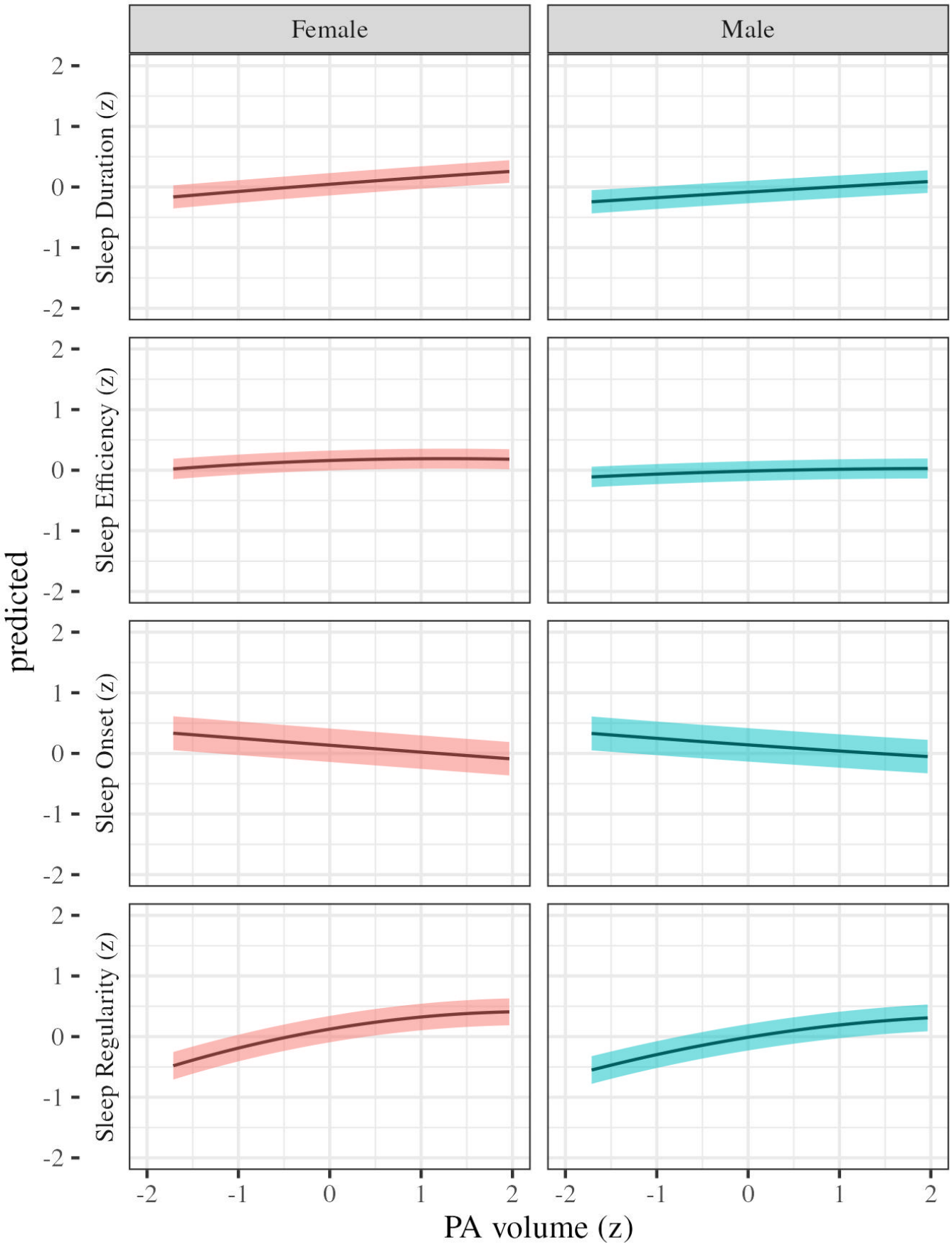

Figure 36. Sleep metrics on Physical activity volume by sex

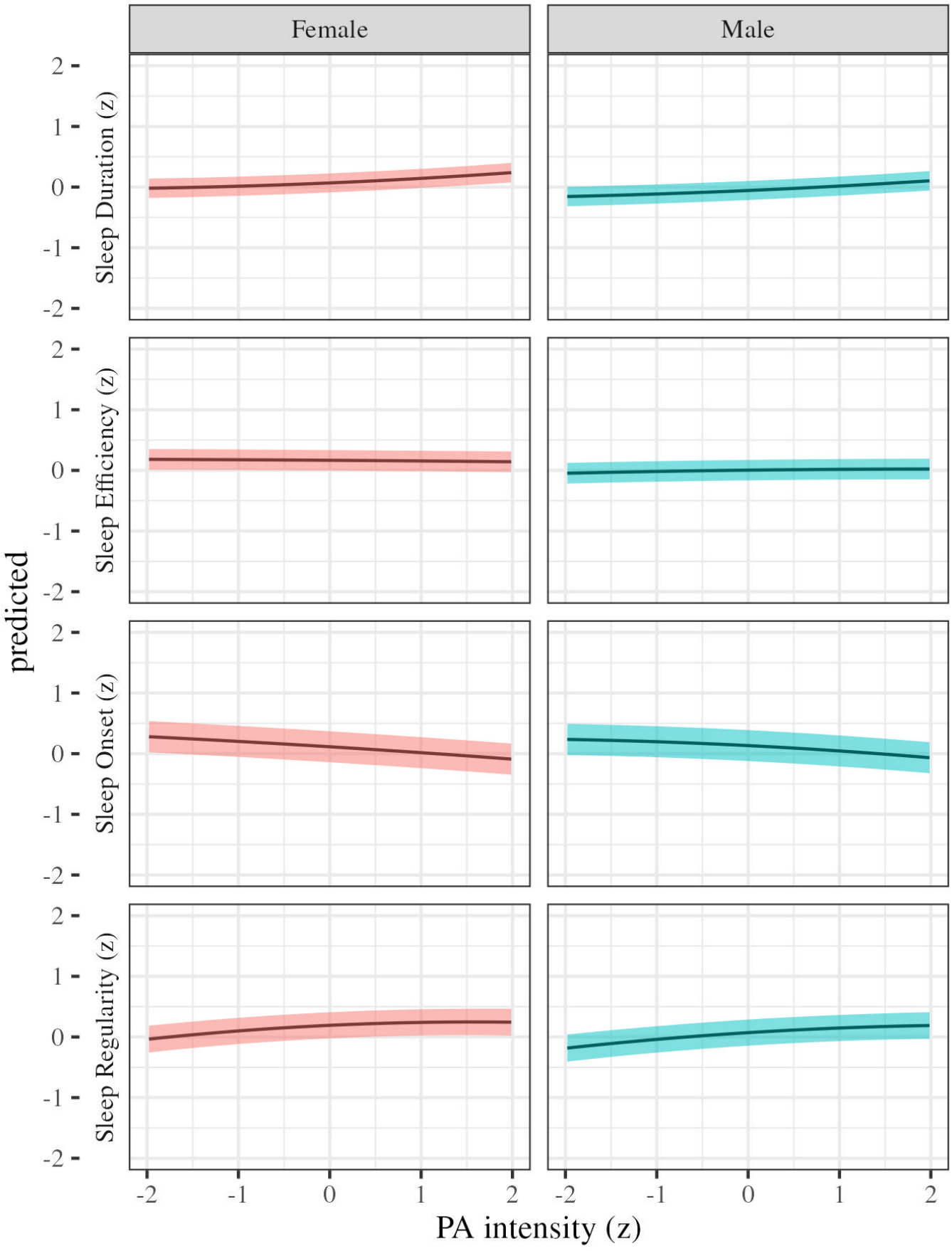

Figure 37. Sleep metrics on Physical activity intensity moderated by sex

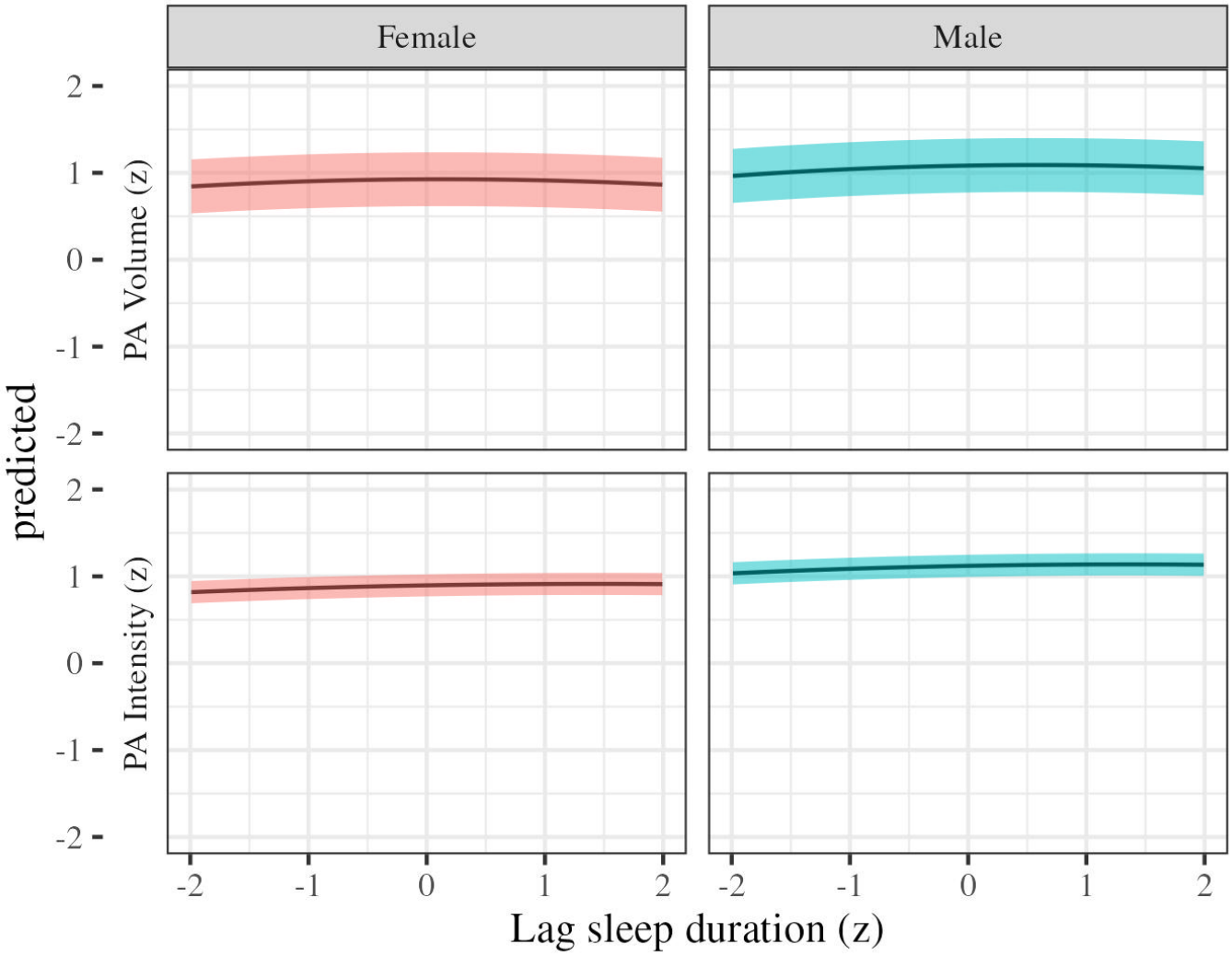

Figure 38. Physical activity by sleep duration moderated by sex

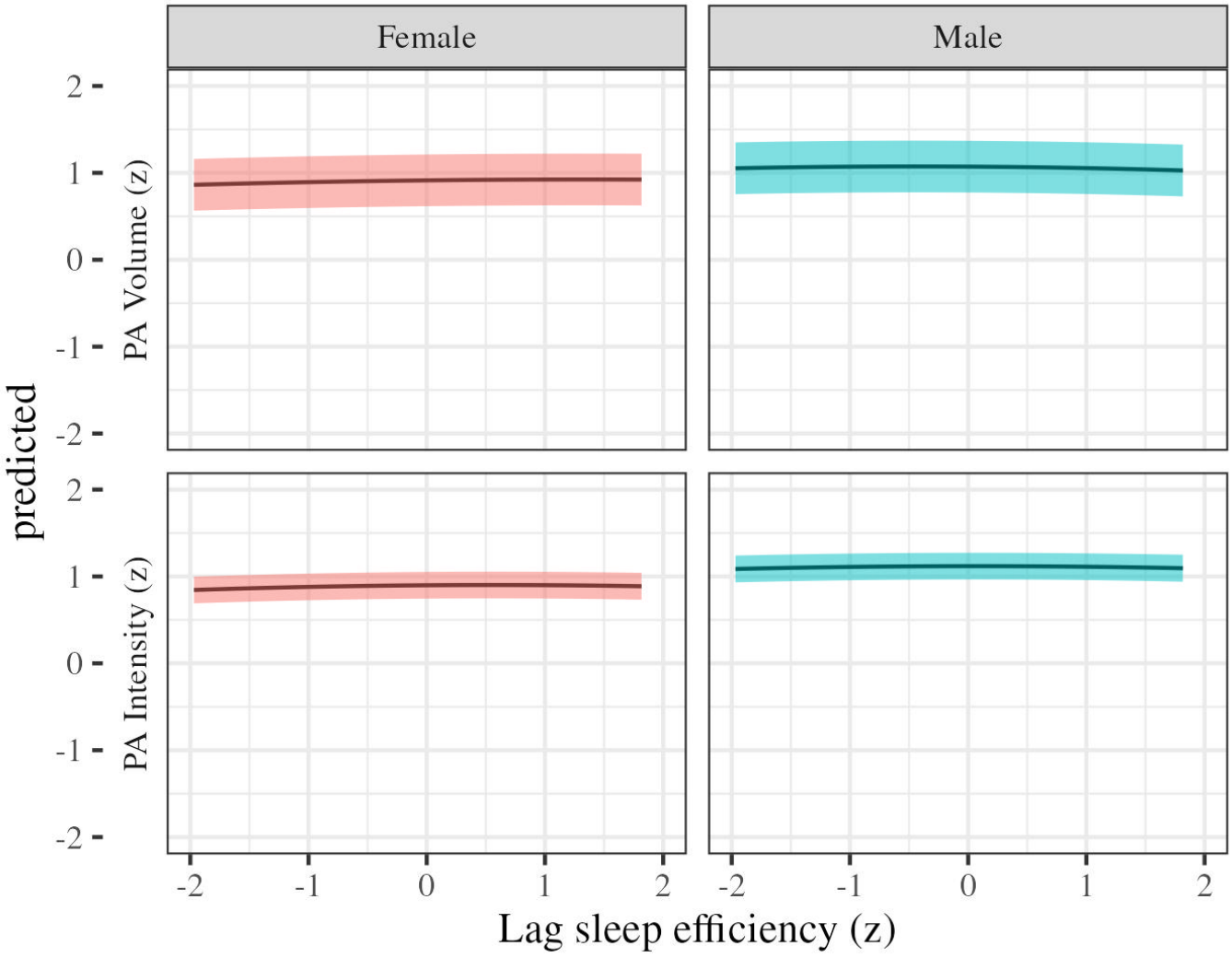

Figure 39. Physical activity by sleep efficiency moderated by sex

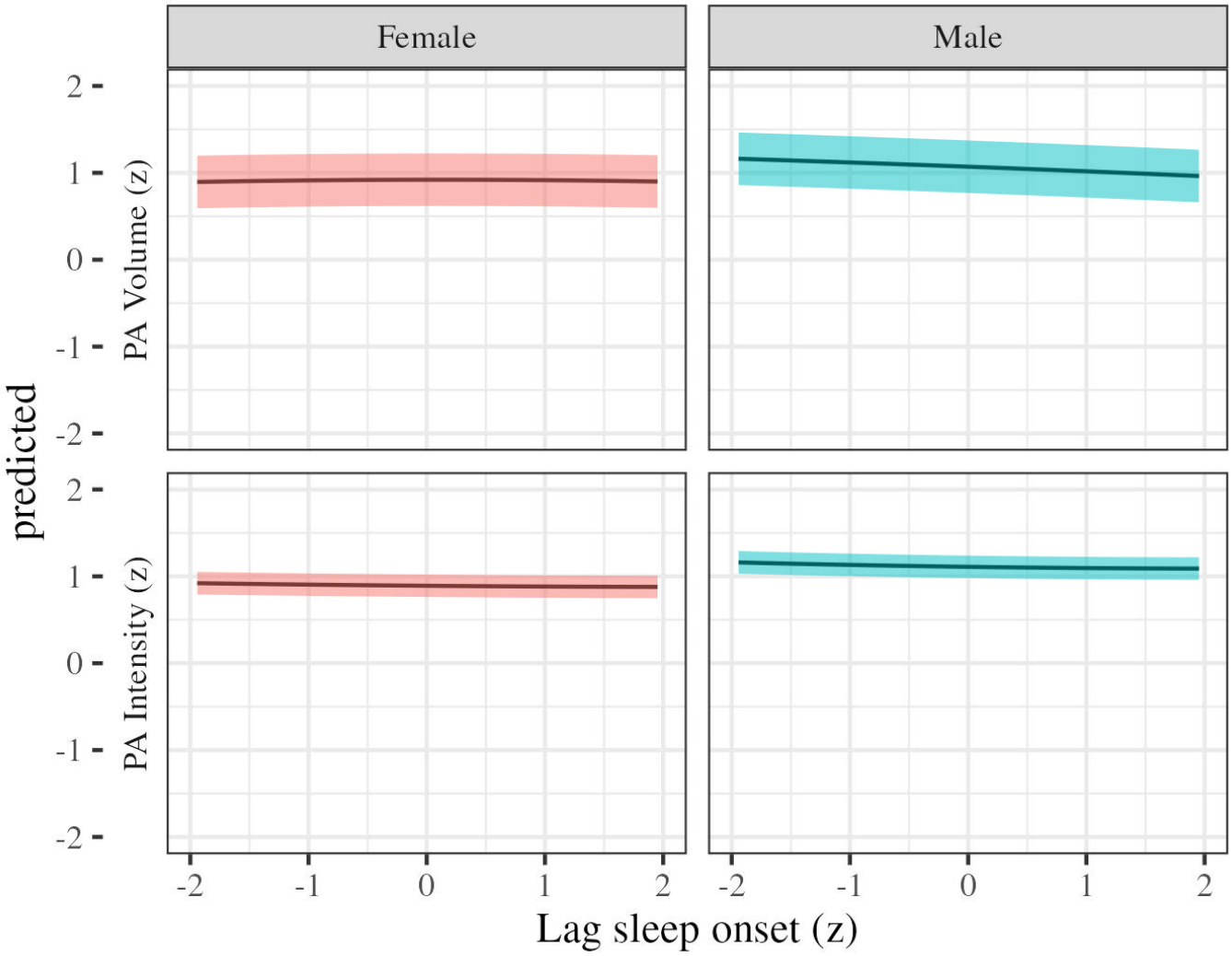

Figure 40. Physical activity by sleep onset moderated by sex

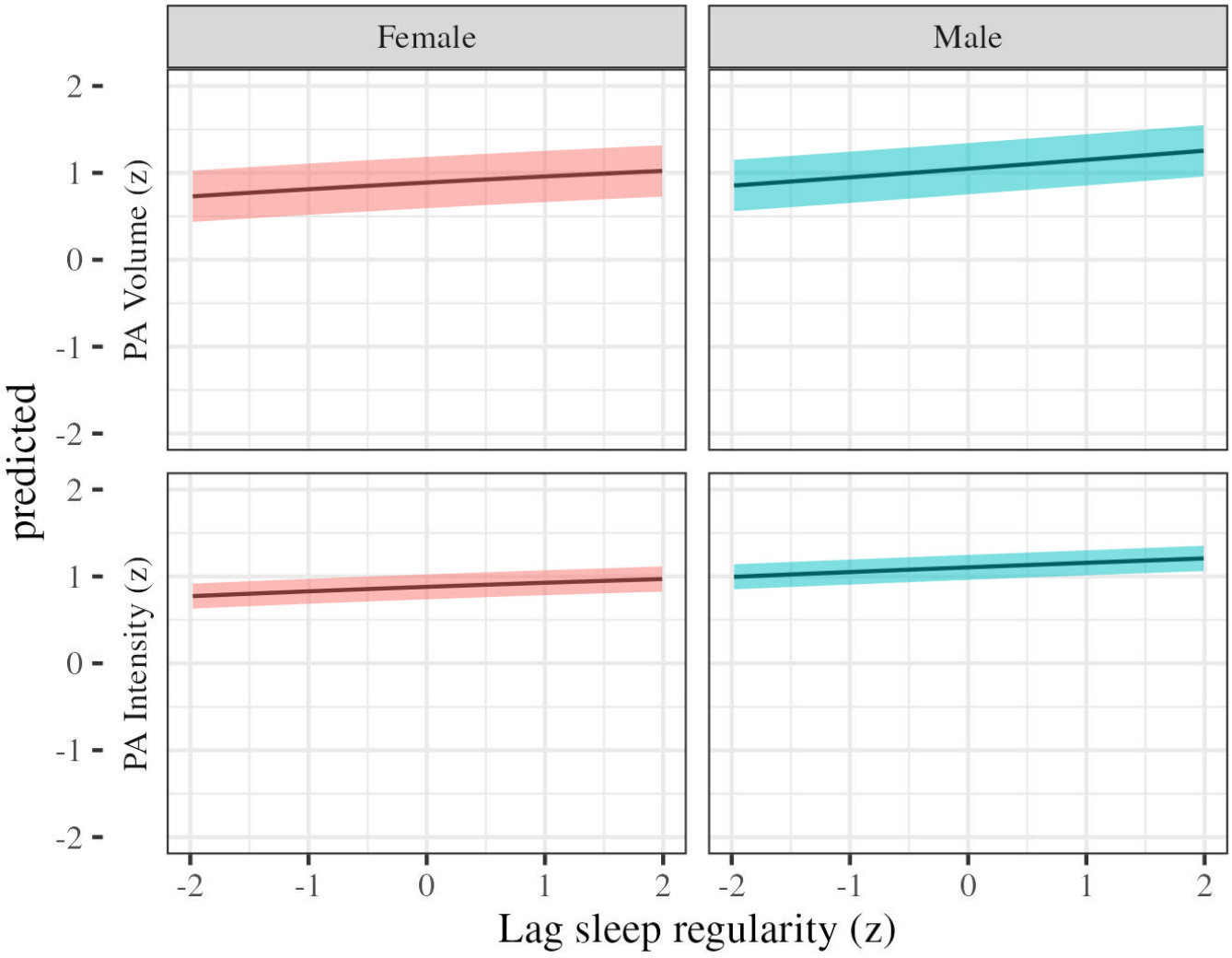

Figure 41. Physical activity by sleep regularity moderated by sex

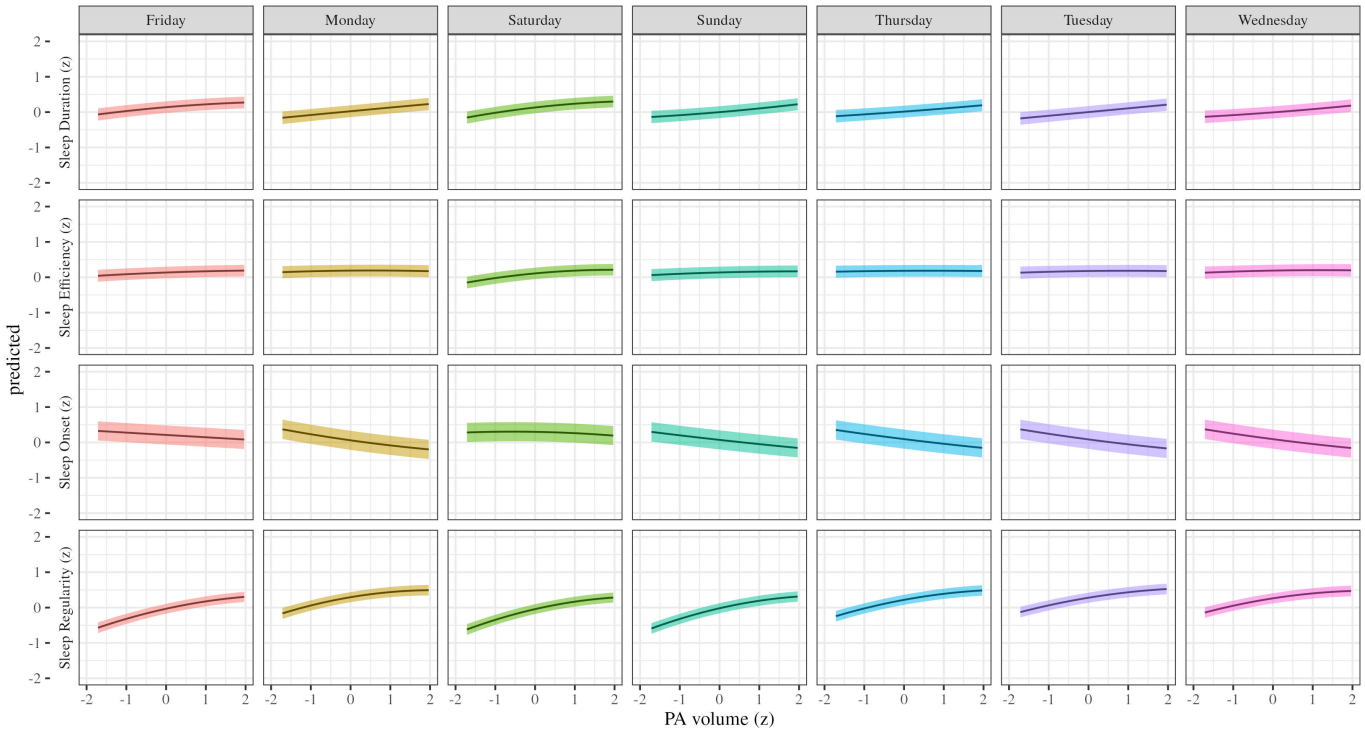

Figure 42. Sleep metrics on Physical activity volume by weekday

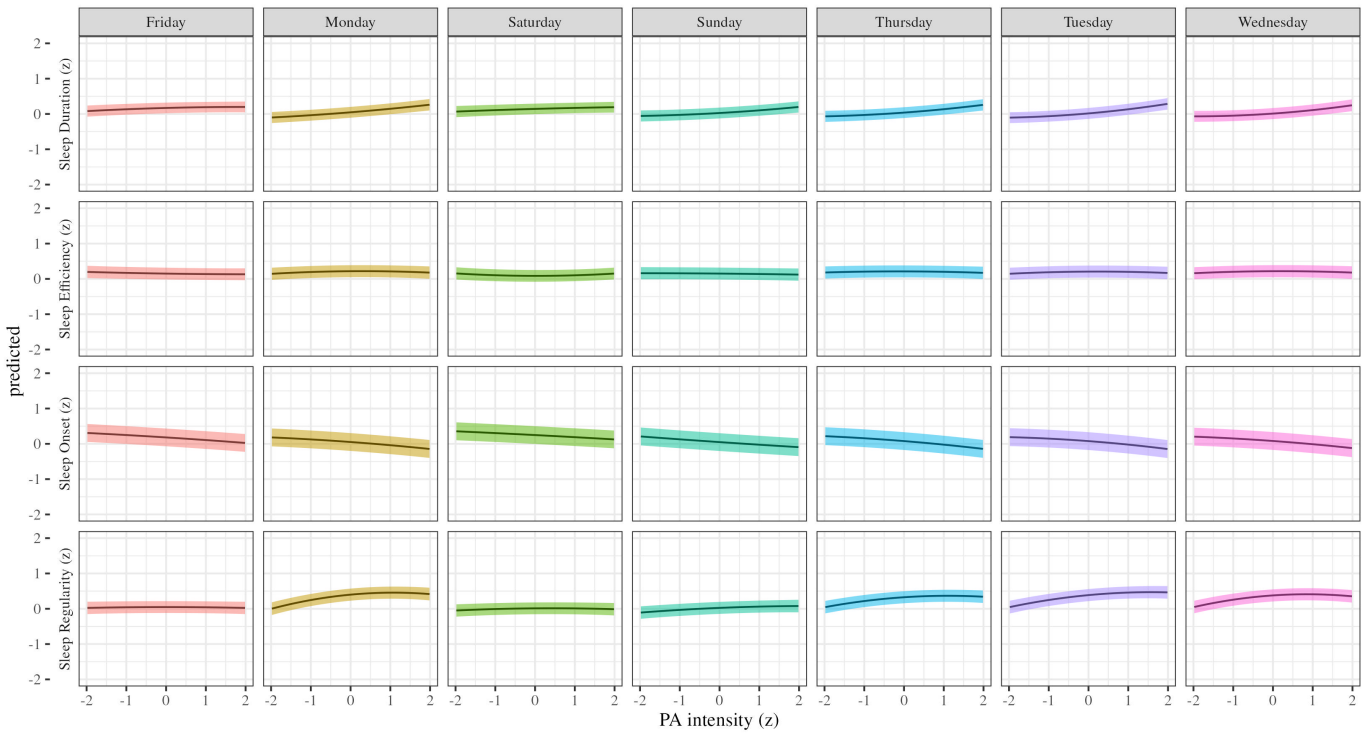

Figure 43. Sleep metrics on Physical activity intensity moderated by weekday

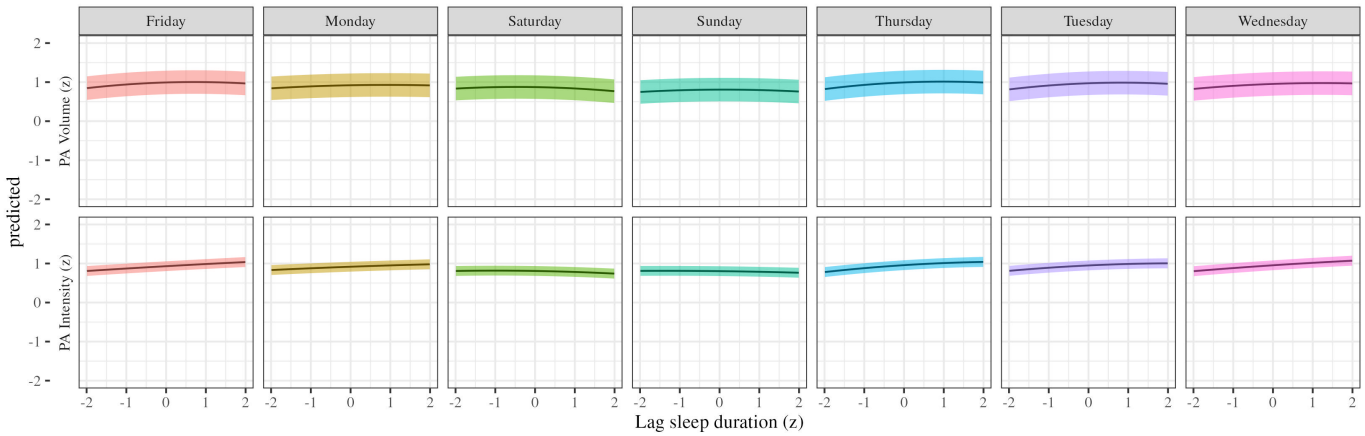

Figure 44. Physical activity by sleep duration moderated by weekday

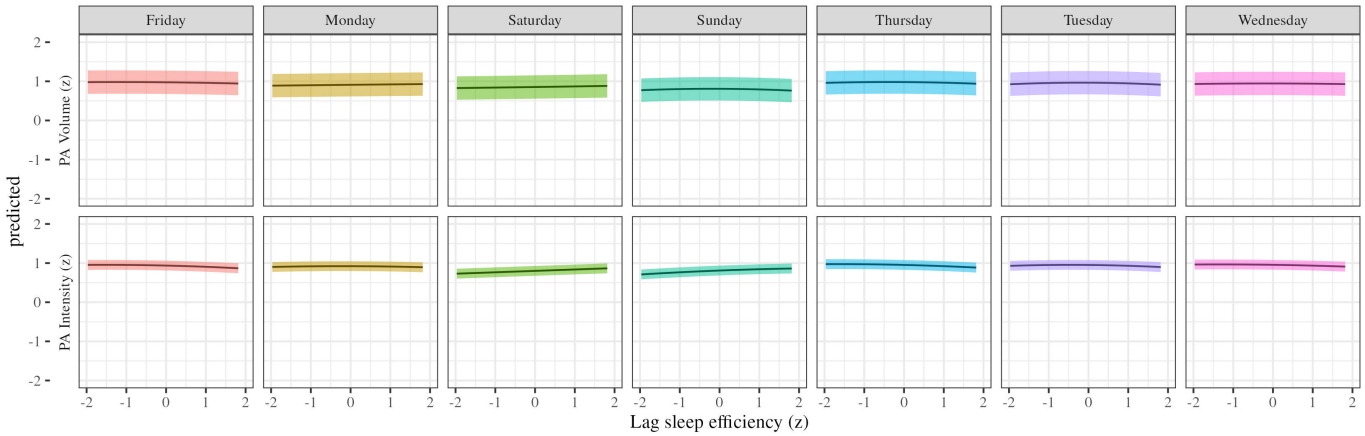

Figure 45. Physical activity by sleep efficiency moderated by weekday

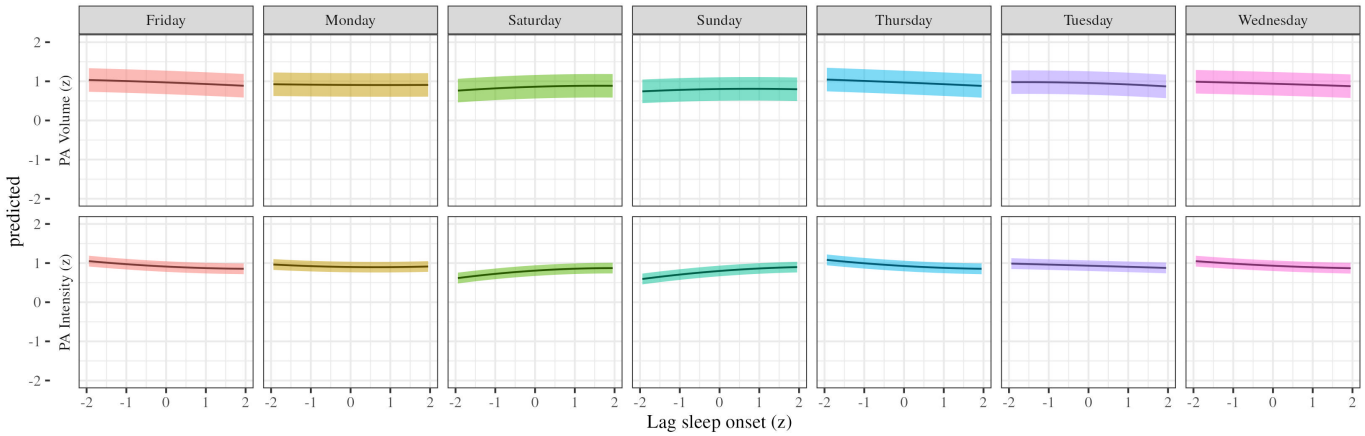

Figure 46. Physical activity by sleep onset moderated by weekday

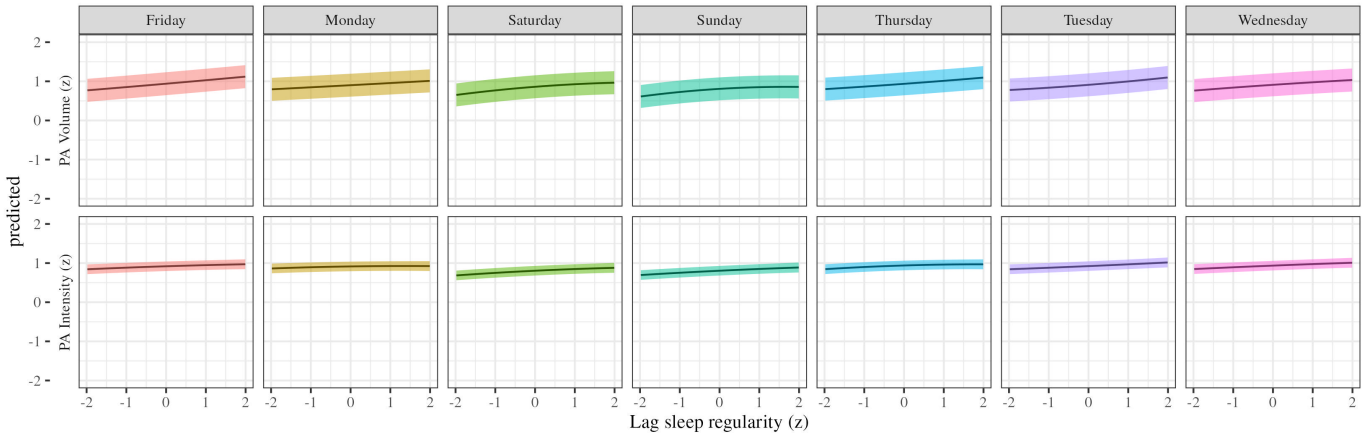

Figure 47. Physical activity by sleep regularity moderated by weekday

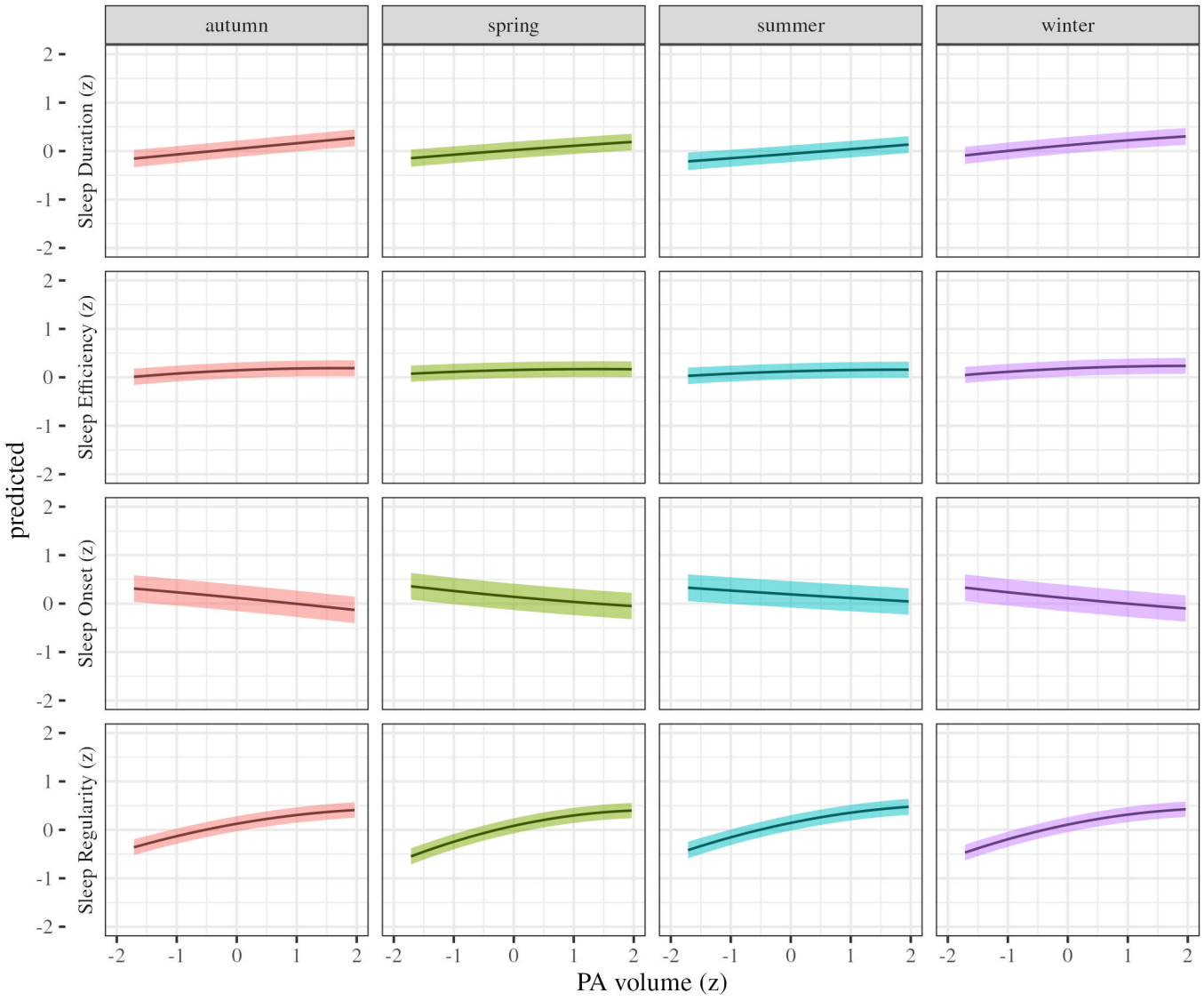

Figure 48. Sleep metrics on Physical activity volume by season

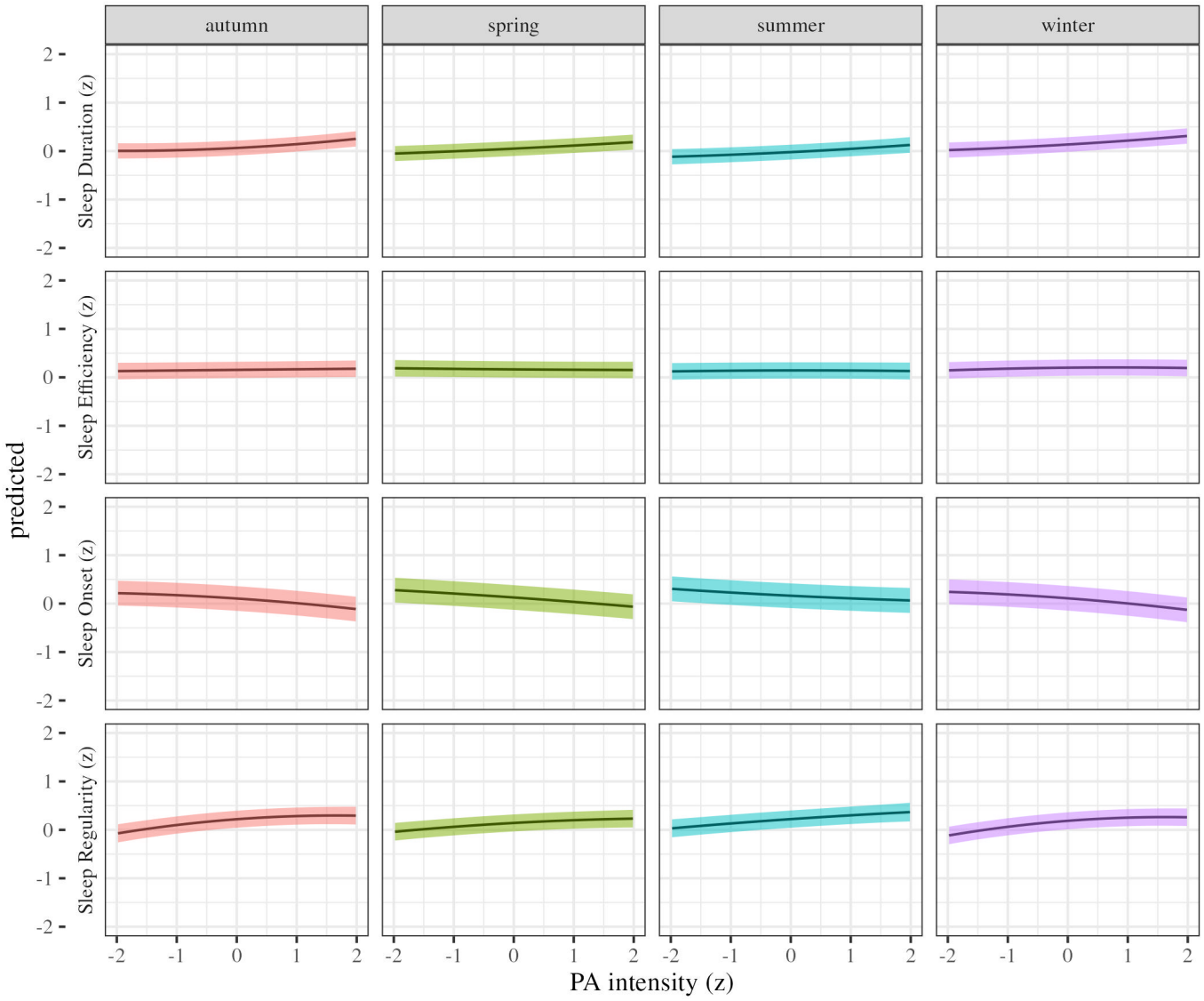

Figure 49. Sleep metrics on Physical activity intensity moderated by season

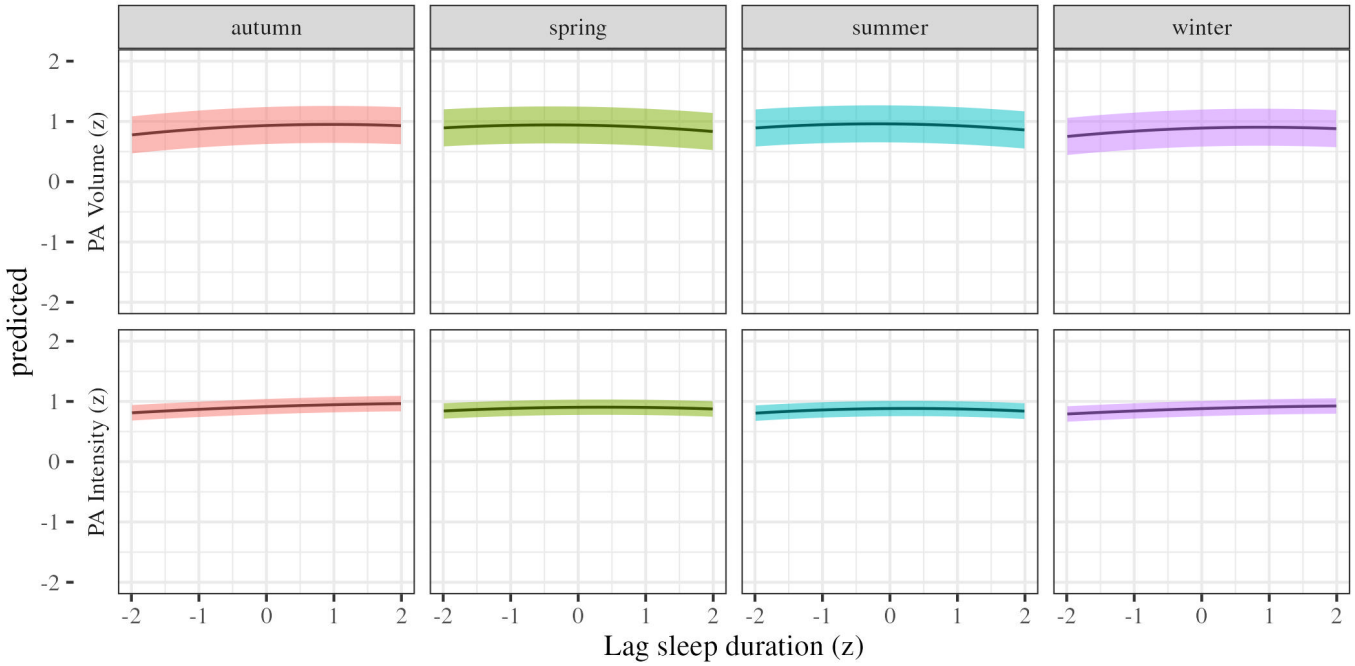

Figure 50. Physical activity by sleep duration moderated by season

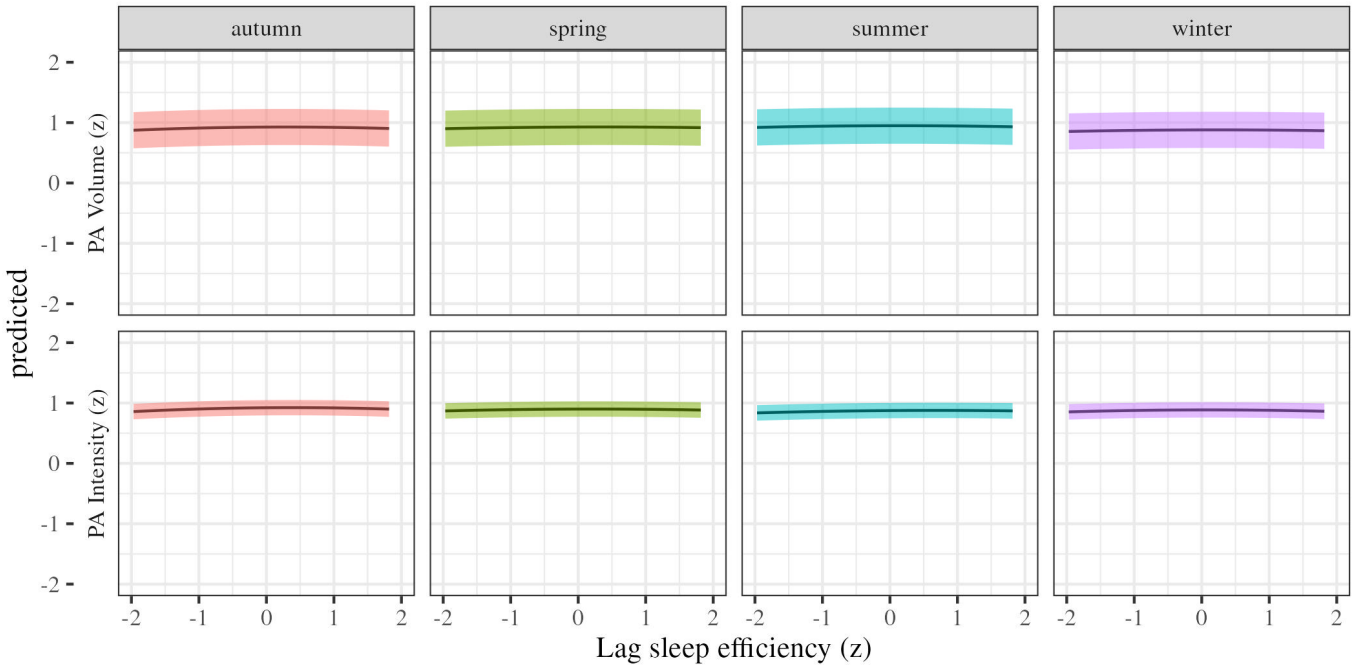

Figure 51. Physical activity by sleep efficiency moderated by season

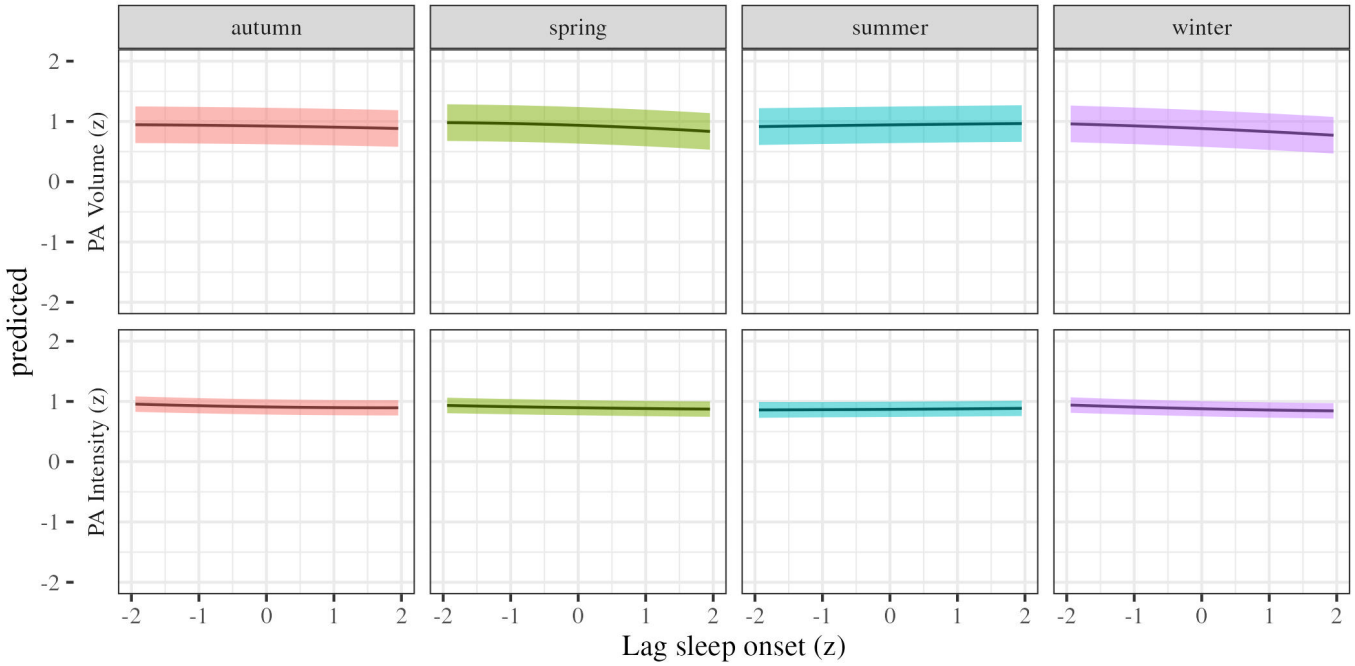

Figure 52. Physical activity by sleep onset moderated by season

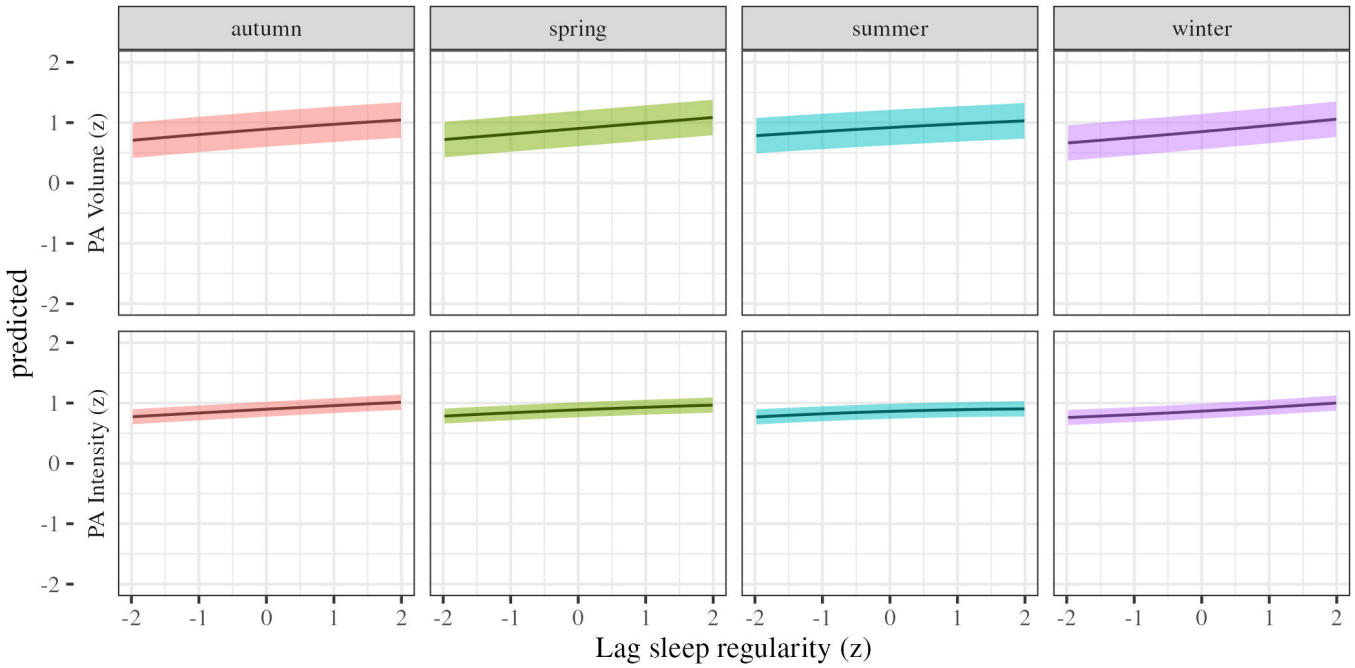

Figure 53. Physical activity by sleep regularity moderated by season

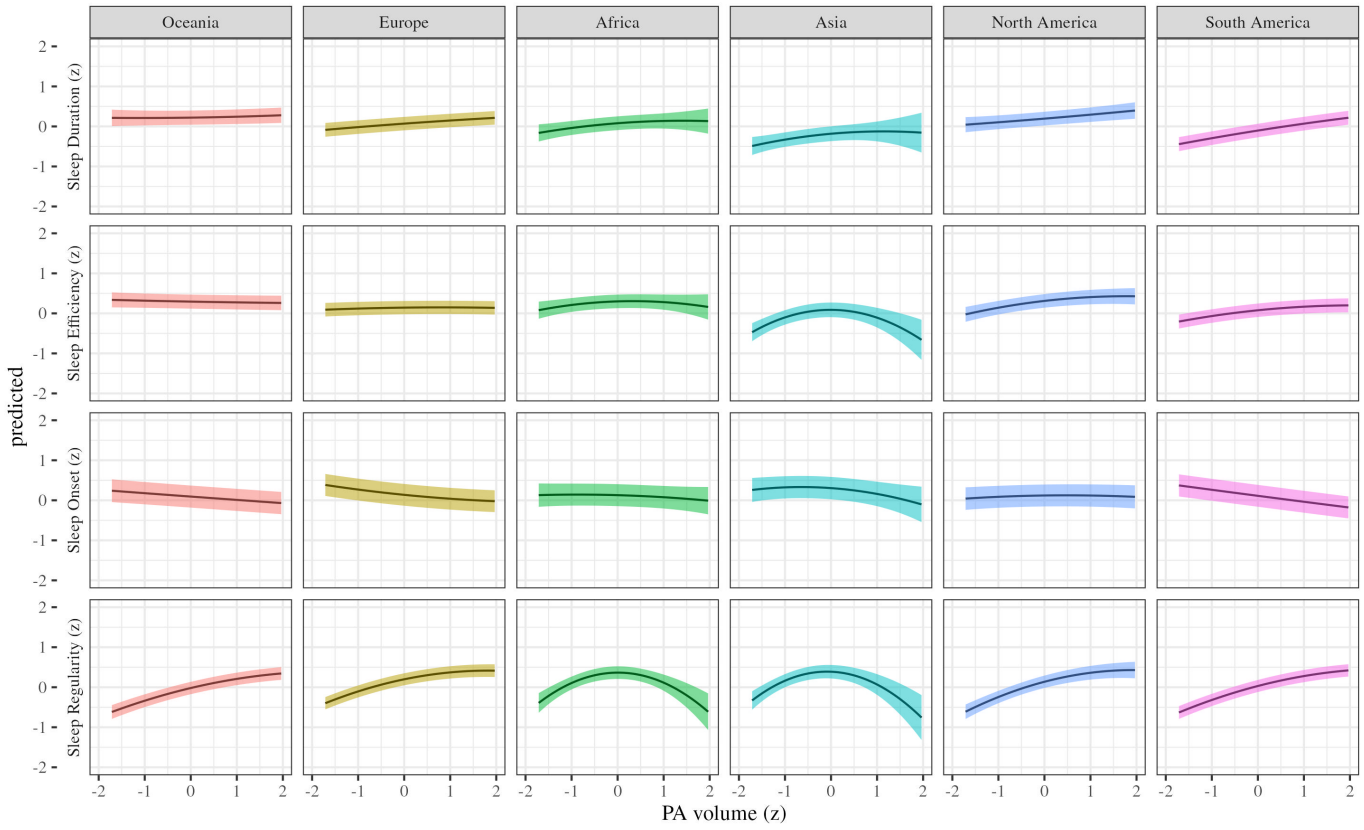

Figure 54. Sleep metrics on Physical activity volume by region

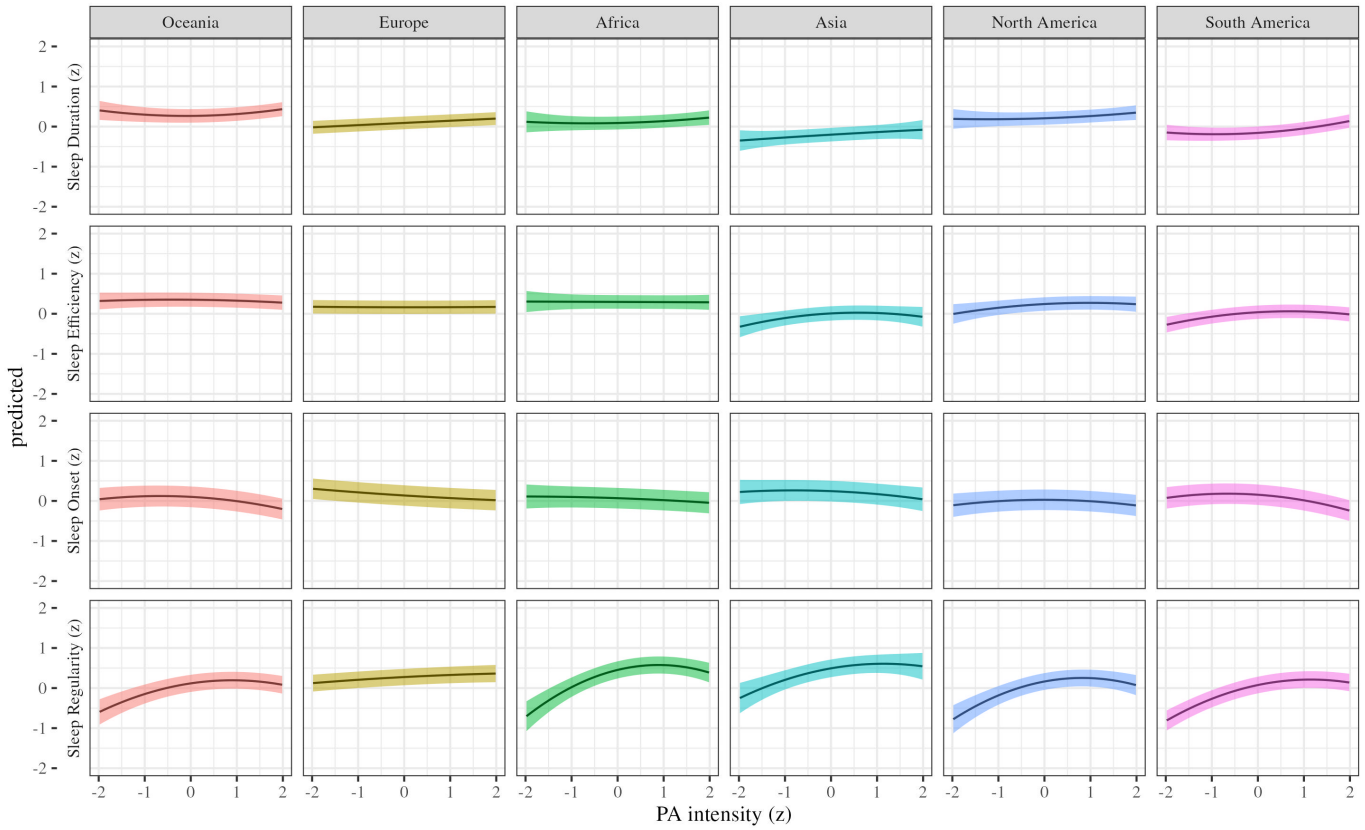

Figure 55. Sleep metrics on Physical activity intensity moderated by region

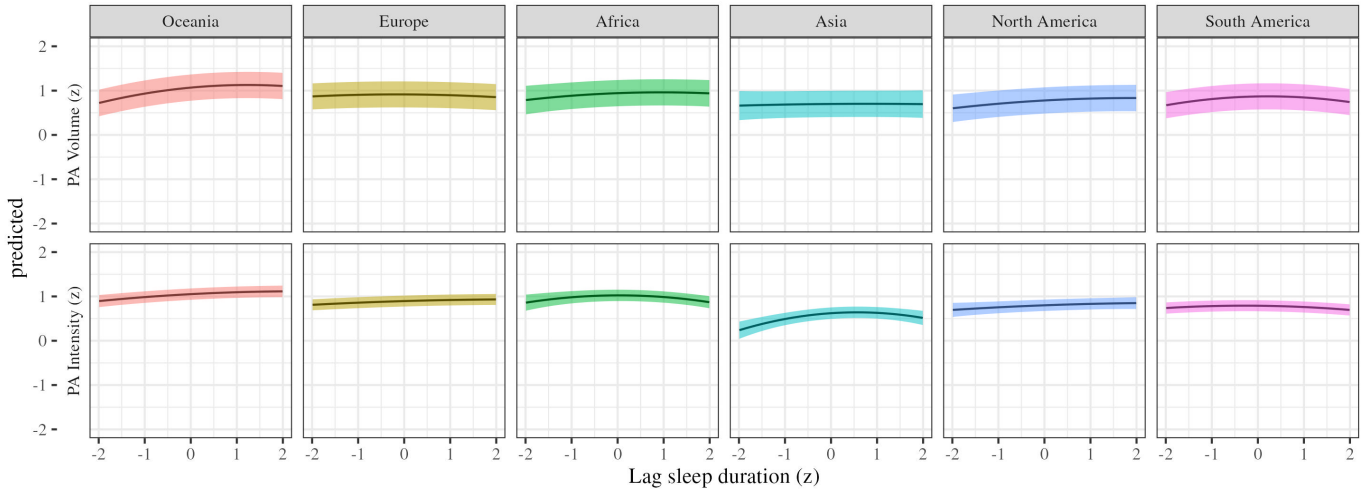

Figure 56. Physical activity by sleep duration moderated by region

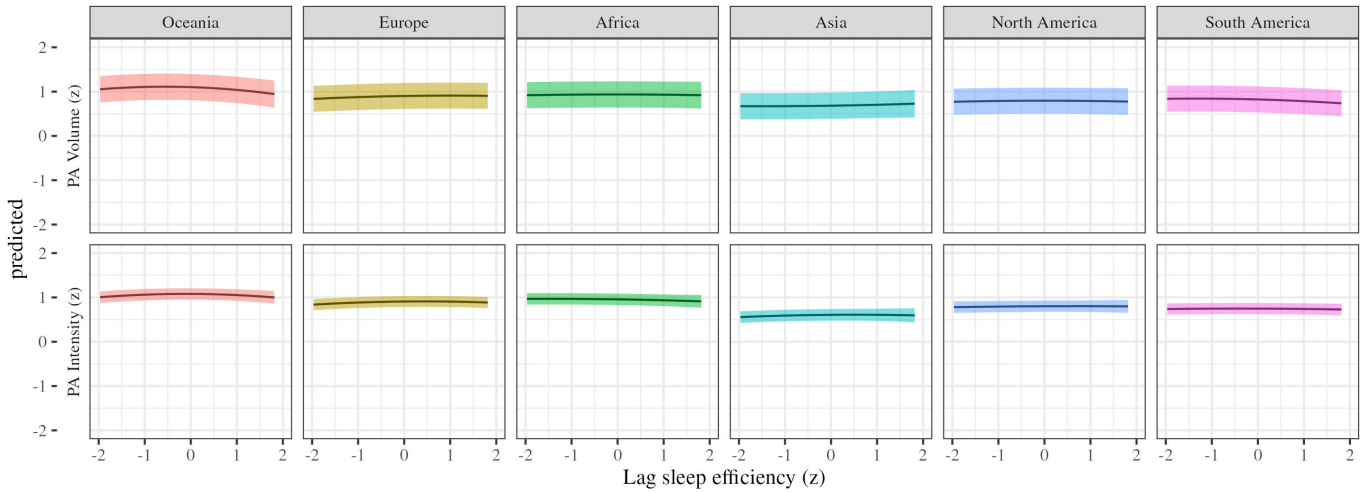

Figure 57. Physical activity by sleep efficiency moderated by region

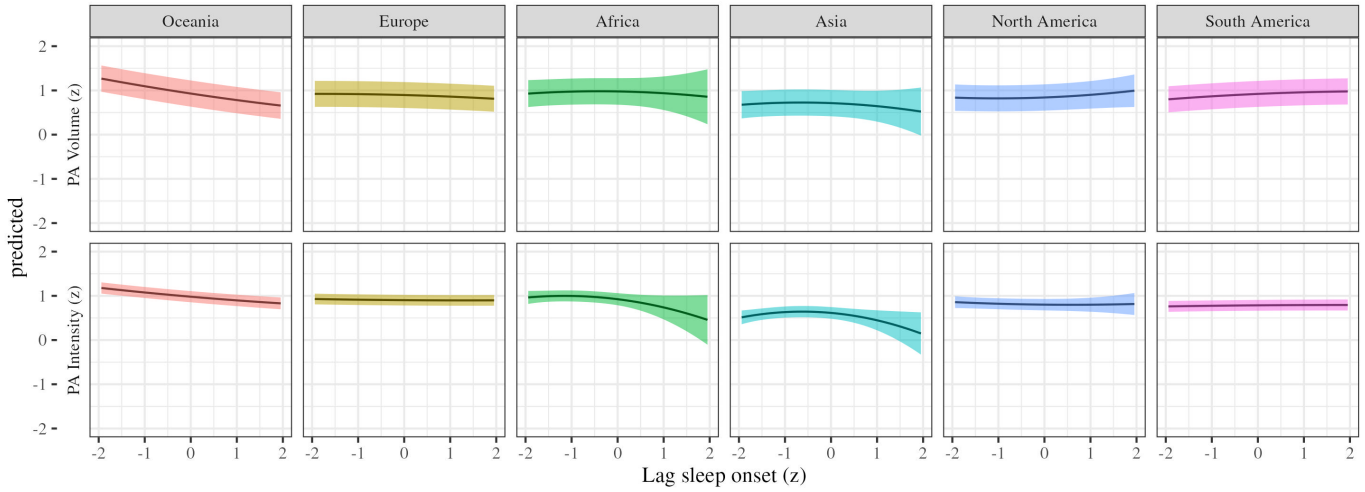

Figure 58. Physical activity by sleep onset moderated by region

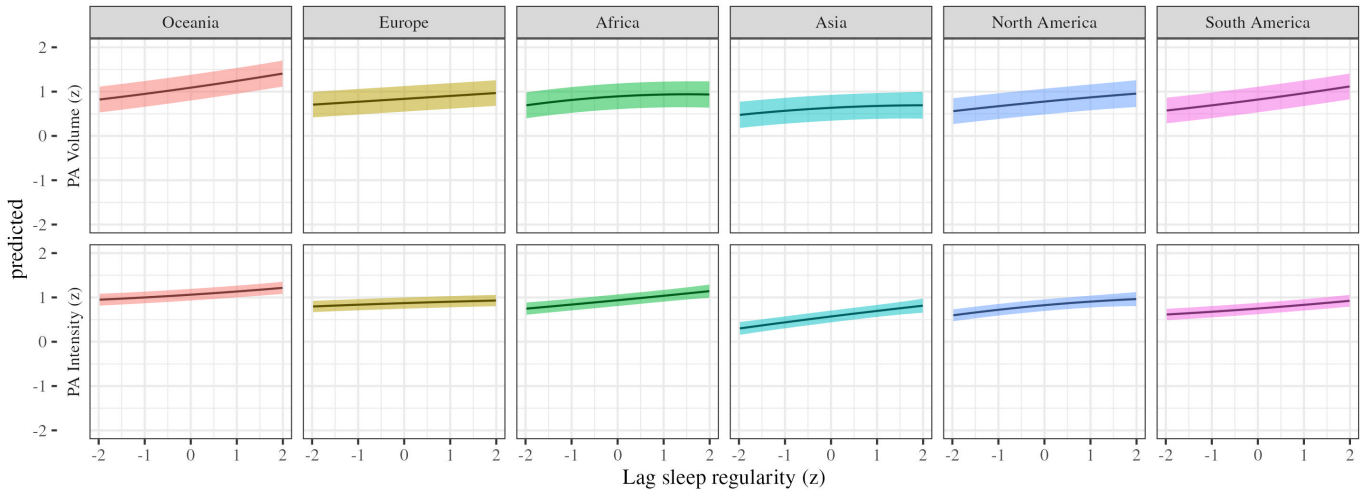

Figure 59. Physical activity by sleep regularity moderated by region

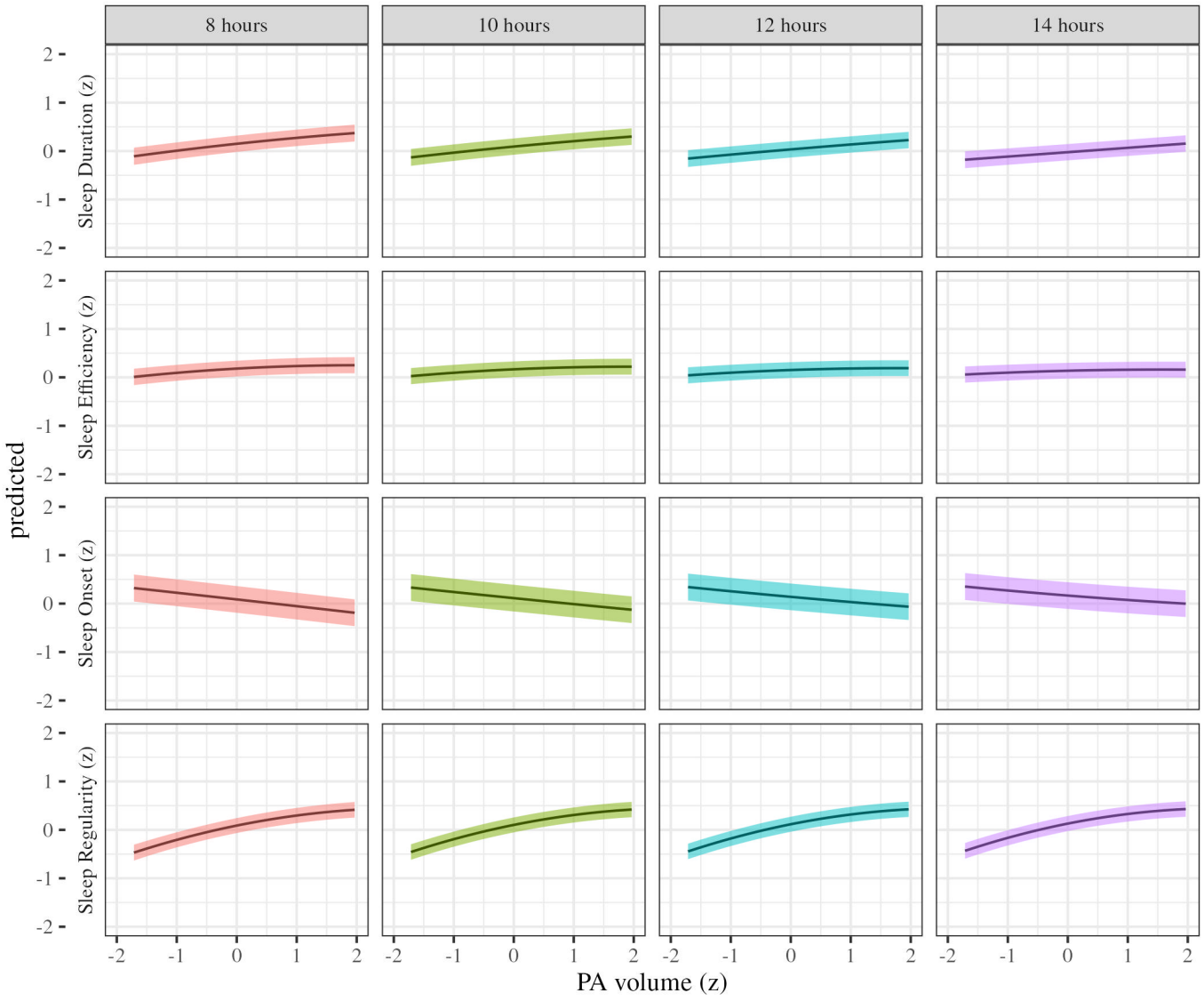

Figure 60. Sleep metrics on Physical activity volume by daylight hours

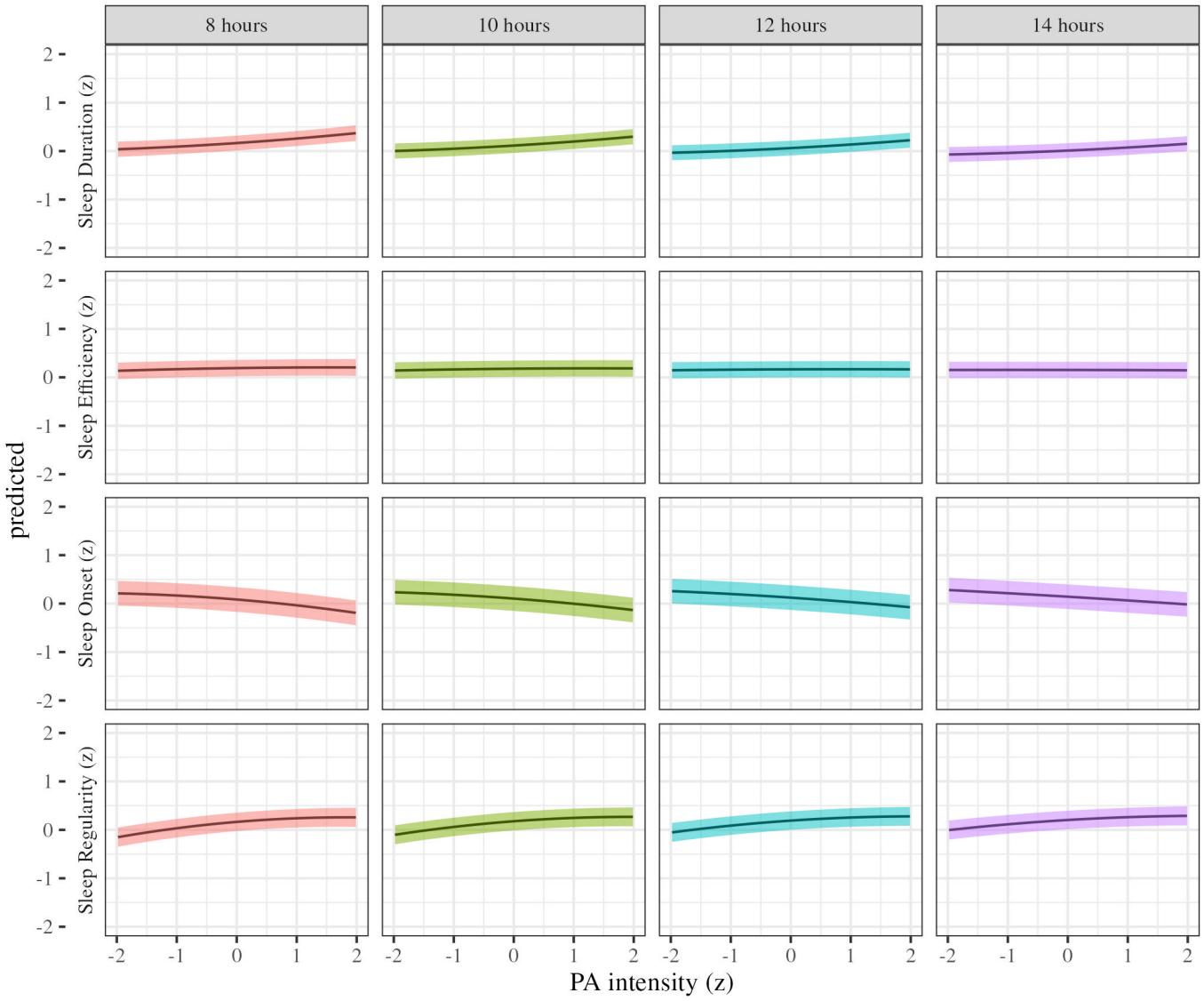

Figure 61. Sleep metrics on Physical activity intensity moderated by daylight hours

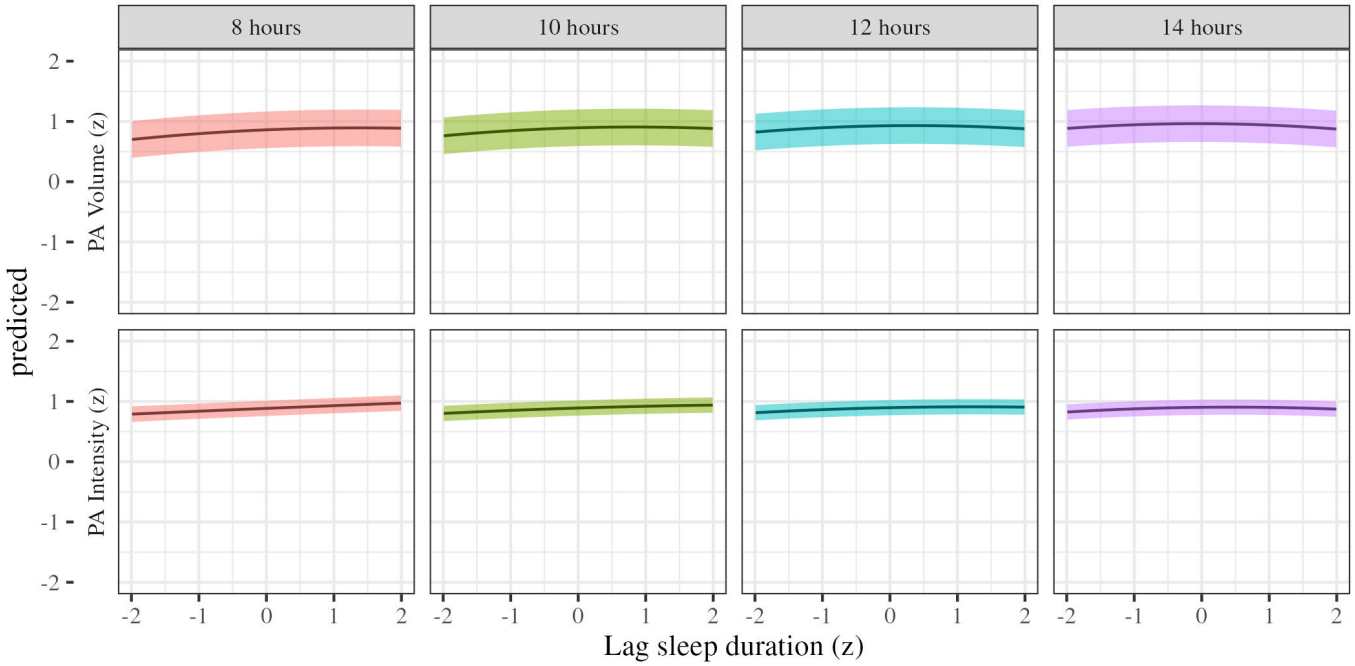

Figure 62. Physical activity by sleep duration moderated by daylight hours

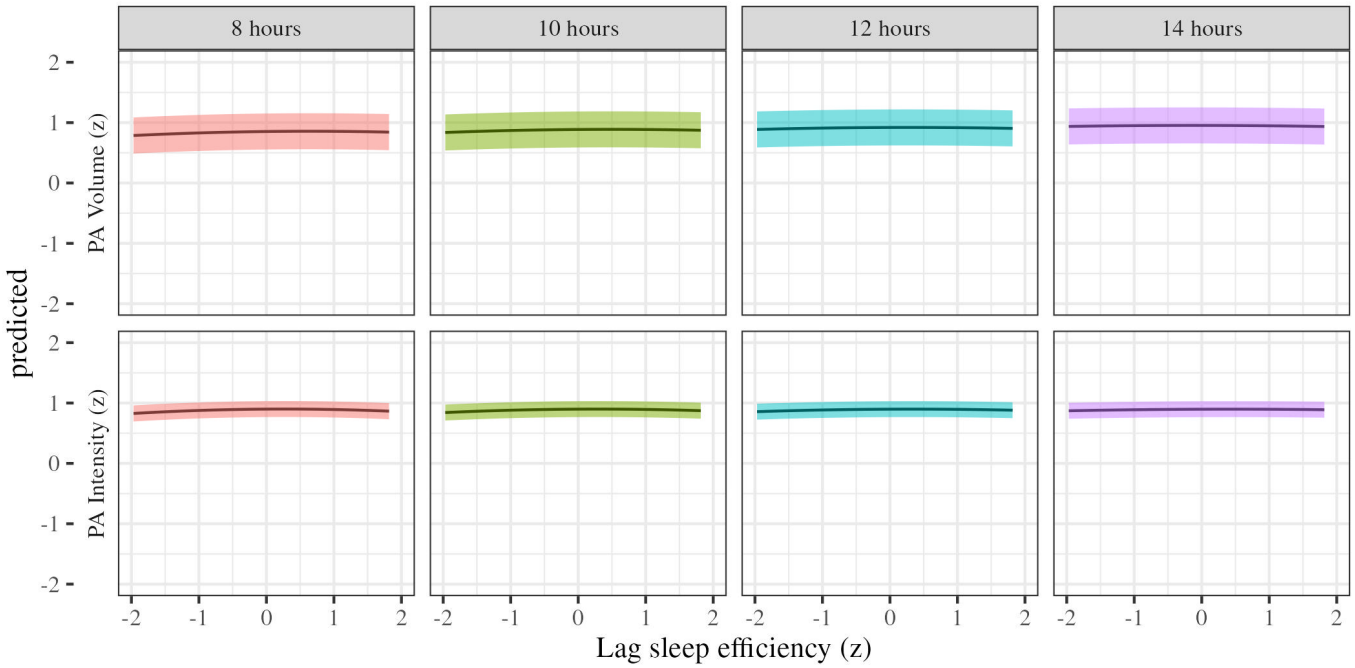

Figure 63. Physical activity by sleep efficiency moderated by daylight hours

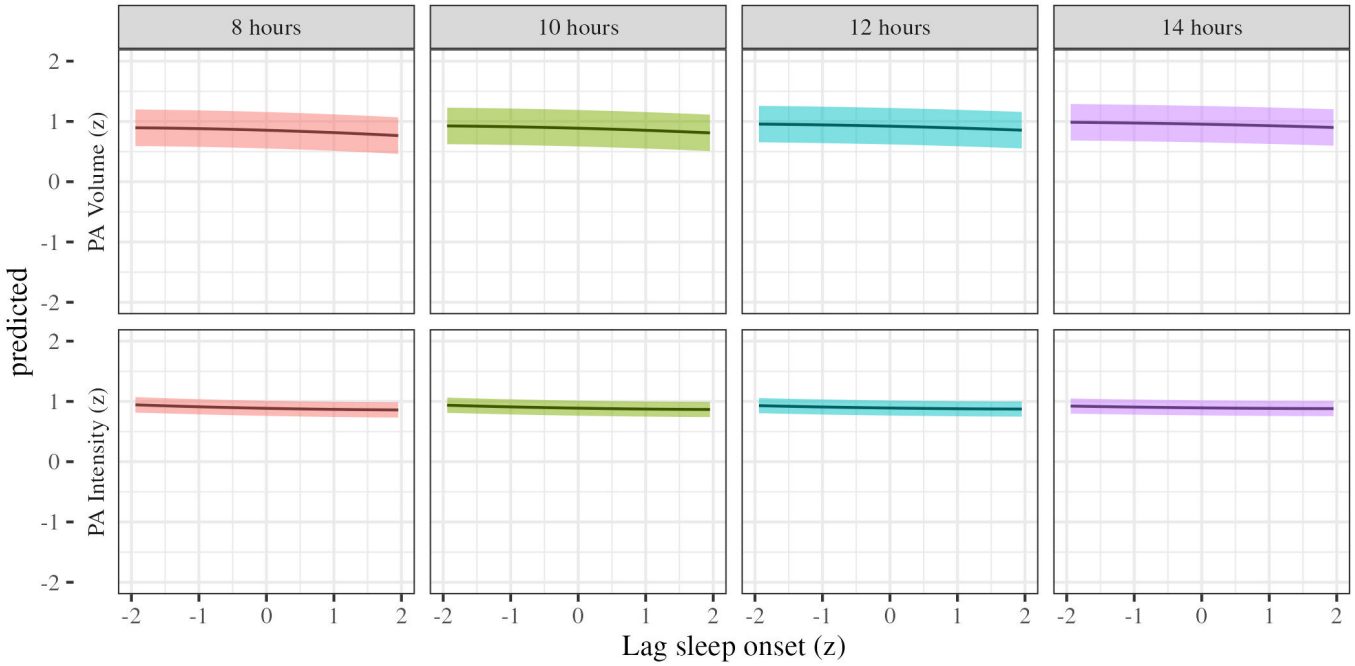

Figure 64. Physical activity by sleep onset moderated by daylight hours

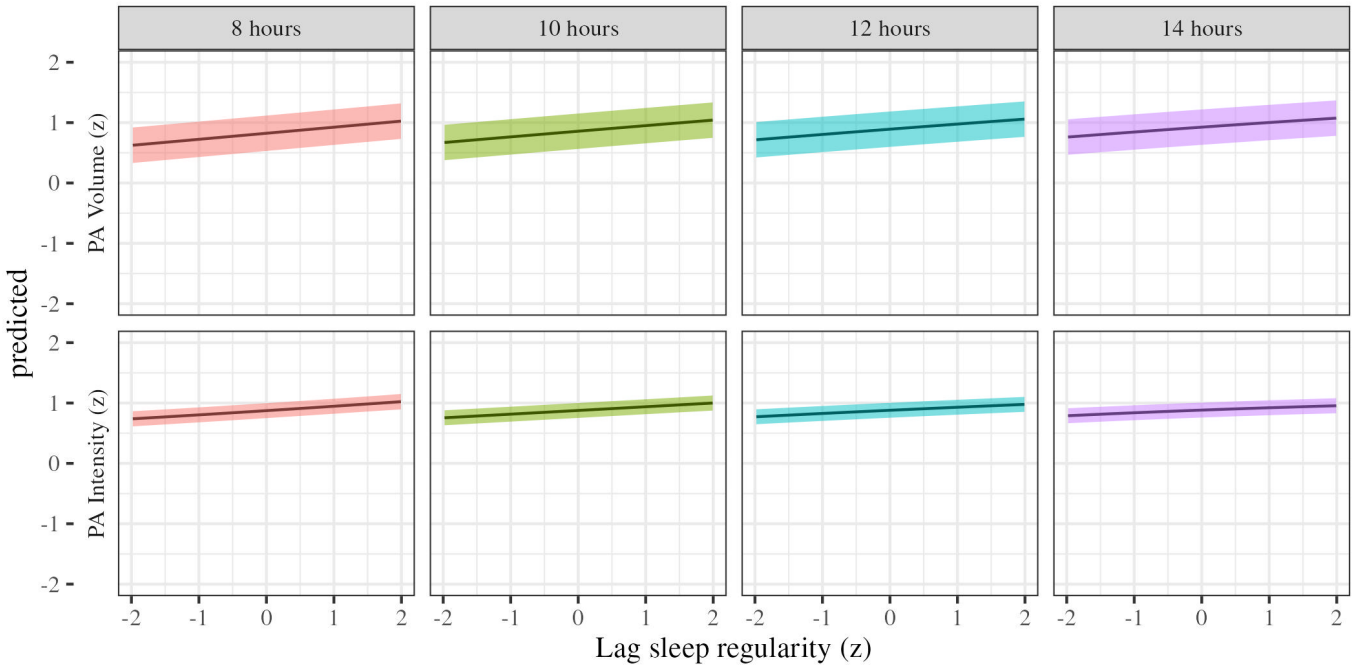

Figure 65. Physical activity by sleep regularity moderated by daylight hours

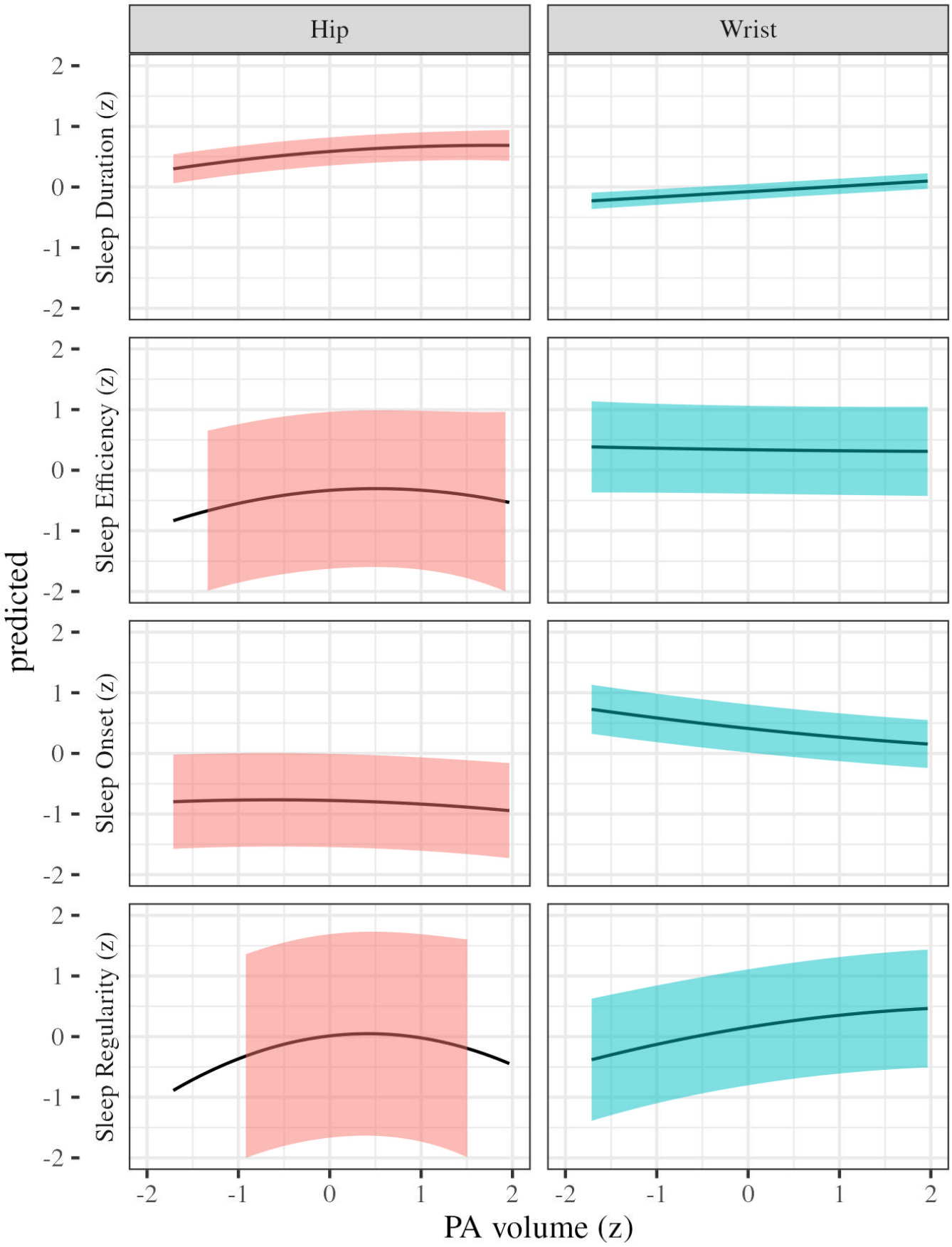

Figure 66. Sleep metrics on Physical activity volume by wear location

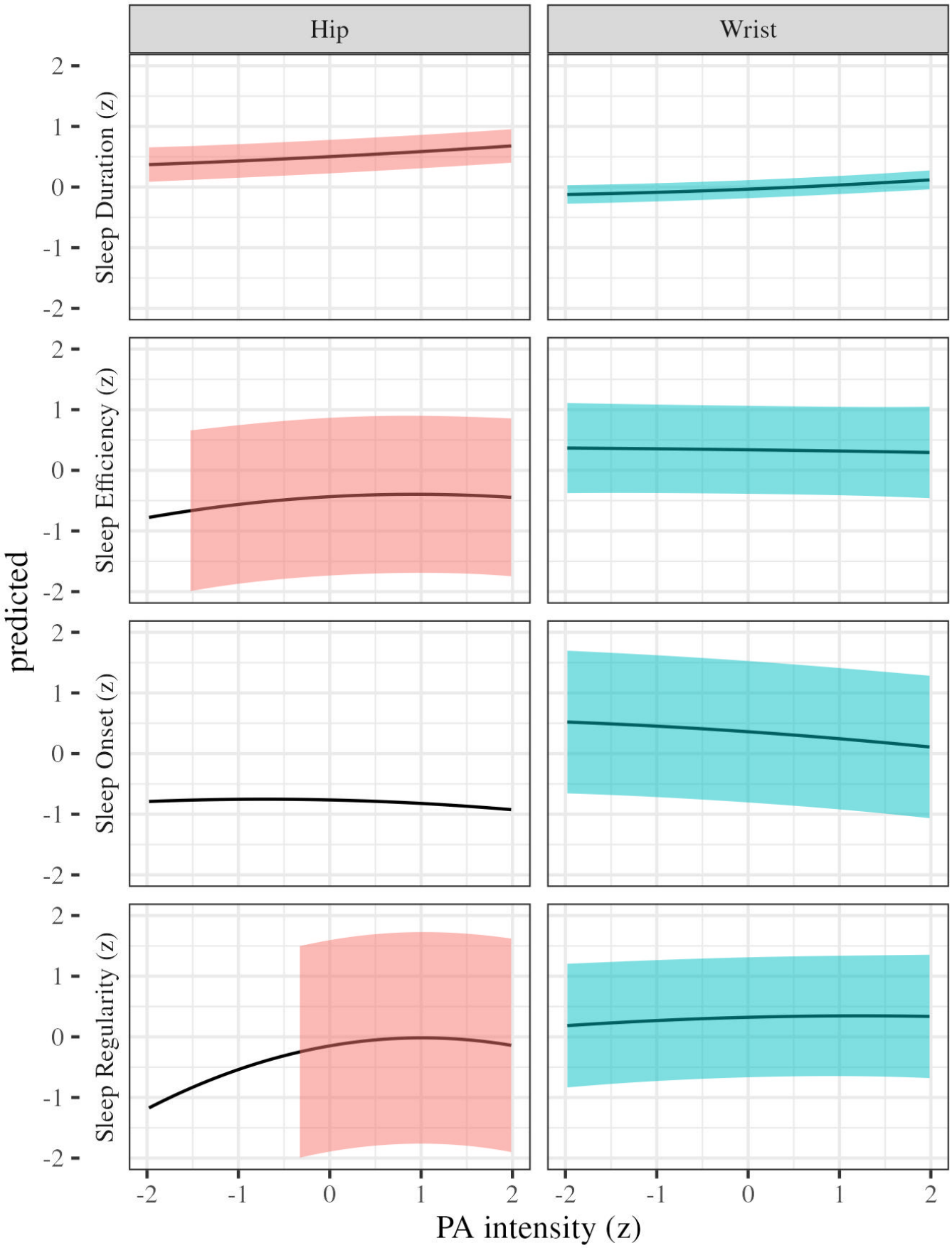

Figure 67. Sleep metrics on Physical activity intensity moderated by wear location

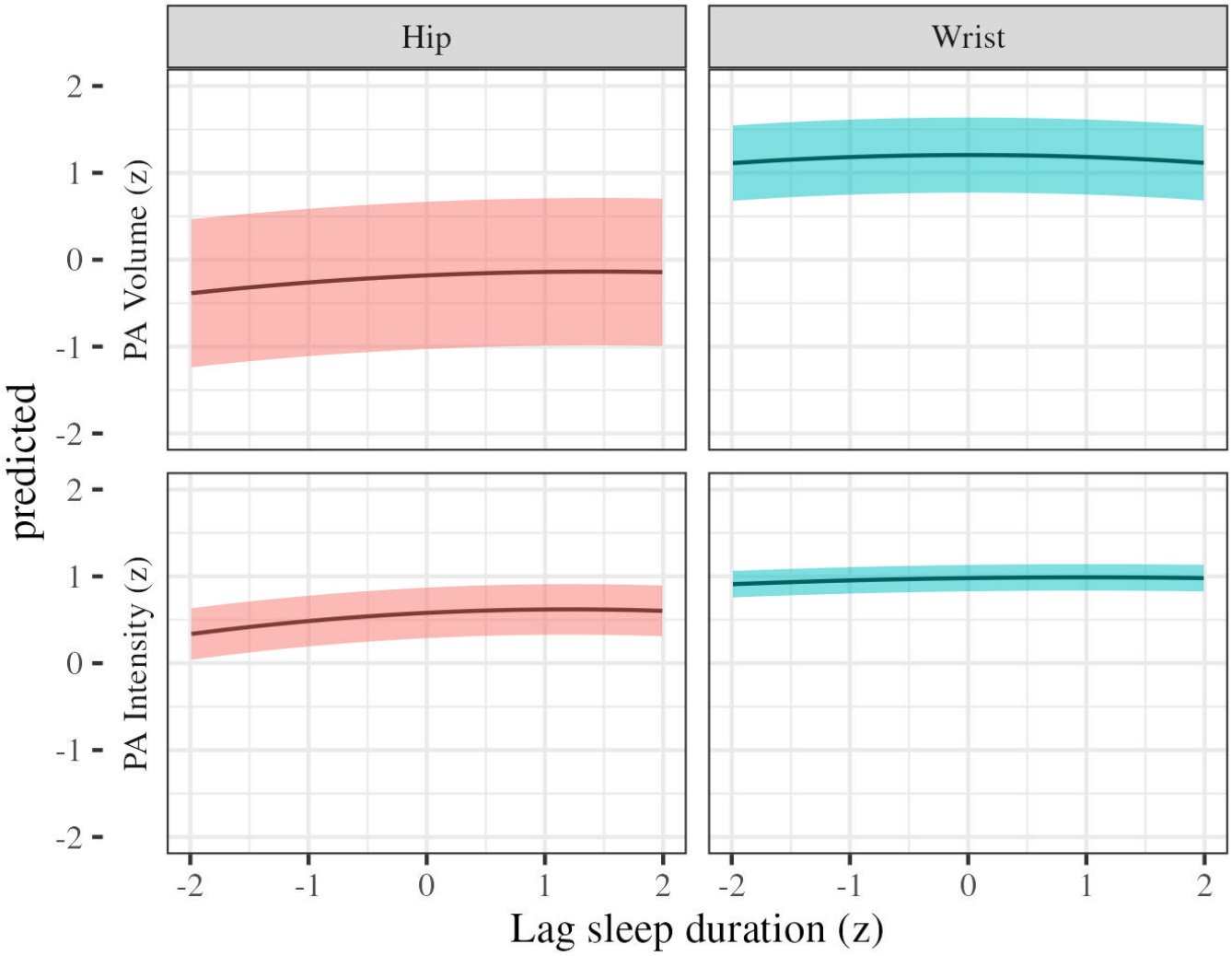

Figure 68. Physical activity by sleep duration moderated by wear location

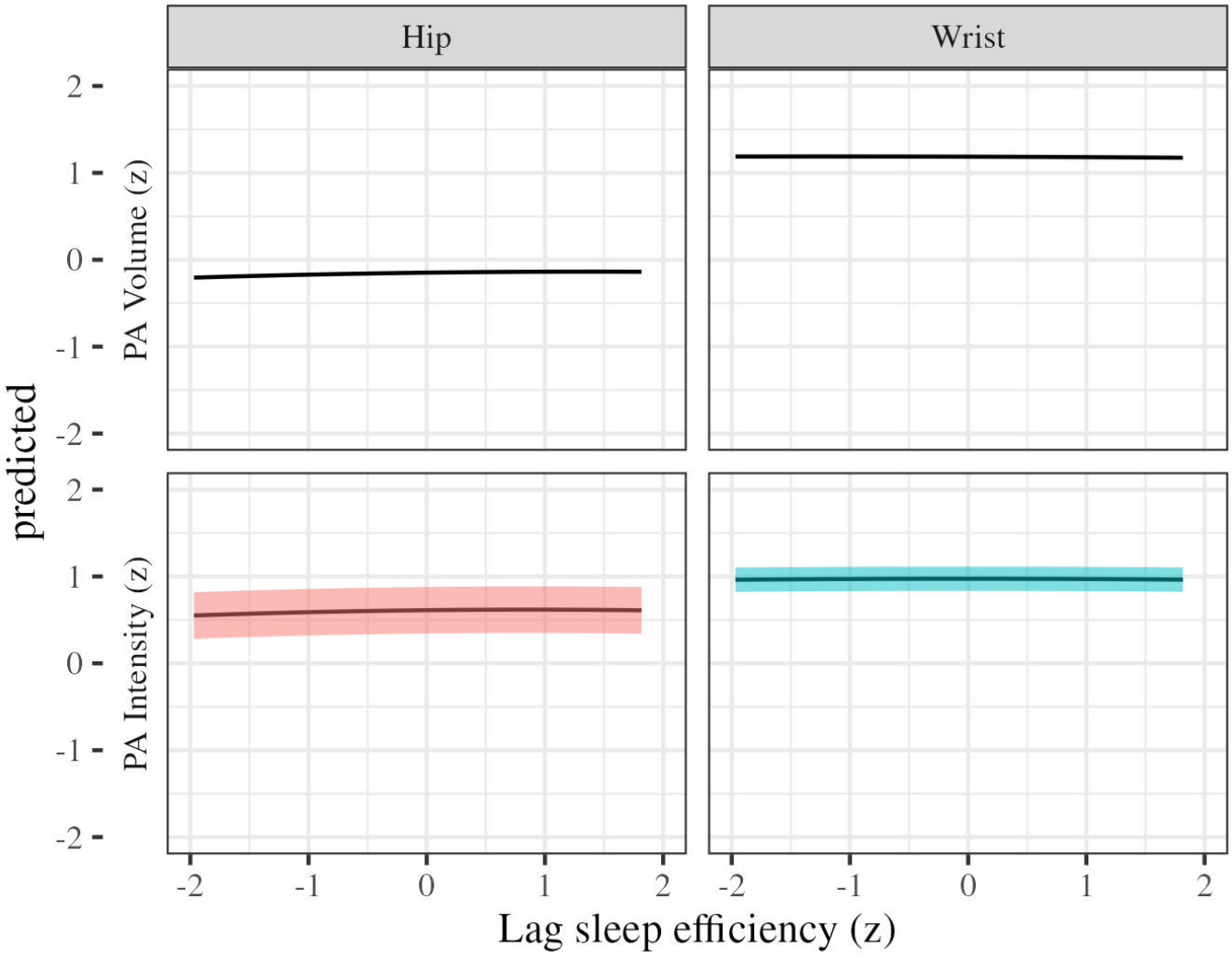

Figure 69. Physical activity by sleep efficiency moderated by wear location

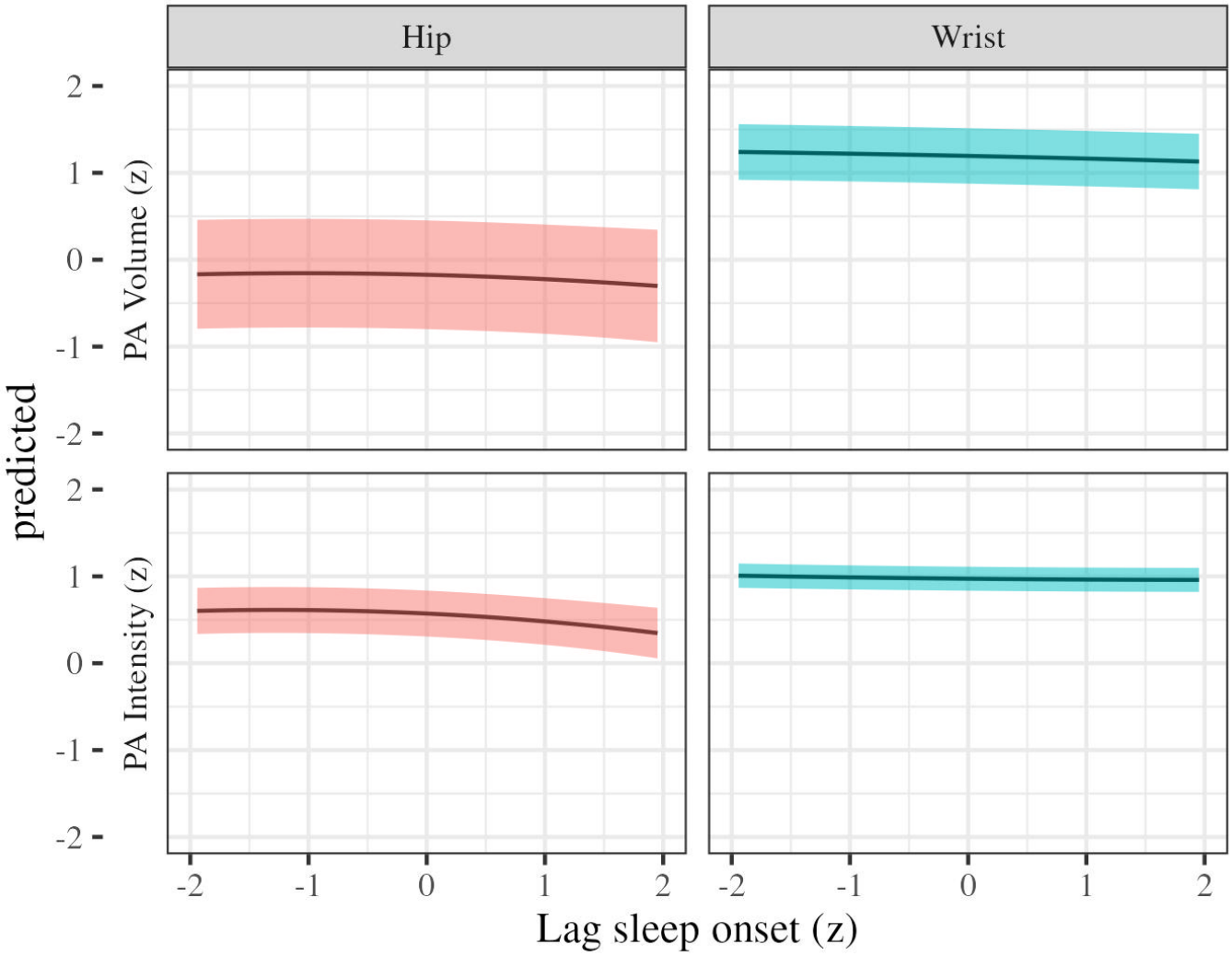

Figure 70. Physical activity by sleep onset moderated by wear location

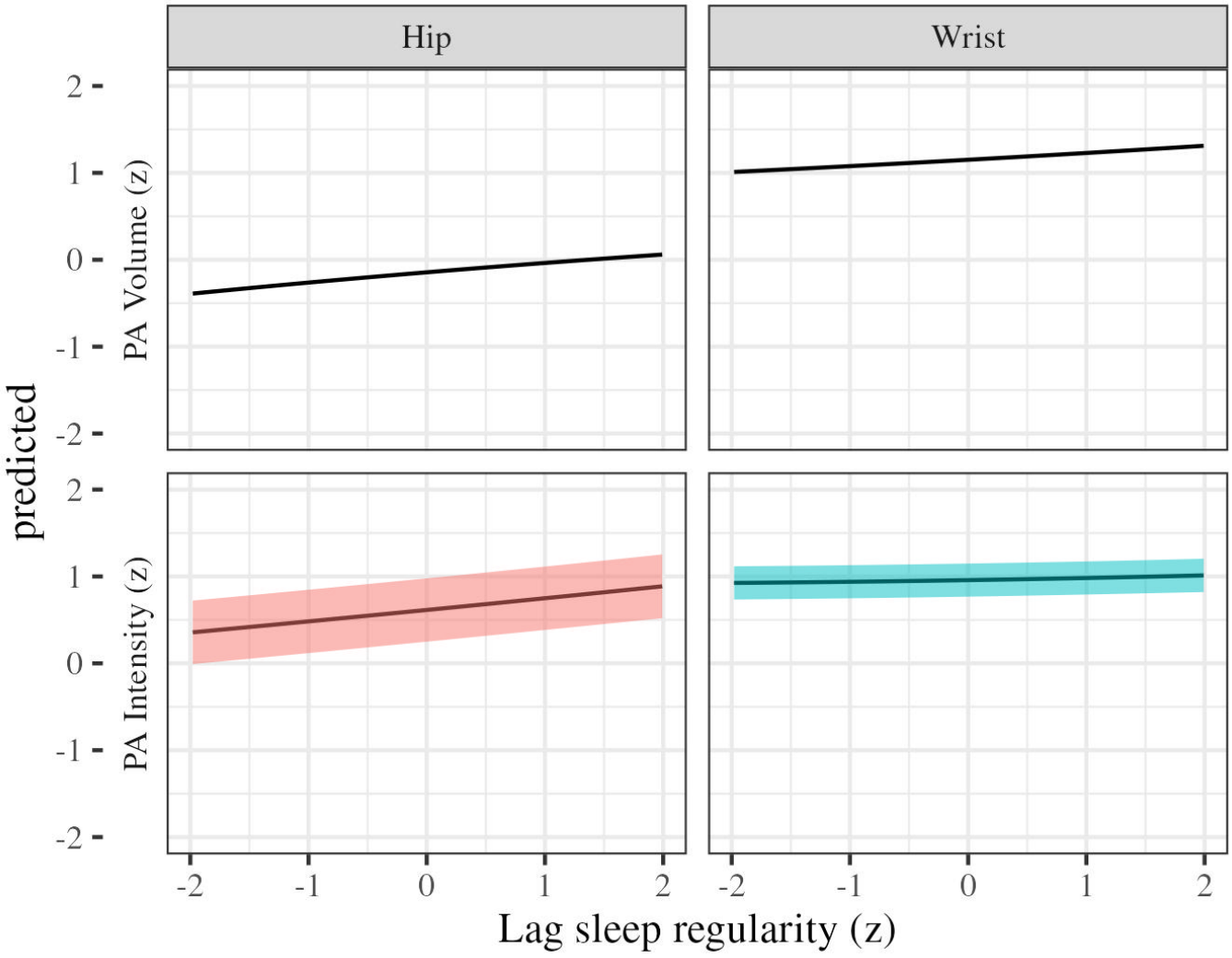

Figure 71. Physical activity by sleep regularity moderated by wear location

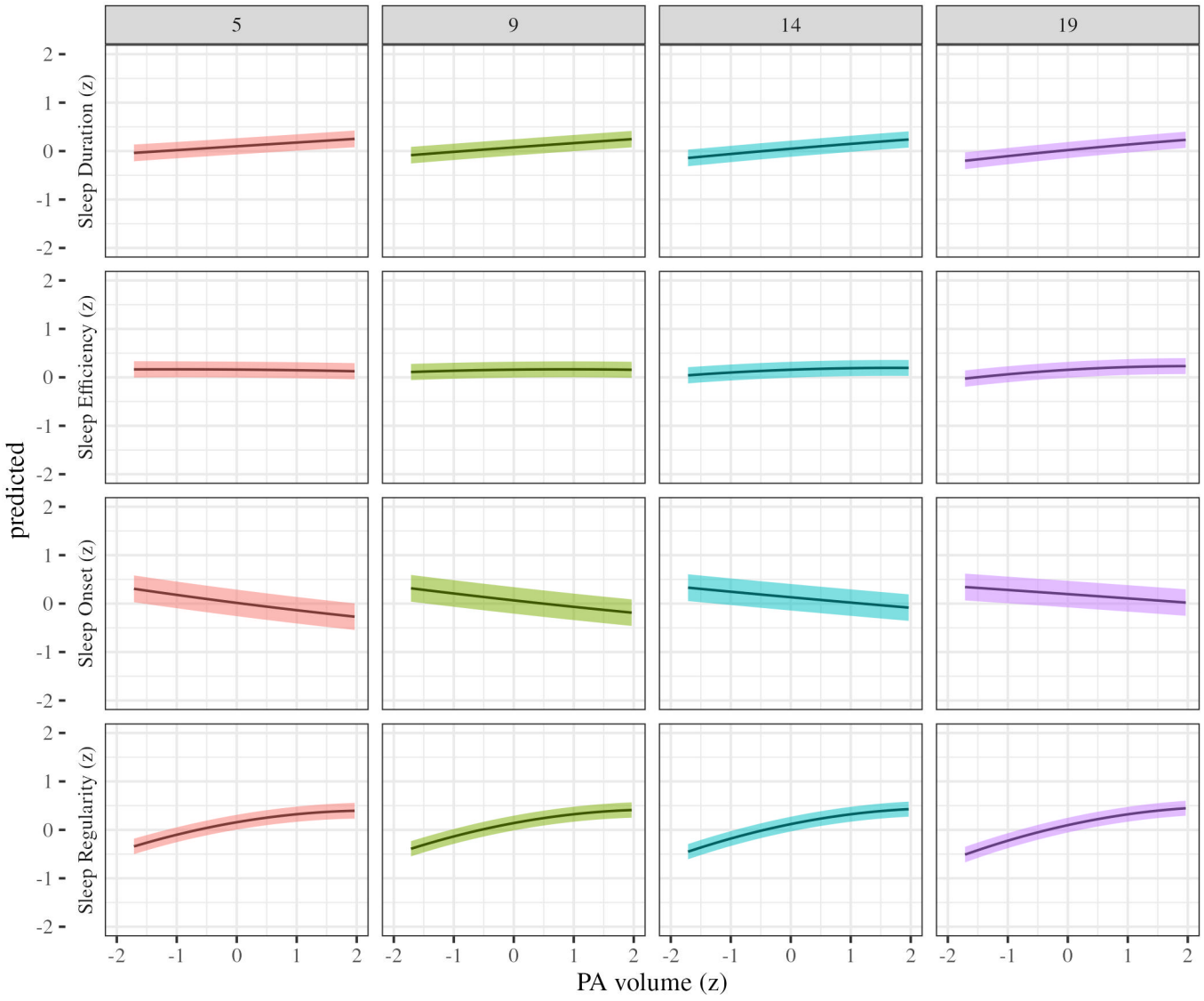

Figure 72. Sleep metrics on Physical activity volume by most active hour

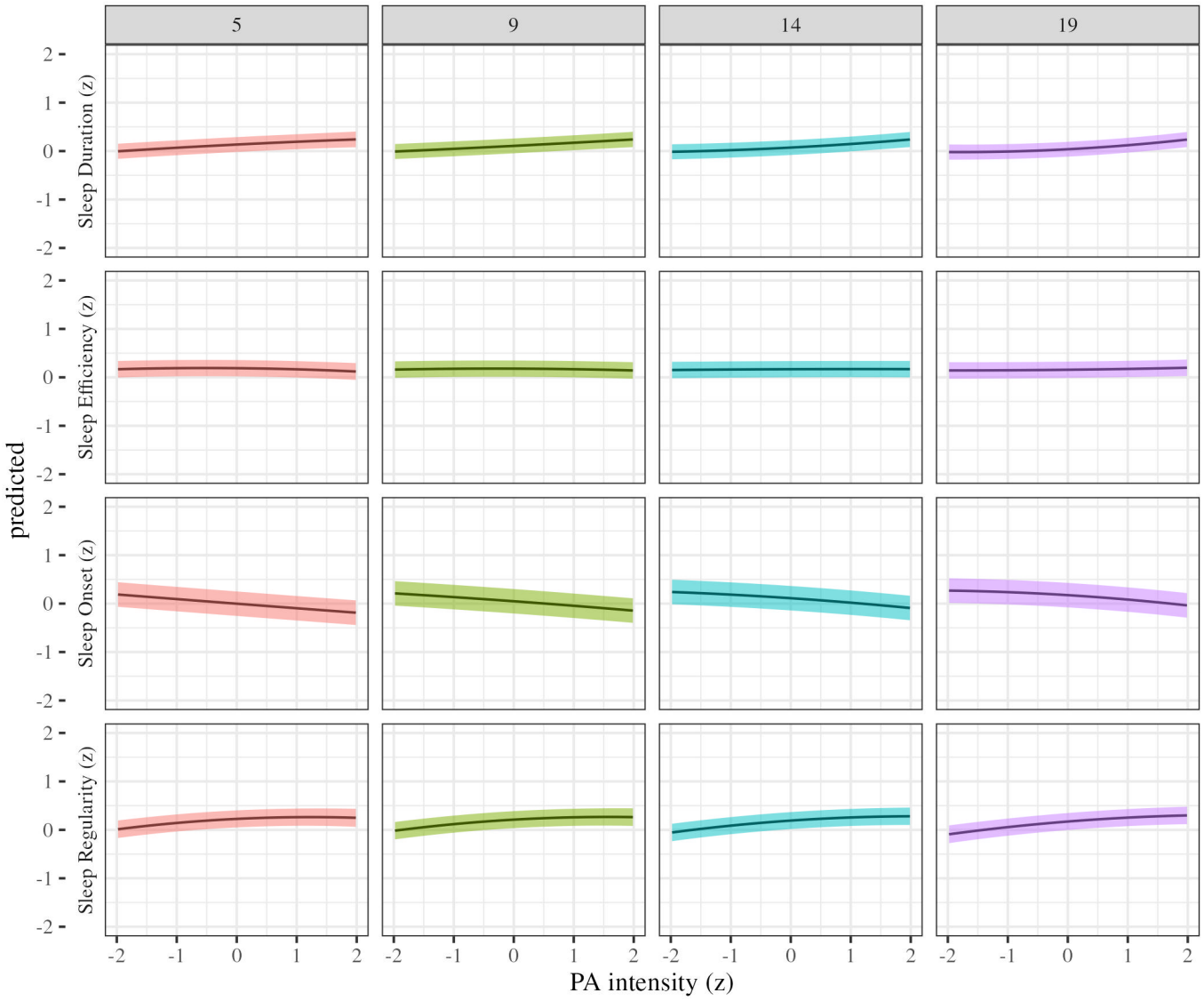

Figure 73. Sleep metrics on Physical activity intensity moderated by most active hour

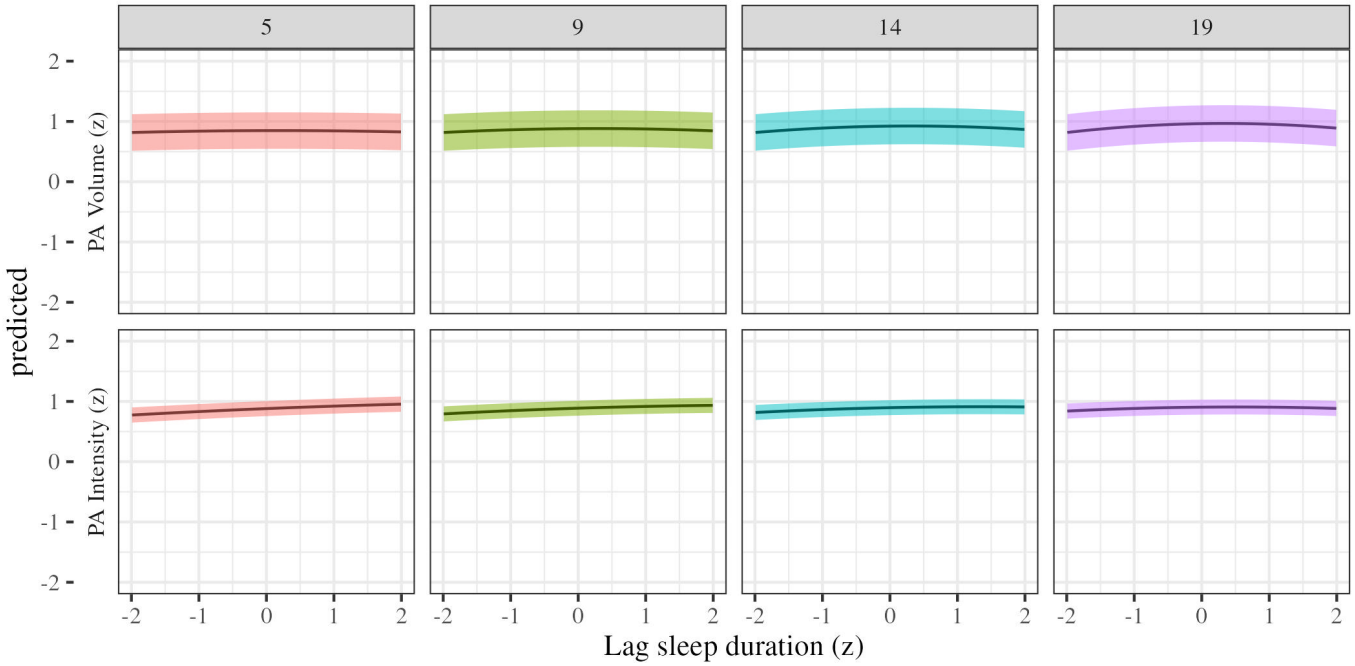

Figure 74. Physical activity by sleep duration moderated by most active hour

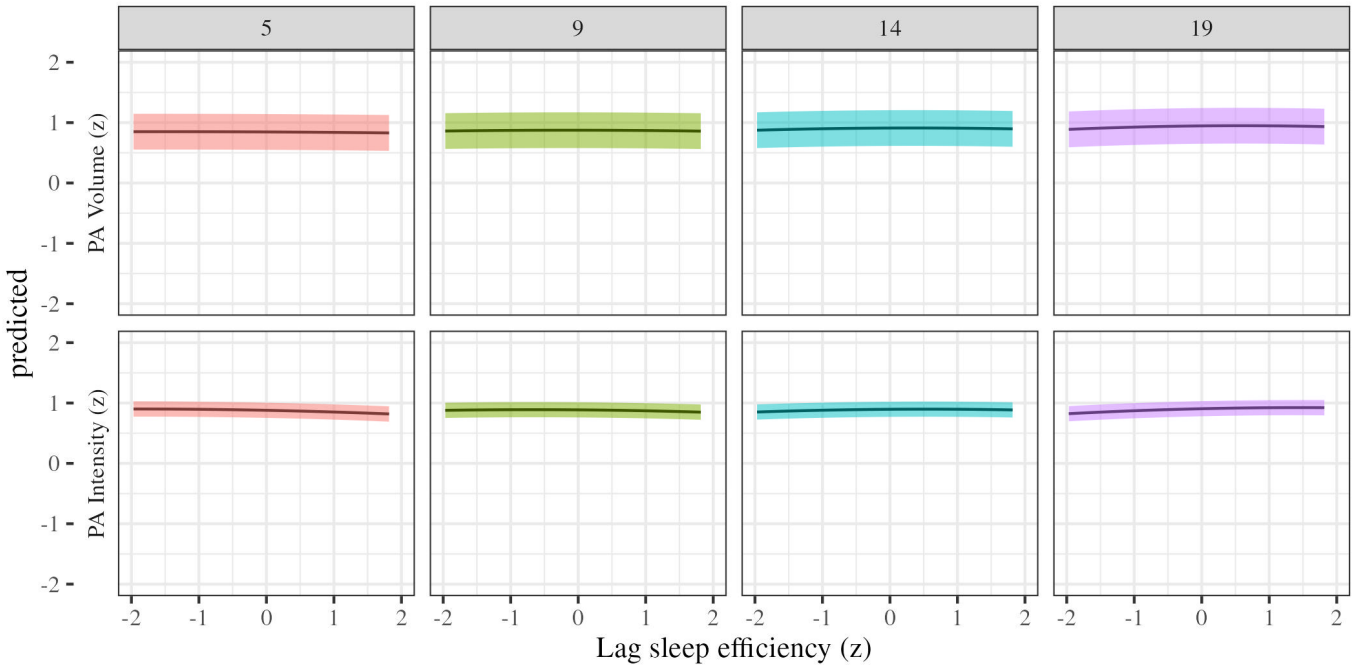

Figure 75. Physical activity by sleep efficiency moderated by most active hour

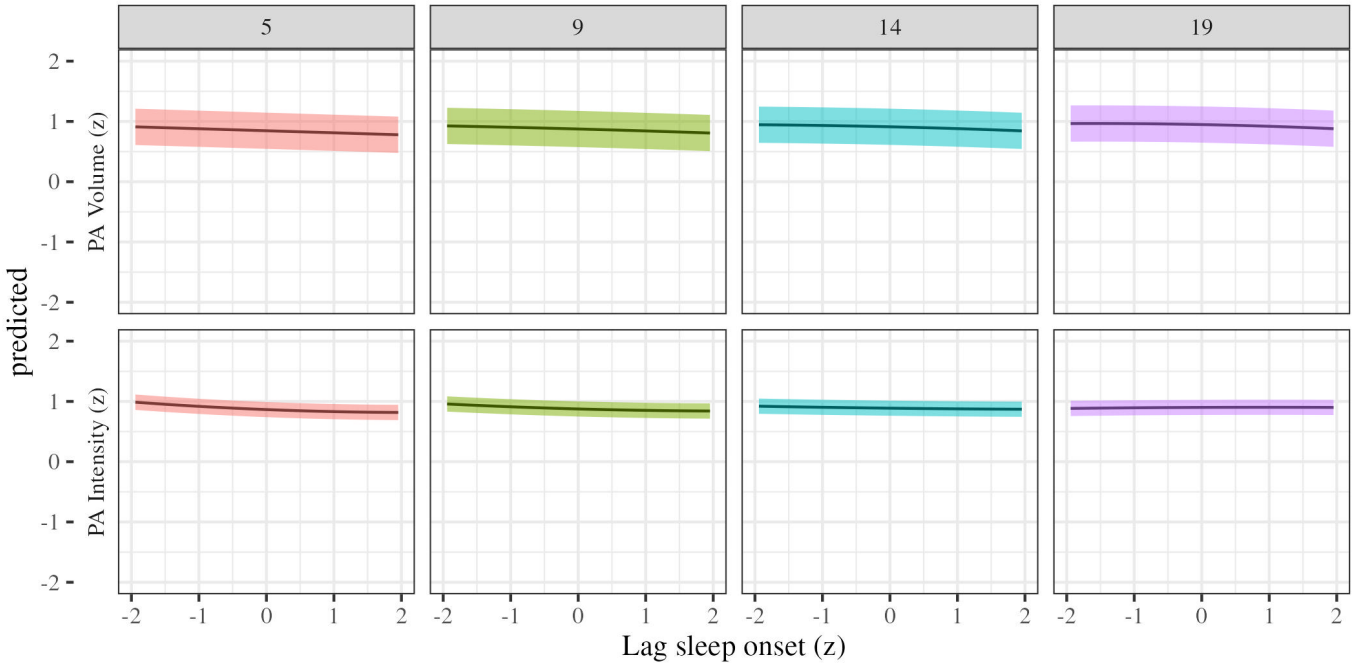

Figure 76. Physical activity by sleep onset moderated by most active hour

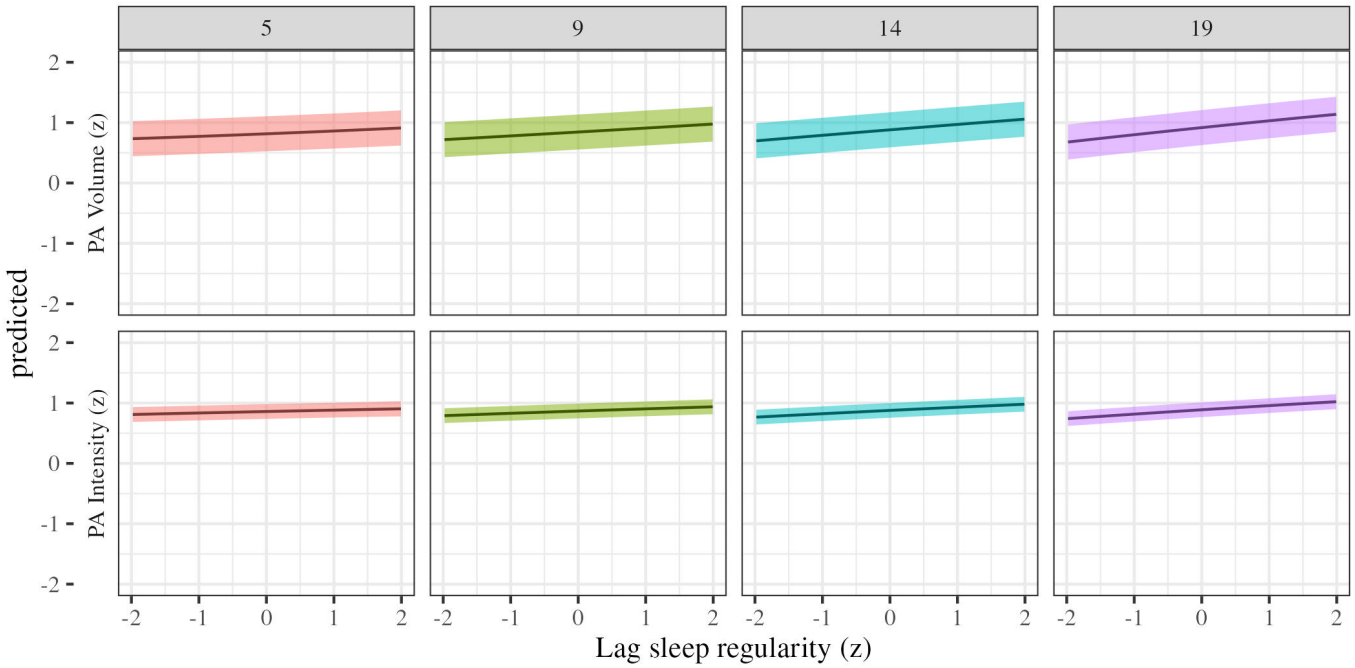

Figure 77. Physical activity by sleep regularity moderated by most active hour

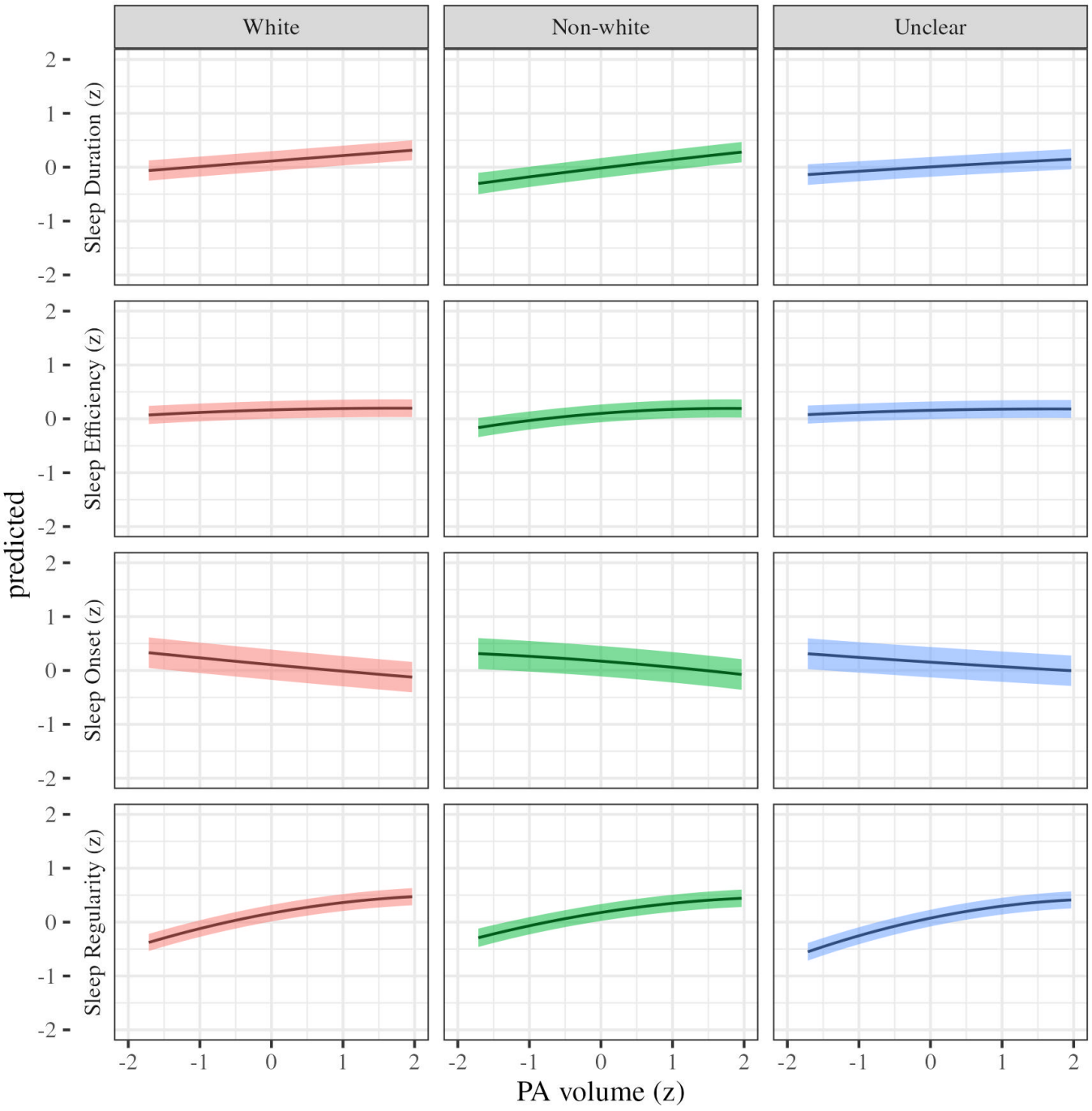

Figure 78. Sleep metrics on Physical activity volume by ethnicity

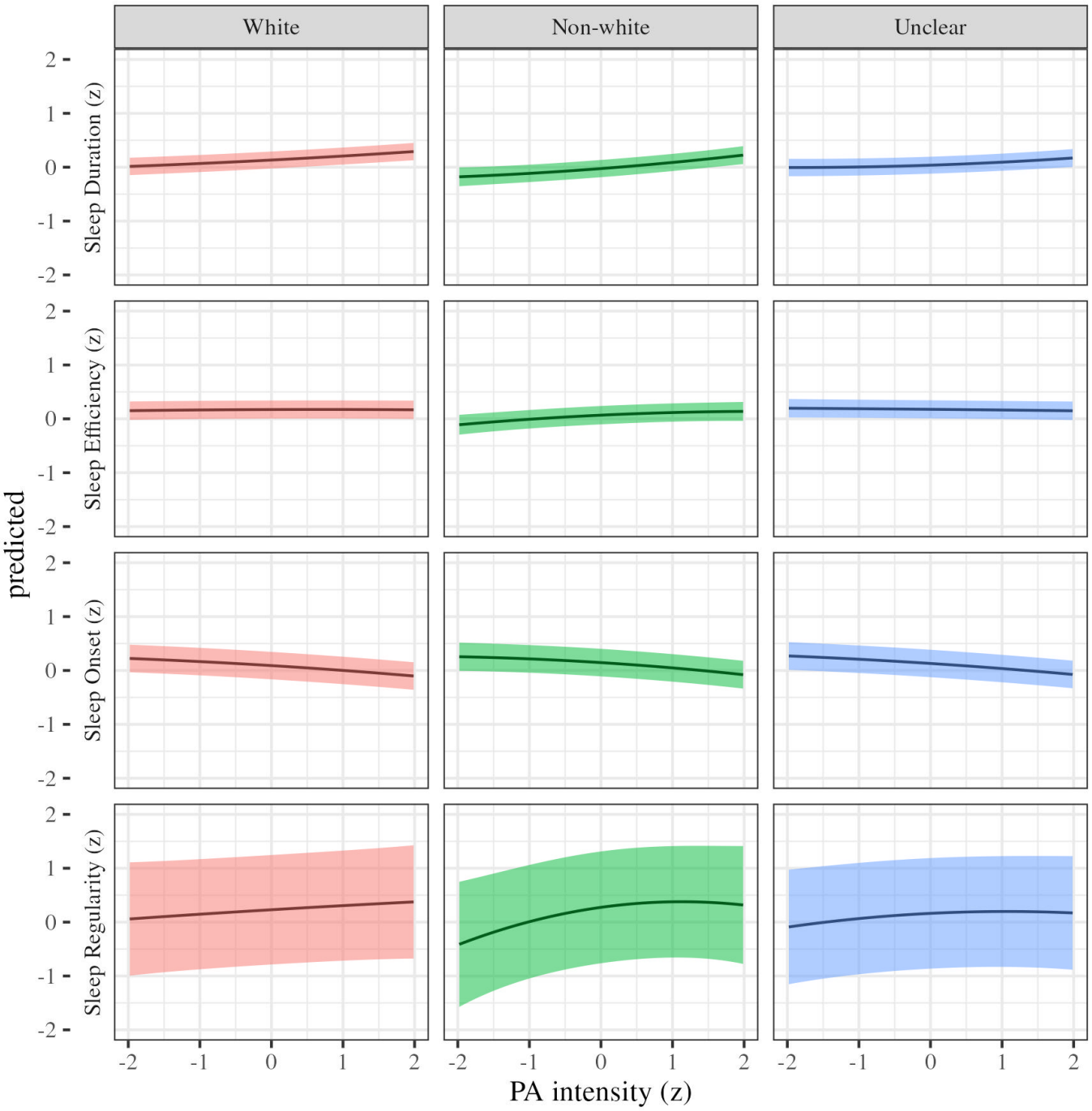

Figure 79. Sleep metrics on Physical activity intensity moderated by ethnicity

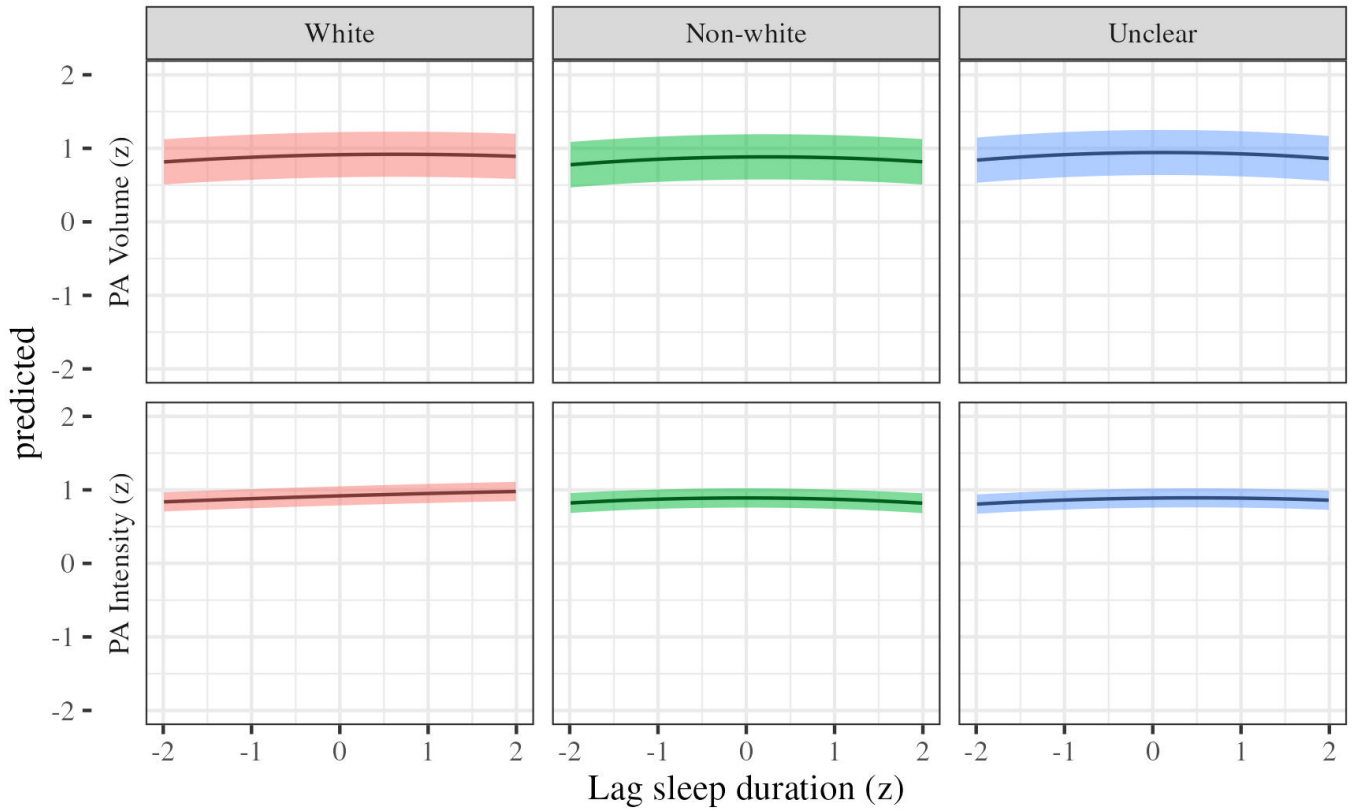

Figure 80. Physical activity by sleep duration moderated by ethnicity

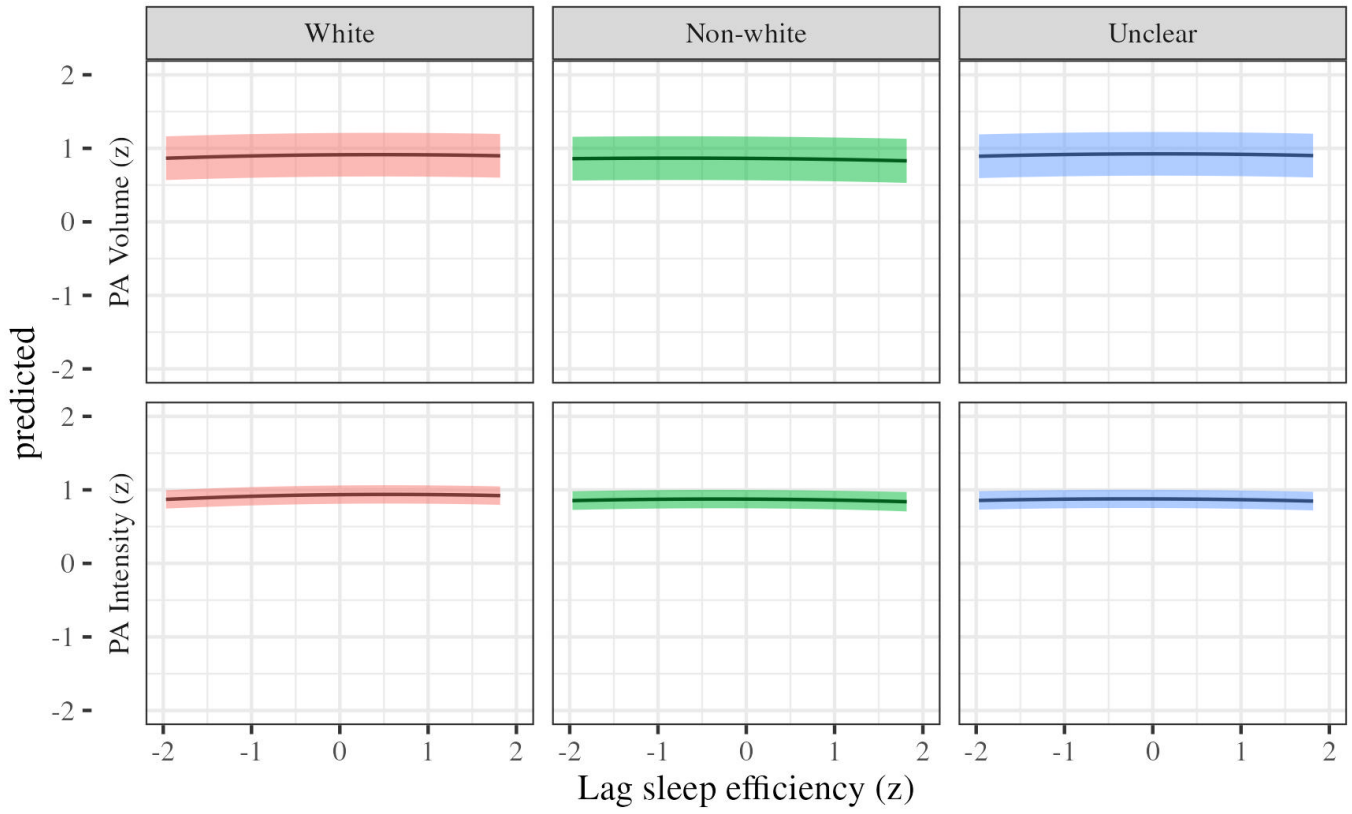

Figure 81. Physical activity by sleep efficiency moderated by ethnicity

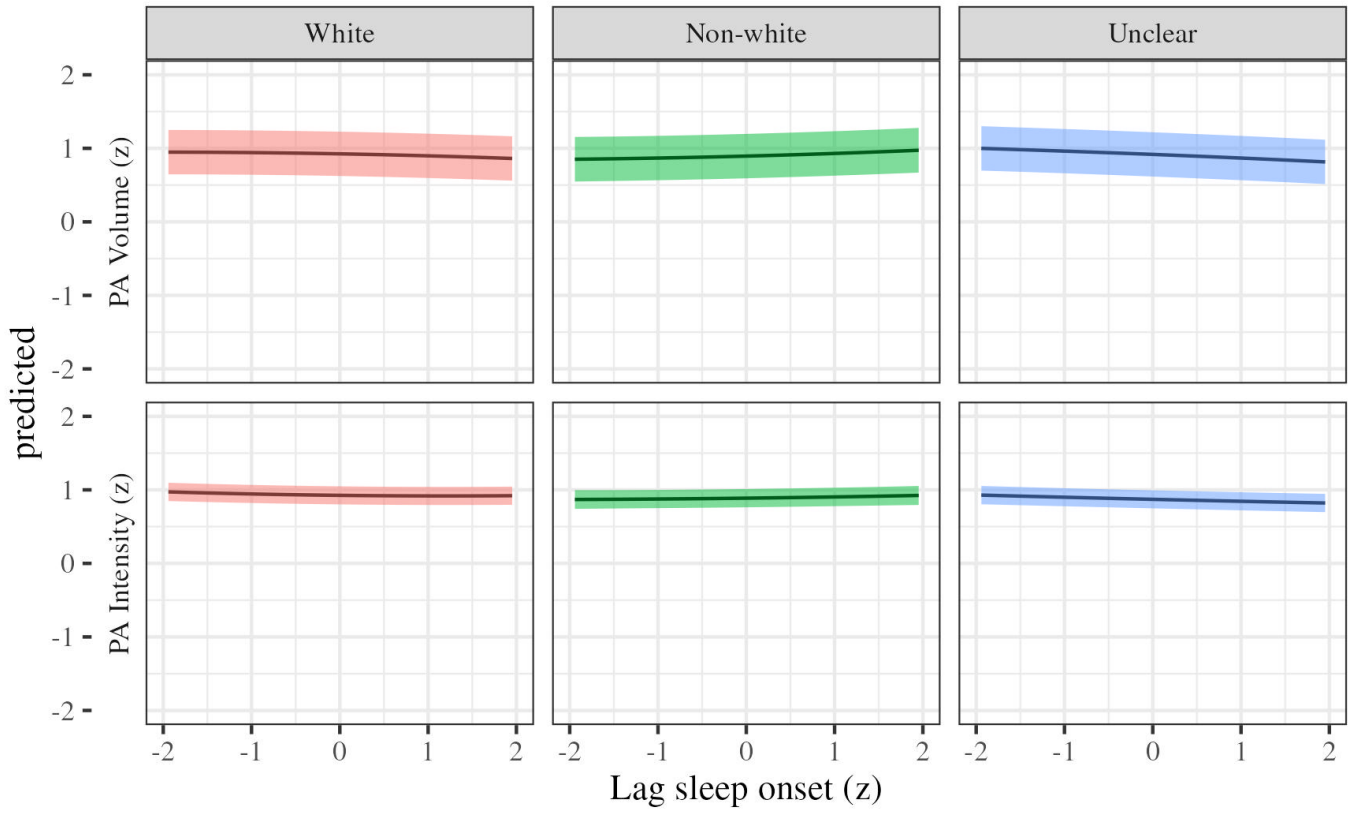

Figure 82. Physical activity by sleep onset moderated by ethnicity

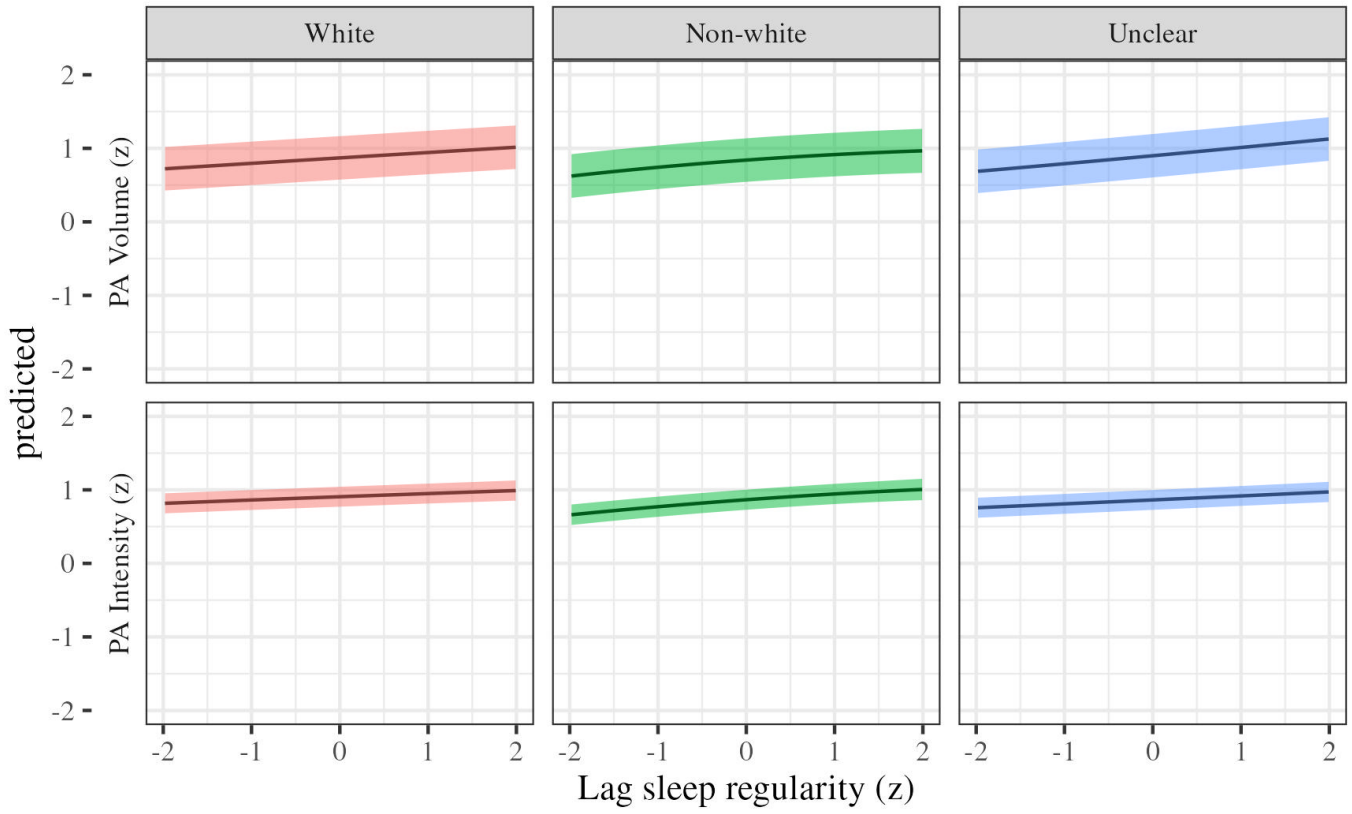

Figure 83. Physical activity by sleep regularity moderated by ethnicity